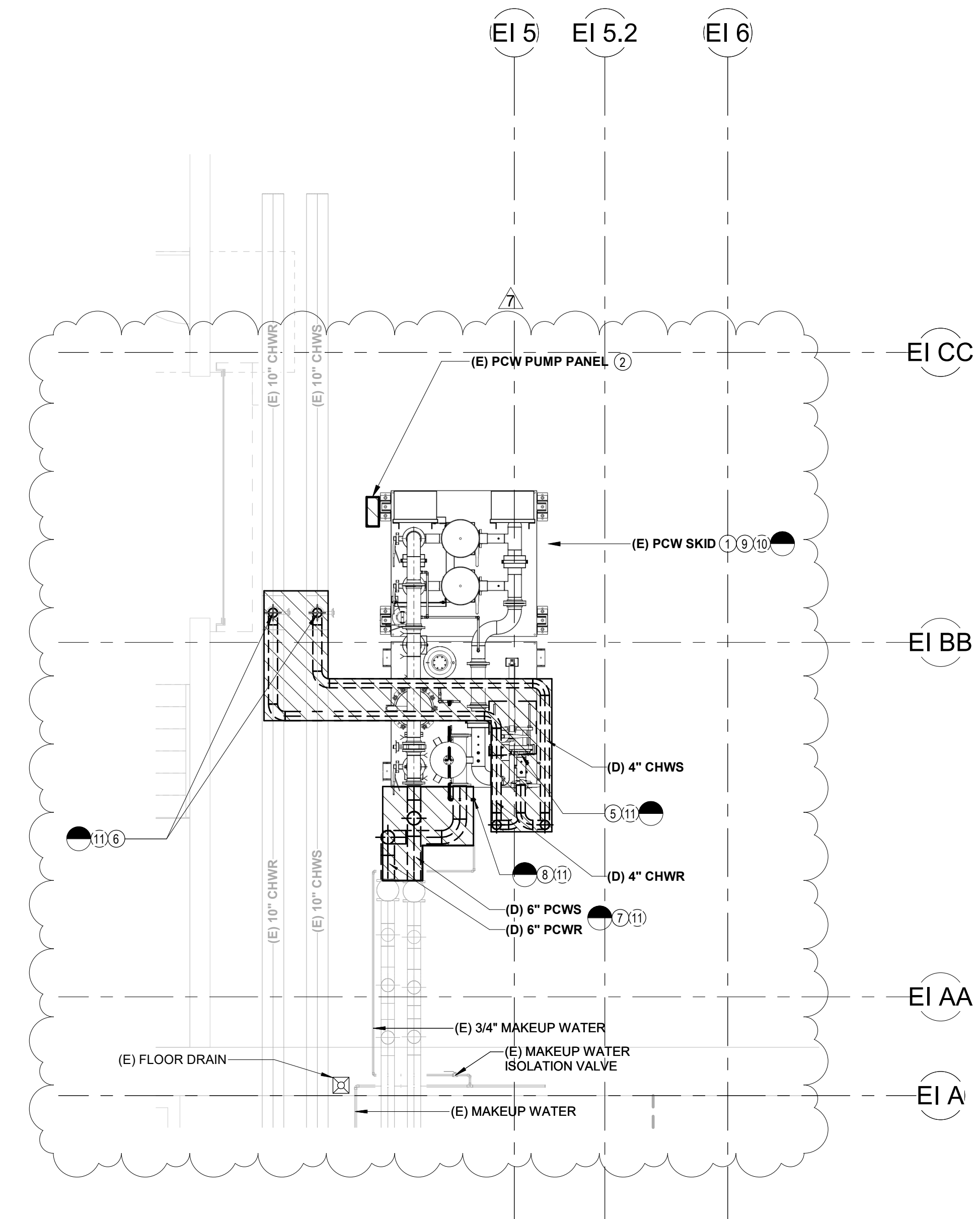
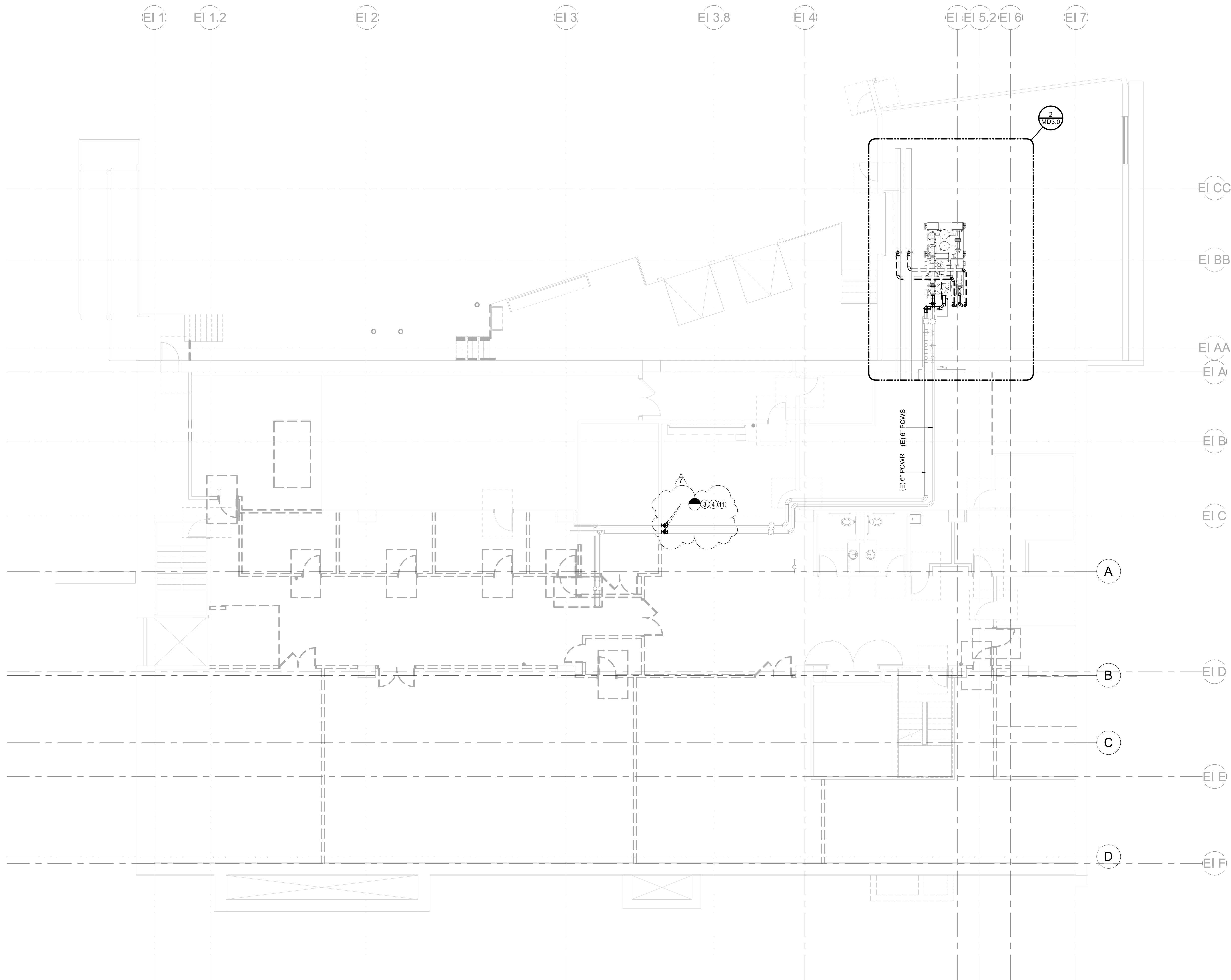


REV	DESCRIPTION	DRN	CHK	APP	DATE
0	ISSUE FOR CONSTRUCTION				25 MAY 14
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



2 BASEMENT (NEH) - PROCESS PIPING OVERALL PLAN - DEMO - ENLARGED
MD3.0 1/4" = 1'-0"

1 BASEMENT (NEH) - PROCESS PIPING OVERALL PLAN - DEMO
MD3.0 1/8" = 1'-0"

GENERAL NOTES

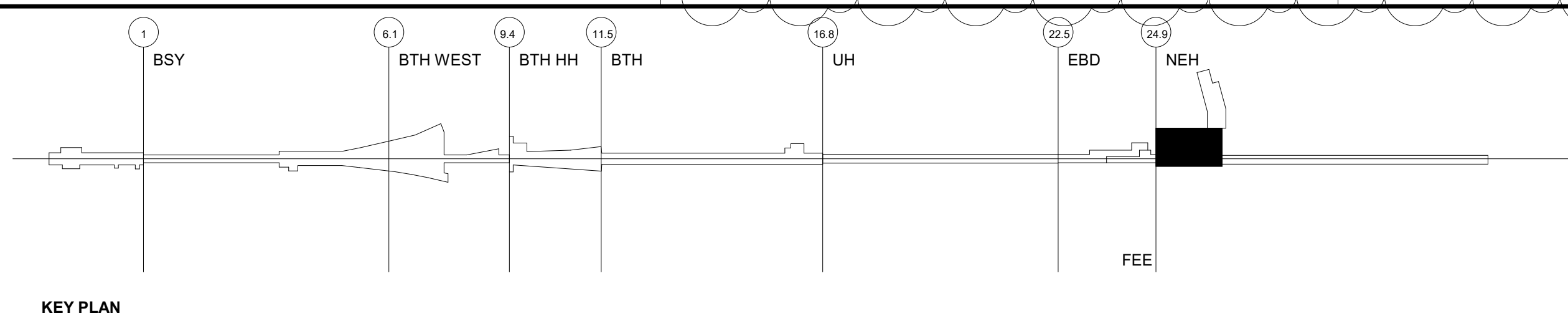
- A. PROVIDE TO SHEET 00 AND 00.1 FOR INDEX OF DRAWINGS & GENERAL NOTES.
- B. DROP LOCATIONS CALLOUT ON PLAN ARE FOR REFERENCE ONLY. USE THE CALLOUT FOR EXACT LOCATION. DROP CALLOUT IS DISTANCE FROM 0.00 DLINE. PROVIDE MO.1 FOR LOCATION. CALLOUT ON PLAN IS SHOWN AS Z. XXX'-X" OR ZXX'-X".
- C. THE (U) PIPING TAG INDICATES THE PIPE IS RUN UNDER THE RAISED FLOOR.
- D. PROVIDE FIRE CULKING BETWEEN ROOMS WHENEVER REQUIRED.
- E. ALL FIRE SPRINKLER MODIFICATIONS REQUIRED SHALL BE DONE BY LICENSED B-3 CONTRACTOR AND BE SLAC FIRE MARSHAL APPROVED. FIRE SPRINKLER CONTRACTOR SHALL PROVIDE MODIFICATIONS AND CALCULATIONS FOR ALL FIRE SPRINKLER MODIFICATIONS.
- F. NO SNAKEBIT/COMPRESSION FITTINGS WITH ALL ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.
- G. ALL MANUAL ALARMS AND DRAINS ON ANY POWWILL SYSTEM SHOULD BE TO PLUMBED BACK TO THE PIPING HOUSING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.
- H. INCLUDE ISOLATION VALVES (AND TEES) FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.
- I. ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC SME THROUGH SHIP DRAWING SUBMITTAL.

KEYED NOTES:

1. DISCONNECT AND CAREFULLY REMOVE AND CAP FOR FUTURE USE. (E) POW SCID INCLUDING ALL EQUIPMENT ON THE SKID, CONTROL PANELS AND INSTRUMENTS, AND ALL PIPING AND INSTRUMENTATION. COORDINATE THE STORAGE OF (E) SKID WITH SLAC PROJECT TEAM.
2. DISCONNECT AND CAREFULLY REMOVE (E) POW PUMP CONTROL PANEL INCLUDING (E) SUPPORTS TO BE STORED FOR FUTURE. COORDINATE STORAGE OF (E) PUMP WITH SLAC PROJECT TEAM.
3. DISCONNECT AND DEMOLISH (E) 4" POW/WR BRANCHES BACK TO THE (E) 6" POW/WR MAINS OVERHEAD. MODIFY AS REQUIRED TO ACCOMMODATE THE NEW BRANCHES TO THE (E) CONSTRUCTION PHASE. SEE M0.0 FOR CONSTRUCTION PHASE SCOPE.
4. DEMOLISH EXISTING 4" POW/WR BRANCHES DOWN TO SUB-BASEMENT LEVEL, BELOW SEE M0.0 FOR CONTINUATION.
5. DISCONNECT AND DEMOLISH 4" CHWSR FROM (E) HK. MODIFY AS REQUIRED TO ACCOMMODATE EXTENSION FROM FLANGE IN CONSTRUCTION PHASE.
6. DEMOLISH 4" CHWSR UP TO (E) VALVE AND TEMPORARILY CAP. MODIFY AS REQUIRED TO ACCOMMODATE EXTENSION FROM FLANGE IN CONSTRUCTION PHASE.
7. DISCONNECT AND DEMOLISH 6" POW/WR UP TO (E) VALVE AND TEMPORARILY CAP. MODIFY AS REQUIRED TO ACCOMMODATE EXTENSION FROM FLANGE IN CONSTRUCTION PHASE.

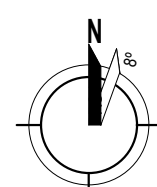
KEYED NOTES, CONT'D:

1. DISCONNECT AND DEMOLISH THE EXISTING MAKEUP WATER LINE AND ASSOCIATED ELEMENTS FROM THE EXPANSION TANK PAST THE EXISTING CONTROL VALVE AND TEMPORARILY CAP NEW MAKEUP LINE TO BE RUN FROM THE EXPANSION TANK TO THE DISCHARGE EVENTS USED AT LEAST ONE WEEK ADVANCE NOTICE AND WILL NEED TO BE SCHEDULED IN COORDINATION WITH THE SEWER AGENCY AT A MUTUALLY AGREEABLE DATE AND TIME.
2. COORDINATE WITH SLAC HVAC AND AC ENGINEER PRIOR TO DEMOLITION. ALL CONTROL PANEL AND PANEL DEVICES MUST REMAIN INTACT AND OPERATIONAL AFTER SKID REPLACEMENT.
3. 10% I&C CONTRACTOR SHALL PROVIDE DEMOLITION DRAWINGS IN TMO/C FORMAT.
4. ANY MATERIAL DEMOLISHED SHALL BE SURVEYED BY SLAC RP PRIOR TO OFFSITE REMOVAL. COORDINATE WITH SLAC PROJECT TEAM AS REQUIRED.
5. 12. CONTRACTOR AND SLAC PROJECT TEAM SHALL COORDINATE ALL FLUSHING, DRAINING, AND OTHER DISCHARGES OF PCW TO SENSITIVE SLURRY CATCH BASIN. ALL DISCHARGE EVENTS USED AT LEAST ONE WEEK ADVANCE NOTICE AND WILL NEED TO BE SCHEDULED IN COORDINATION WITH THE SEWER AGENCY AT A MUTUALLY AGREEABLE DATE AND TIME.



SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: <u>crayder</u>	
DATE: <u>3/9/2026</u>	PRSR: <u>29-058</u>
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and IFBA program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or IFBA programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.



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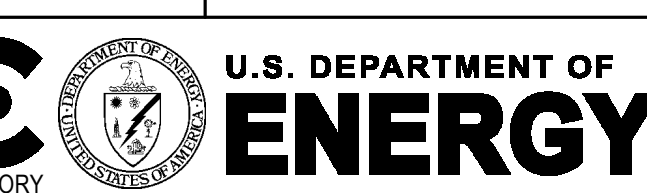


STAMP:

SCALE:	As indicated	DO NOT SCALE DRAWING	B0950.x
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GRAPHIC SCALE: 1/8" = 1'

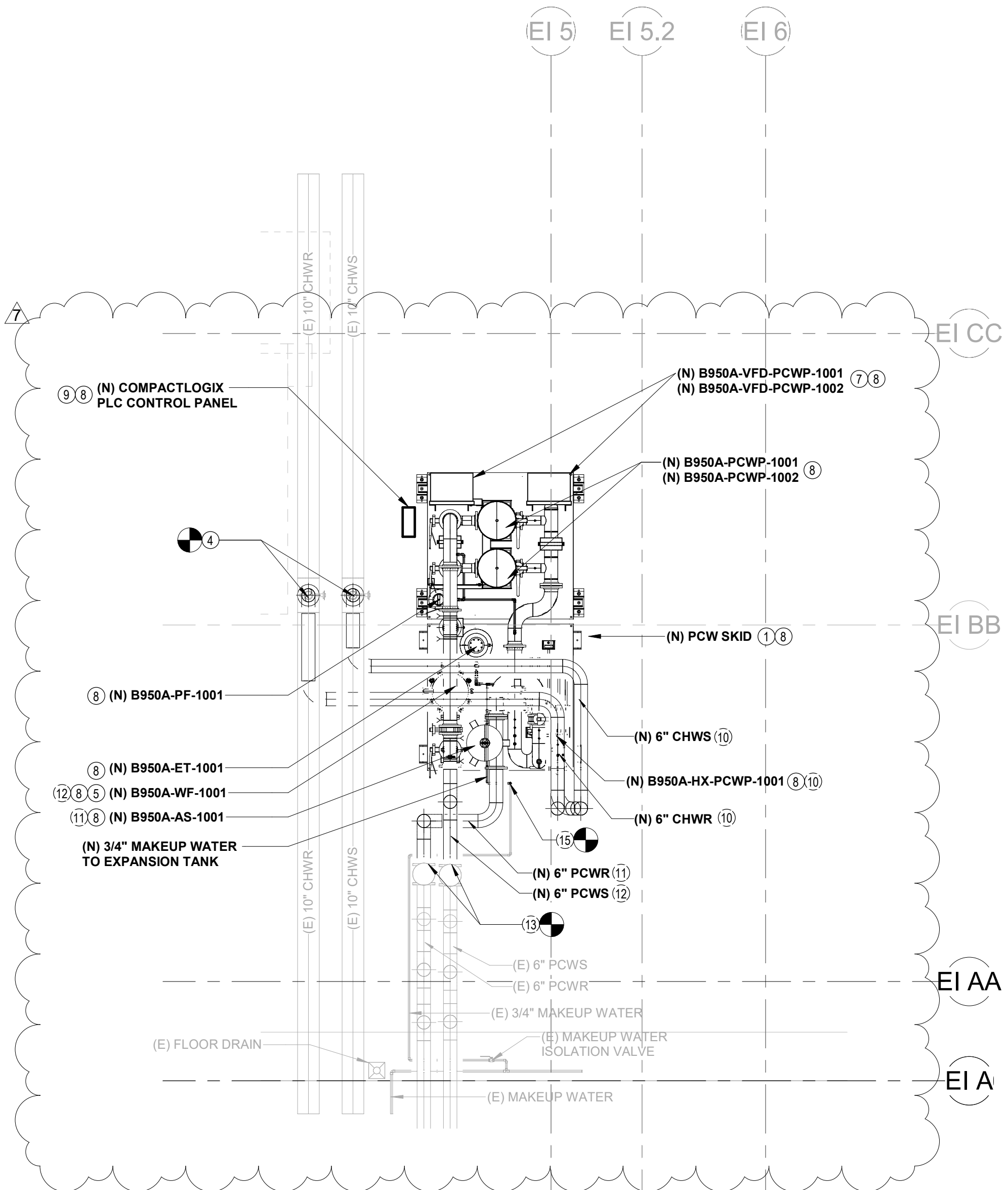
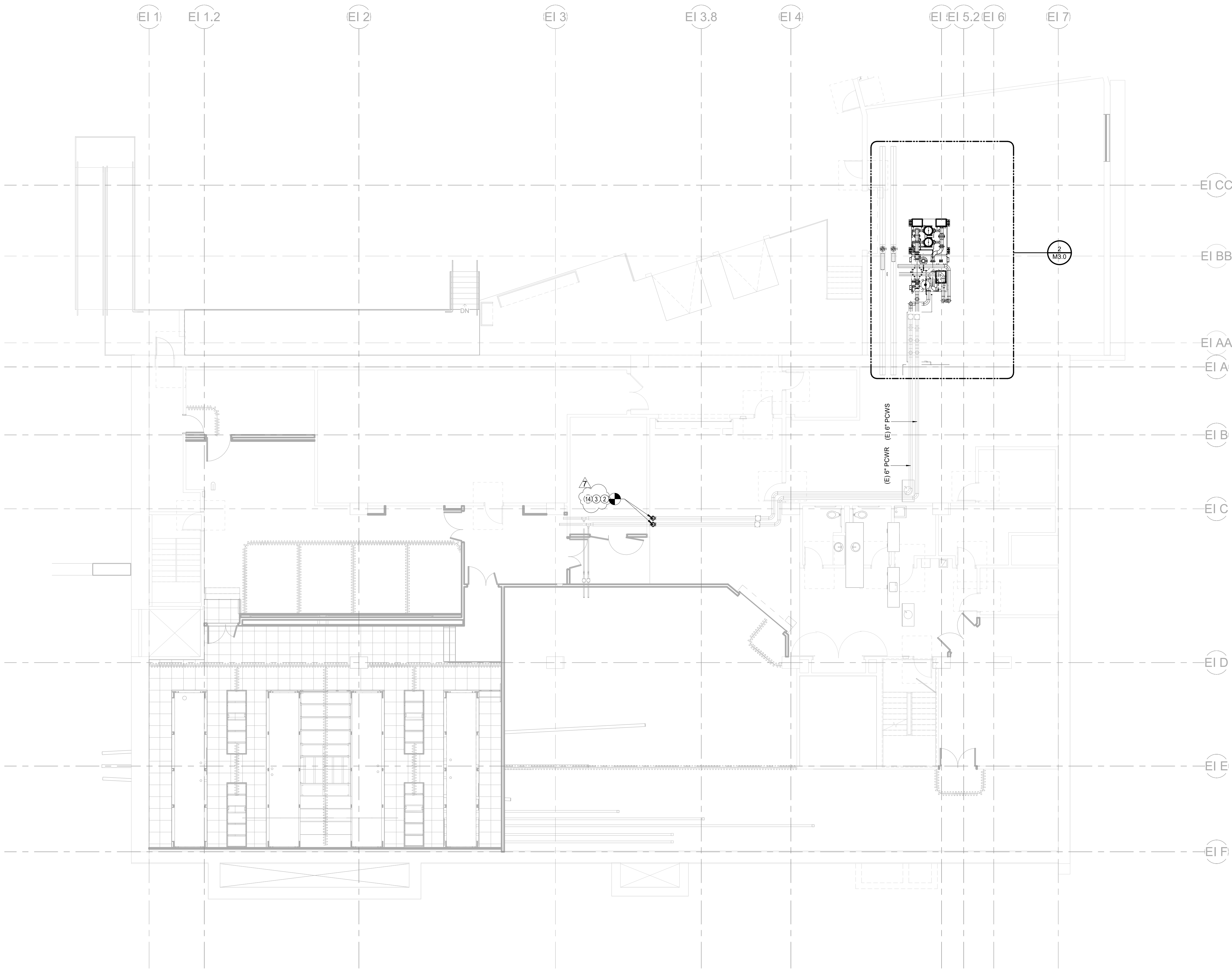
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MD3.0

**B950/NEH
LCLS-II-HE:EI
PROCESS
BSMT (NEH) - OVERALL PLAN - DEMO**

DRAWING NUMBER B0950-MD0-0323	REVISION 7	E
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REV	DESCRIPTION	DRN	CHK	APP	DATE
0	ISSUE FOR CONSTRUCTION				25 MAY 14
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



2
M3.0 BASEMENT (NEH) - PROCESS PIPING OVERALL PLAN - ENLARGED
1/4" = 1'-0"

1
M3.0 BASEMENT (NEH) - PROCESS PIPING OVERALL PLAN
1/8" = 1'-0"

GENERAL NOTES

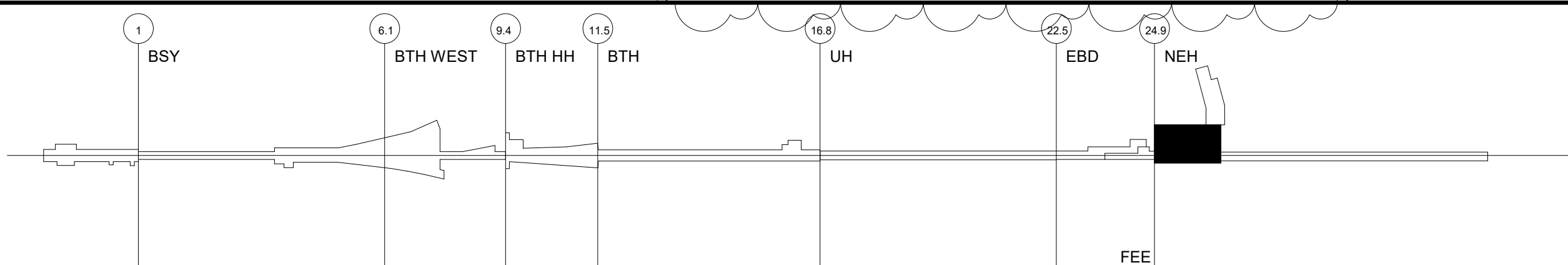
- REFER TO SHEET 00.0 AND 00.1 FOR INDEX OF DRAWINGS & GENERAL NOTES.
- DROP LOCATIONS SHOWN ON PLANS ARE FOR REFERENCE ONLY. USE THE CALLOUT FOR EXACT LOCATION OF DROP. CALLOUT IS DISTANCE FROM 00 GRIDLINE. REFER TO M0.1 FOR LOCATION. CALLOUT ON PLAN IS SHOWN AS 2, 300" X" OR 2, 300.X".
- THE (UP) PIPING TAG INDICATES THE PIPE IS RUN UNDER THE RAISED FLOOR.
- PROVIDE FIRE CAULKING BETWEEN ROOMS WHENEVER REQUIRED.
- ALL FIRE SPRINKLER MODIFICATIONS REQUIRED SHALL BE DONE BY LICENSED C16 CONTRACTOR AND BE SLAC FIRE MARSHAL APPROVED. FIRE SPRINKLER CONTRACTOR SHALL PROVIDE SHOP DRAWINGS AND CALCULATIONS FOR ALL FIRE SPRINKLER MODIFICATIONS.
- NO SHAVE/BITE COMPRESSION FITTINGS WITH ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.
- ALL MANUAL AIR VENTS AND DRAINS ON ANY PCW/LOV SYSTEM NEED TO BE PLUMBED BACK TO SUMP OR POLY HOLDING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.
- INCLUDE ISOLATION VALVE(S) AND TEE(S) FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.
- ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC S&E THROUGH SHOP DRAWING SUBMITTAL.

KEYED NOTES:

- PROVIDE AND INSTALL (N) PCW SKID IN SIMILAR LOCATION AS EXISTING PCW SKID BUT OFF-SET TO MISS EXISTING ANCHOR PENETRATIONS. SEE STRUCTURAL DETAILS 5-7/55.1 FOR SPRING ISOLATION AND MOUNTING INFORMATION.
- CONNECT (N) 6" PCWS/R MAINS OFF EXISTING (E) 6" PCWS/R MAINS OVERHEAD. COORDINATE SYSTEM SHUTDOWN AND DRAINING WITH SLAC BUILDING OWNER AND PM/CM TEAM.
- ROUTE (N) 6" PCWS/R DOWN THROUGH FLOOR TO SUB-BASEMENT LEVEL BELOW. RE-USE EXISTING 8"Ø PENETRATIONS THROUGH SLAB AND EXISTING SUPPORTS. PROVIDE FIRE CAULKING PROTECTION AT FLOOR PENETRATION PER UL C-A-1153. PROVIDE LINE-SIZE BUTTERFLY VALVES IN VERTICAL.
- EXTEND (N) 6" CHWS/R FROM (E) VALVE AND ROUTE TO (N) HX. RE-USE EXISTING SUPPORTS.
- PROVIDE (N) BACKWASH DRAIN VALVE ON BOTTOM OF WATER FILTER. SEE 2/M6.3 FOR MORE.
- NOT USED.
- SEE ELECTRICAL DRAWING E1.3 FOR POWER TO VFD'S AND COORDINATE AS REQUIRED.
- PCW SKID AND ALL INCLUDED EQUIPMENT AND DEVICES SHALL BE INSTALLED PER MANUFACTURER RECOMMENDATIONS. SEE DETAILS 3-S&M.1 FOR PLAN VIEW AND ISOMETRIC DETAILS, AND 6.3/6.4 FOR I&C AND FLOW DIAGRAMS.

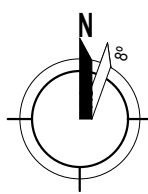
KEYED NOTES (CONT'D):

- INSTALL PLC CONTROL PANEL ON NEW UNISTRUT SUPPORTS TIED TO NEW SKID IN SAME LOCATION AS EXISTING PANEL. ENSURE 36"Wx80"H CLEAR ACCESS FOR PLC CONTROL PANEL IS PROVIDED.
- CONNECT (N) 6" CHWS/R TO (N) HEAT EXCHANGER. SEE DETAILS 1/55.2 THROUGH 5/55.2 FOR CHW PIPE SUPPORT DETAIL. INFORMATION.
- EXTEND FROM TEMPORARY CAP AND CONNECT (N) 6" PCWR TO (N) AIR SEPARATOR.
- EXTEND FROM TEMPORARY CAP AND CONNECT (N) 6" PCWS TO (N) PCW SKID JUST UPSTREAM OF THE (N) CARTRIDGE FILTER.
- EXTEND (N) 6" PCWS/R FROM (E) VALVES AND ROUTE TO (N) PCW SKID AS SHOWN. RE-USE EXISTING SUPPORTS.
- DRAIN FROM AAV ON PCWS/R SHALL BE PIPED TO SPILL TO FLOOR 6" ABOVE FINISHED FLOOR.
- REMOVE TEMPORARY CAP AND EXTEND (N) 3/4" MAKEUP WATER TO EXPANSION TANK INCLUDING ALL I&C ELEMENTS AS SHOWN ON M6.3. SEE SCHEMATIC 2/M6.3 FOR MORE.



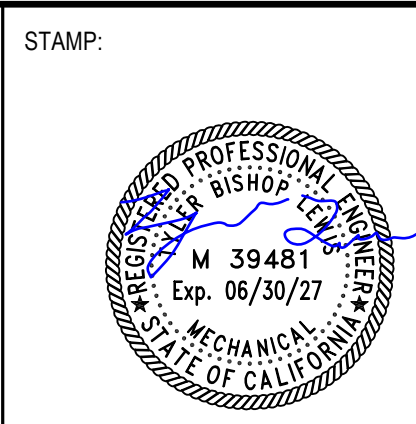
SLAC BUILDING INSPECTION OFFICE Construction Authorization (BIC)	
BY: cinder	DATE: 09/20/20
PROB: 26-050	Revised Authorization: ☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and all plan requirements.	
The authorization of these plans, and not the contract, is a permit or approval for violation of any applicable codes, standards, or CSM programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.



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U.S. DEPARTMENT OF ENERGY

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ENGR	GP	DATE	APPROVALS	DATE
DES	GP	02/28/2025	Approver	
CHK	GP	02/28/2025		
CHR	GP	02/28/2025		

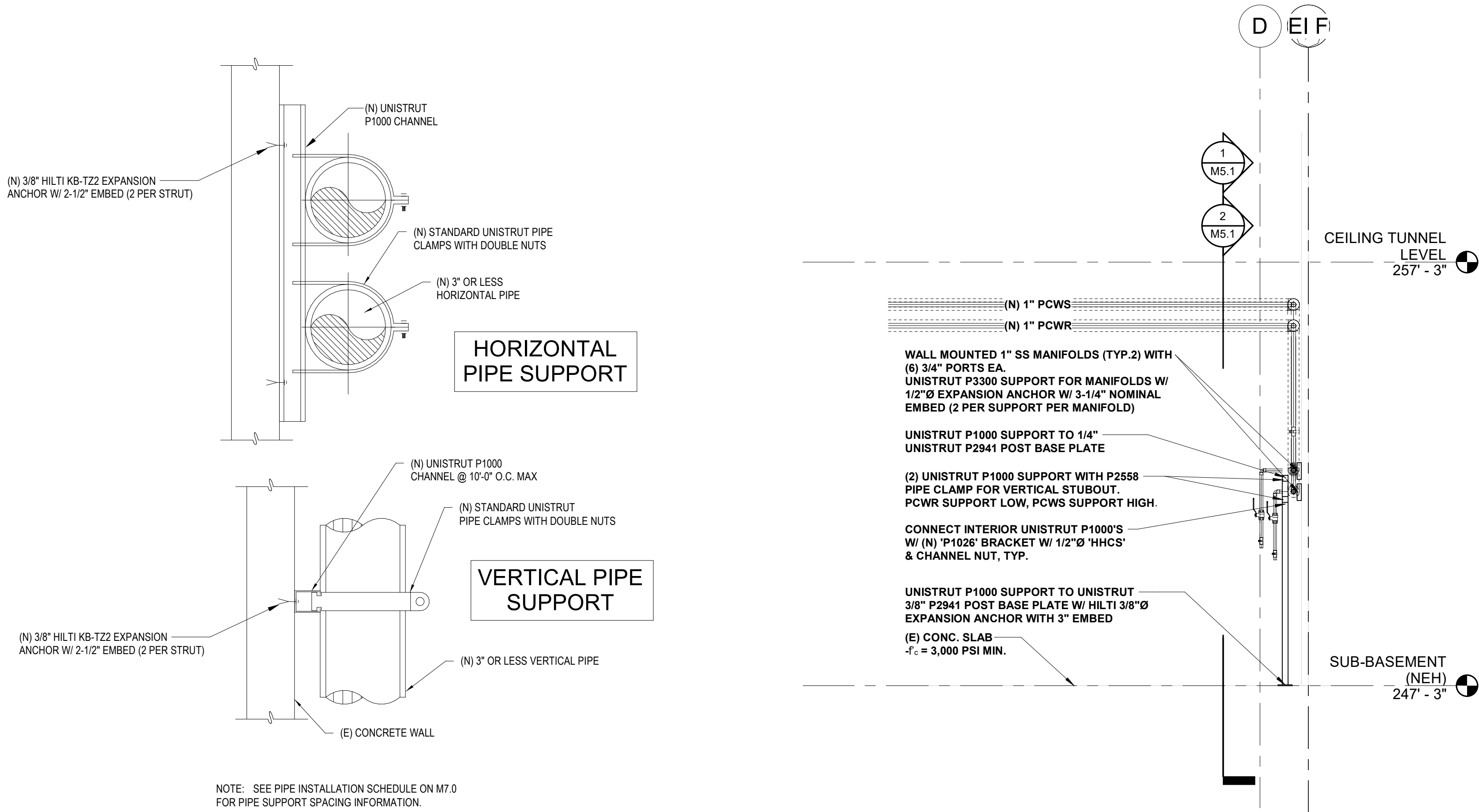
GRAPHIC SCALE: 1/8" = 1'
0 8 16 24 32 40

B950/NEH
LCLS-II-HE-EI
PROCESS
BSMT (NEH) - OVERALL PLAN

DRAWING NUMBER
B0950-MP1-0331

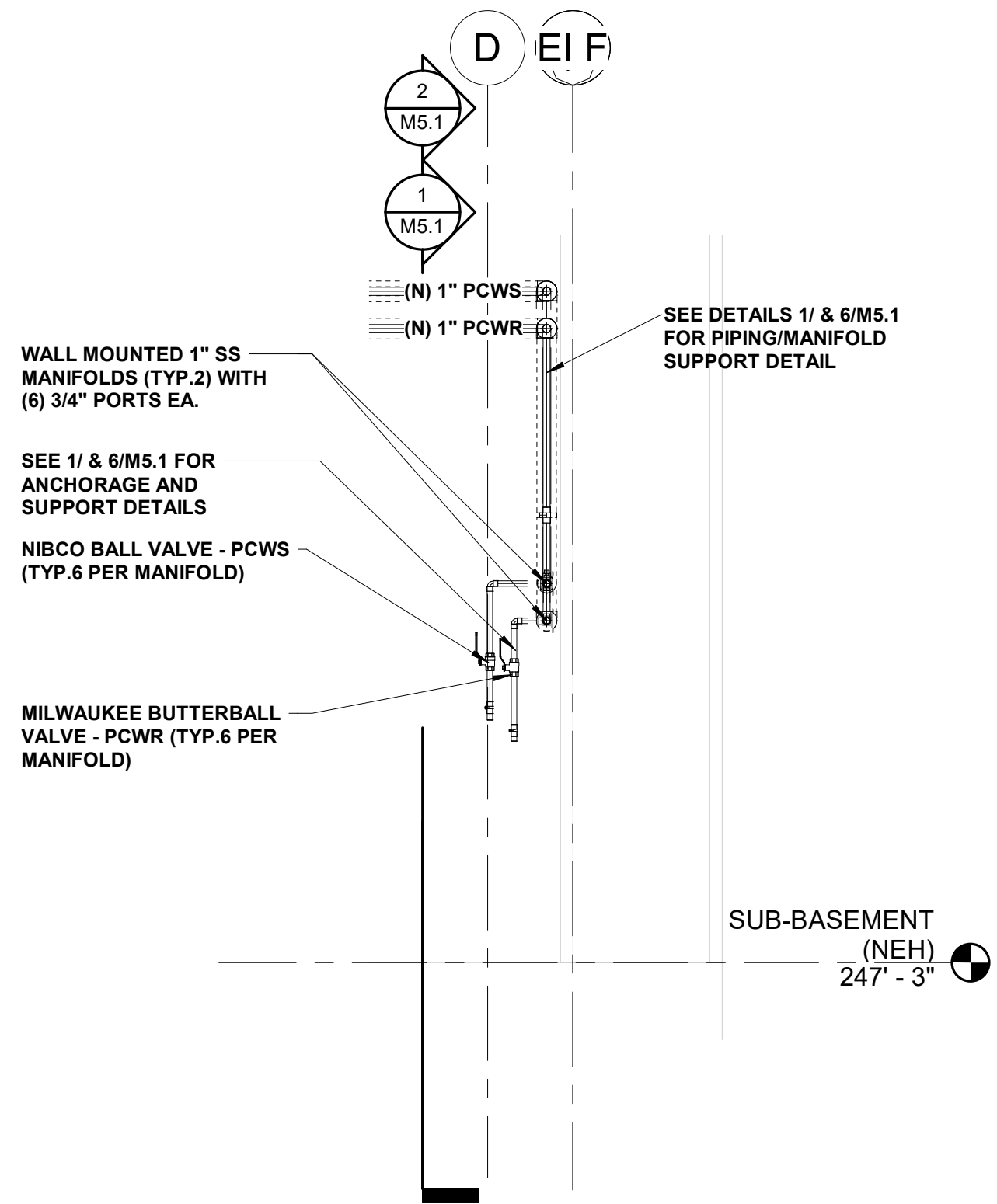
REVISION
7

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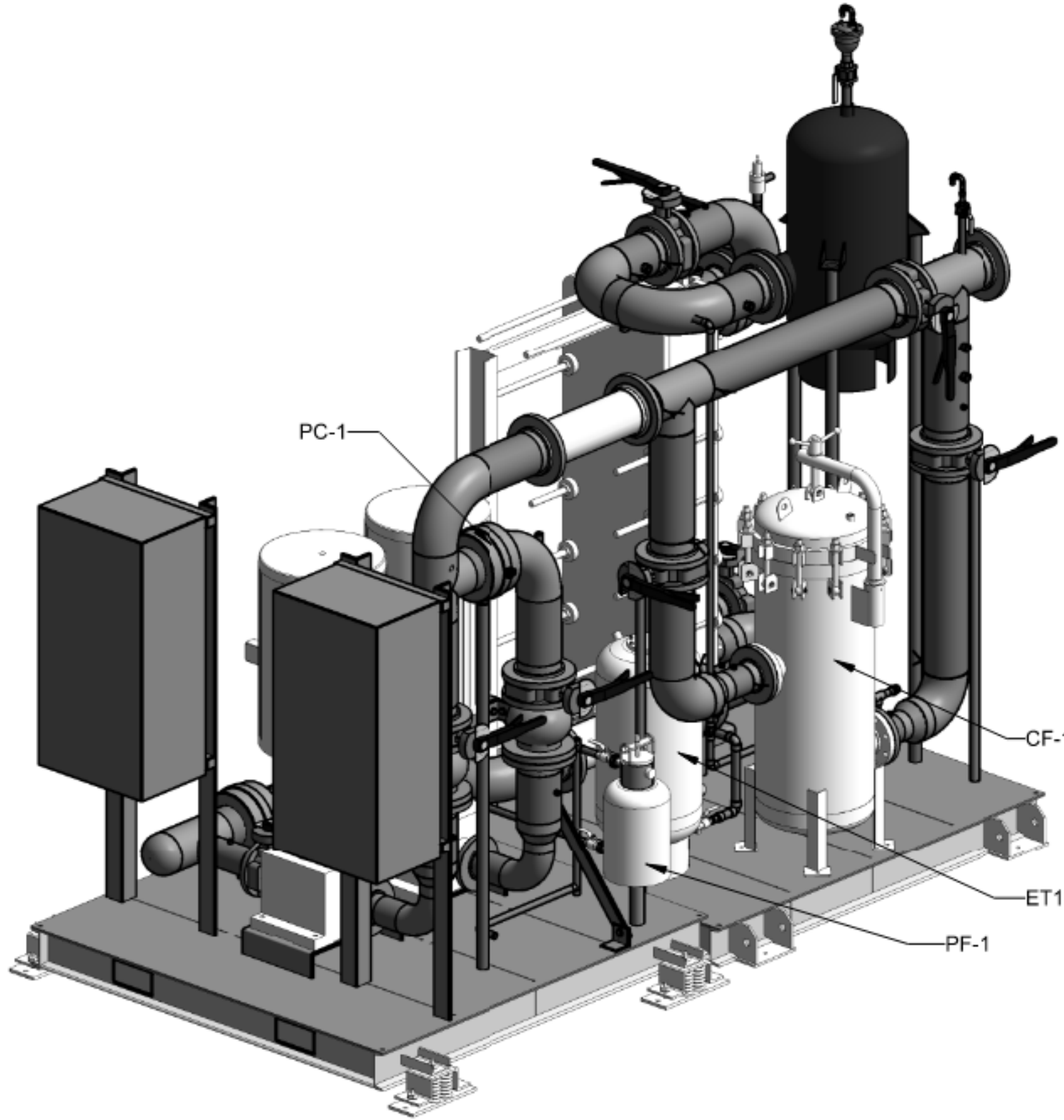


8 PIPE WALL SUPPORT DETAIL
M5.1 NTS

6 XRT 1" PCW CUSTOM MANIFOLD W/ (6) 3/4" STUBOUT ANCHORAGE DETAIL (E-W)
M5.1 1/2" = 1'-0"

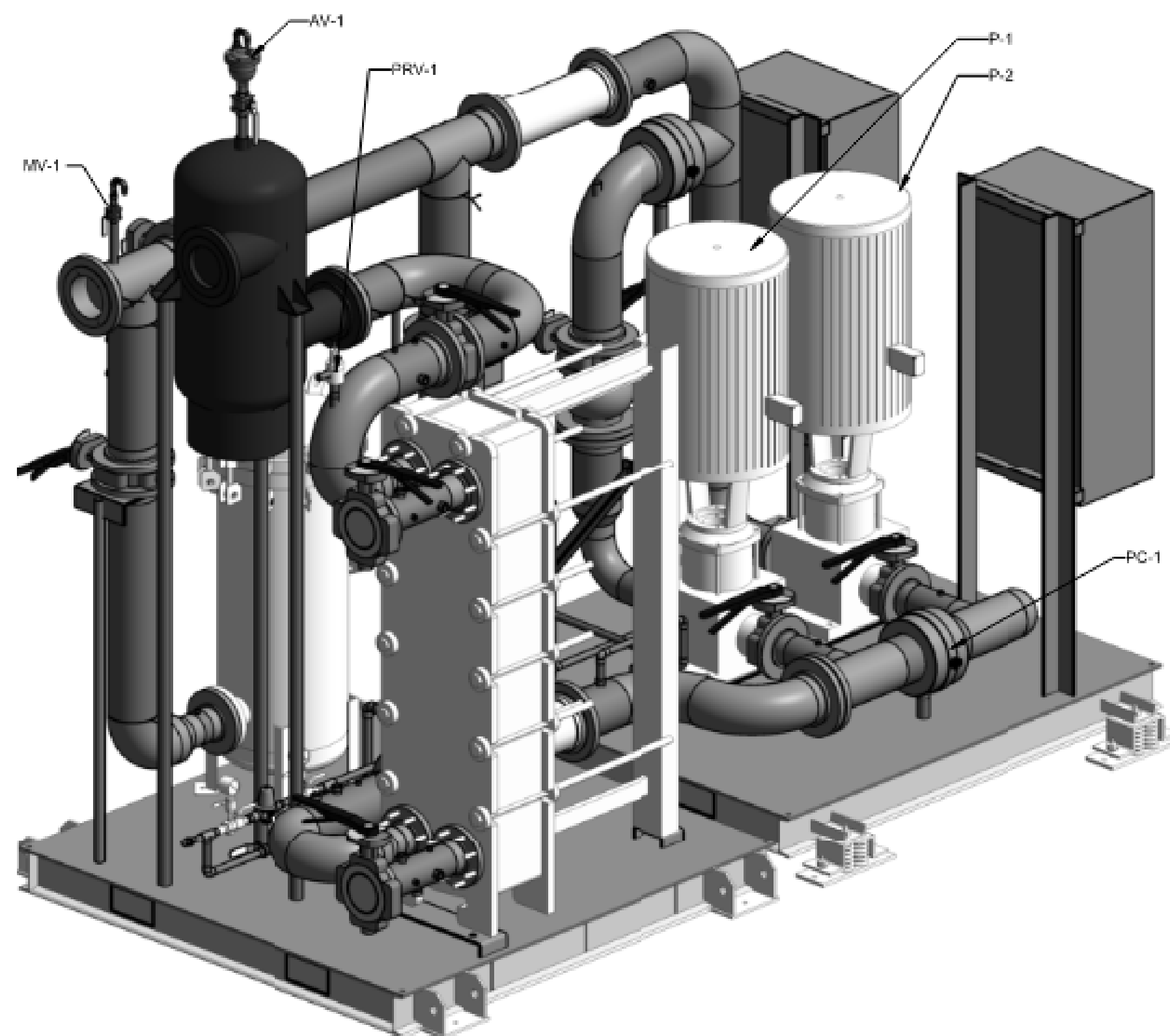


7 XRT 1" PCW CUSTOM MANIFOLD W/ (6) 3/4" STUBOUT SECTION (E-W)
M5.1 1/2" = 1'-0"



NOTE: MANUFACTURER SHALL RECONFIGURE THE EQUIPMENT COMPONENTS DURING SUBMITTAL PHASE IN COORDINATION WITH SLAC PROJECT TEAM TO RE-ORIENT THE PUMPS AND MOTORS FOR EASIER ACCESS DURING REMOVAL/MAINTENANCE. EXISTING INFRASTRUCTURE CONFIGURATION MAKES IT DIFFICULT TO REMOVE.

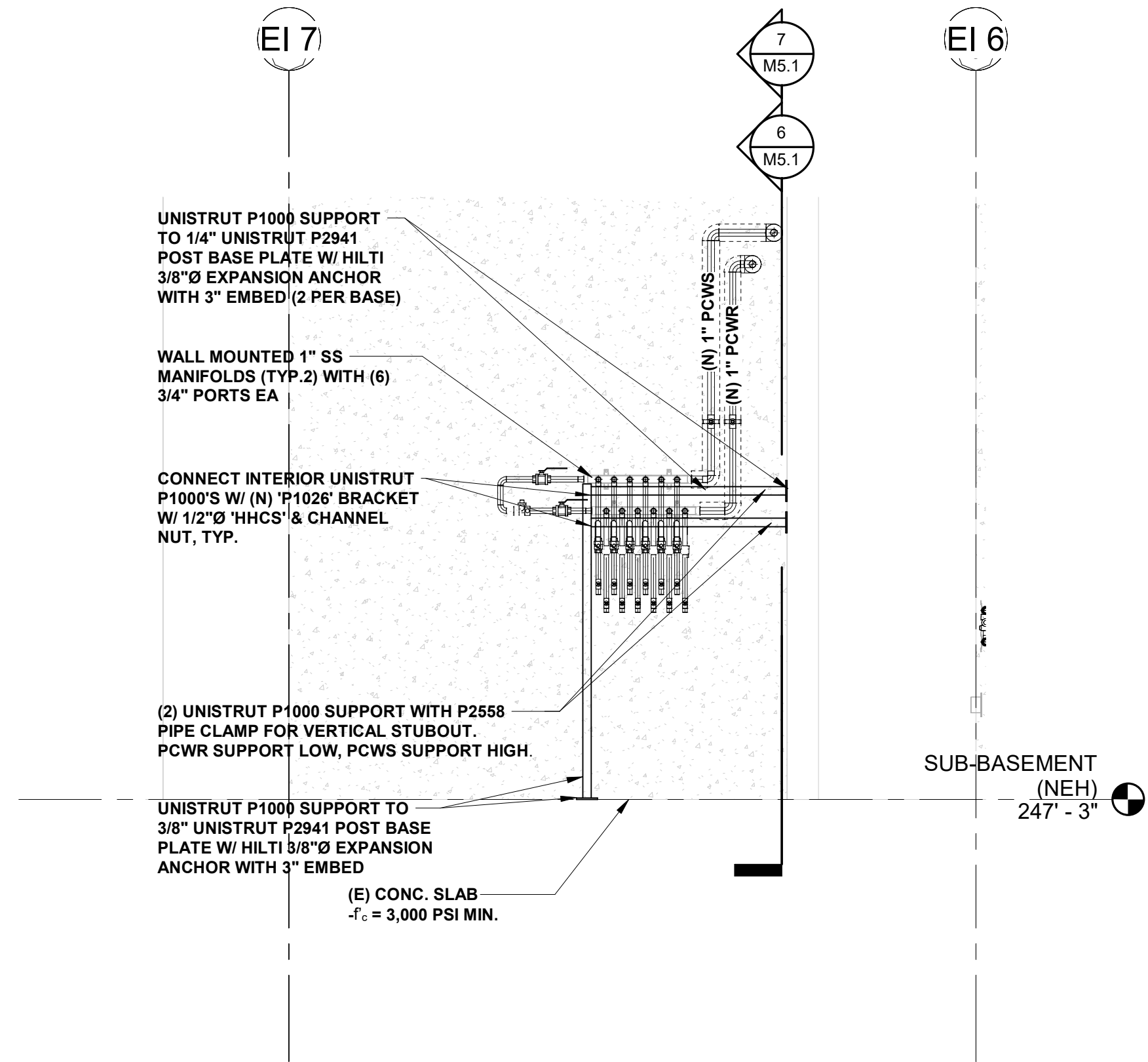
4 (N) PCW SKID ISOMETRIC 1 DETAIL
M5.1 NTS



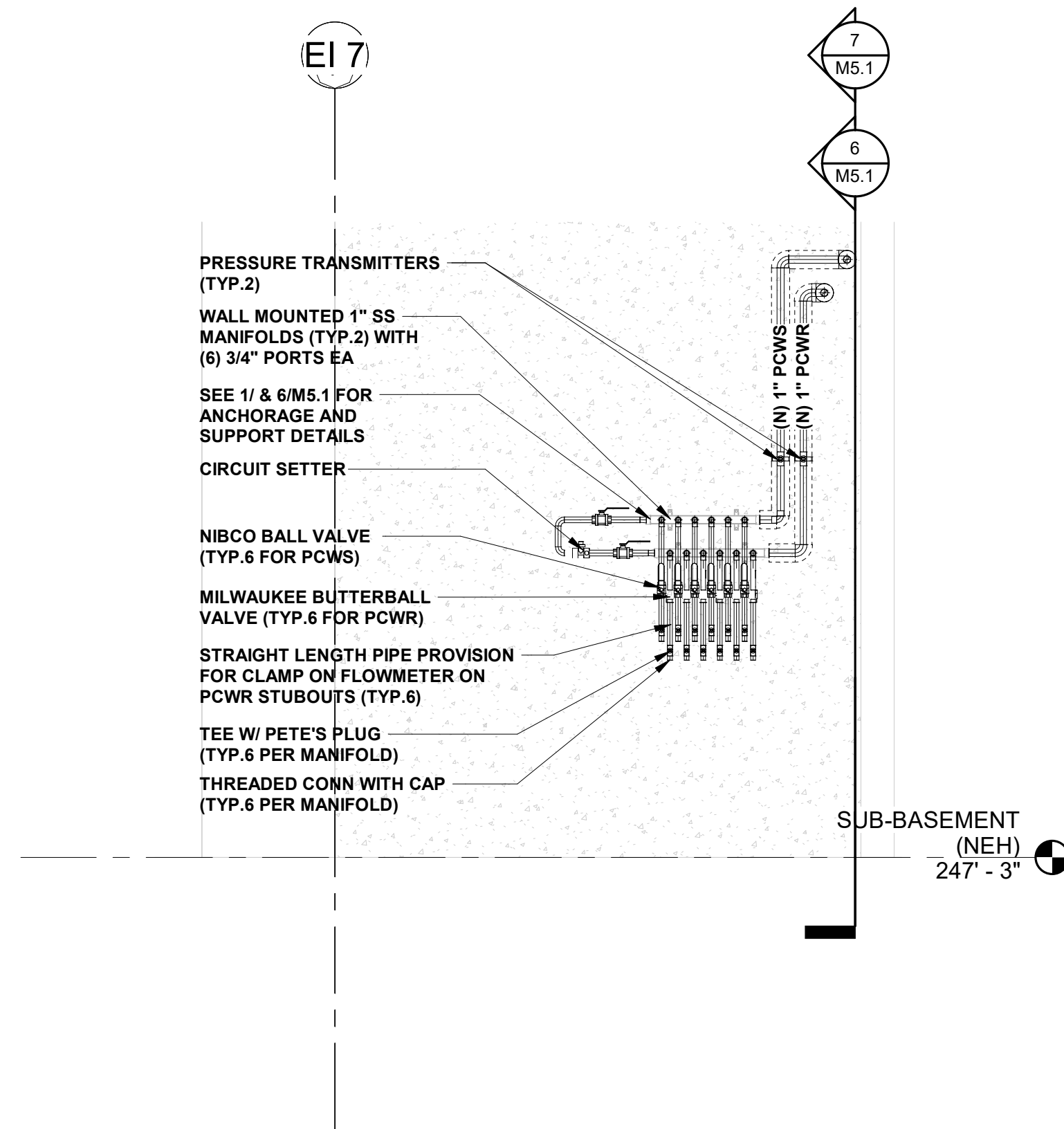
NOTE: MANUFACTURER SHALL RECONFIGURE THE EQUIPMENT COMPONENTS DURING SUBMITTAL PHASE IN COORDINATION WITH SLAC PROJECT TEAM TO RE-ORIENT THE PUMPS AND MOTORS FOR EASIER ACCESS DURING REMOVAL/MAINTENANCE. EXISTING INFRASTRUCTURE CONFIGURATION MAKES IT DIFFICULT TO REMOVE.

5 (N) PCW SKID ISOMETRIC 2 DETAIL
M5.1 1/2" = 1'-0"

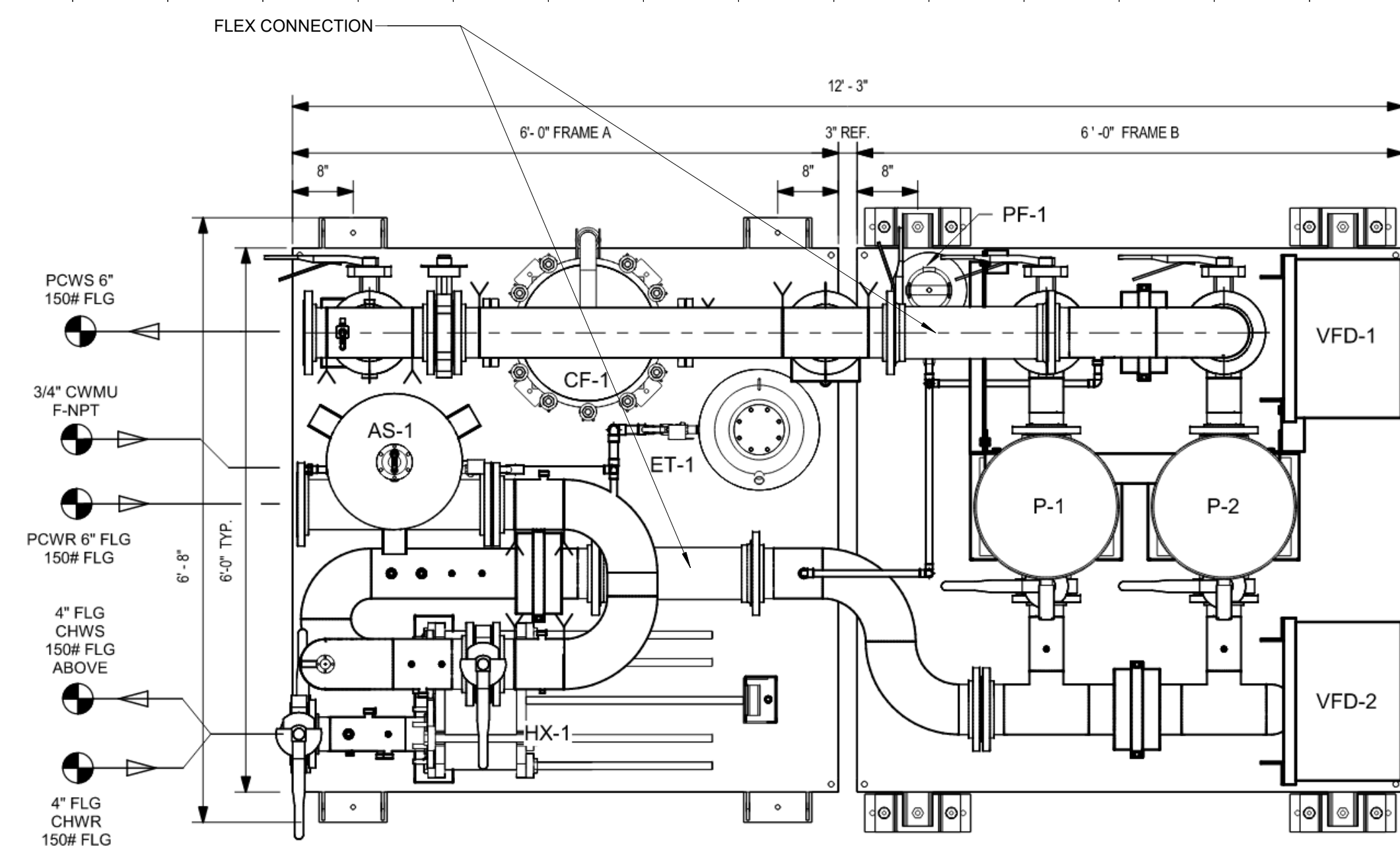
REV	DESCRIPTION	DRN	CHK	APP	DATE
5	BULLETIN 3 - XRT SCOPE				25 OCT 28
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



1 XRT 1" PCW CUSTOM MANIFOLD W/ (6) 3/4" STUBOUT ANCHORAGE DETAIL (N-S)
M5.1 1/2" = 1'-0"



2 XRT 1" PCW CUSTOM MANIFOLD W/ (6) 3/4" STUBOUT SECTION (N-S)
M5.1 1/2" = 1'-0"



NOTE: MANUFACTURER SHALL RECONFIGURE THE EQUIPMENT COMPONENTS DURING SUBMITTAL PHASE IN COORDINATION WITH SLAC PROJECT TEAM TO RE-ORIENT THE PUMPS AND MOTORS FOR EASIER ACCESS DURING REMOVAL/MAINTENANCE. EXISTING INFRASTRUCTURE CONFIGURATION MAKES IT DIFFICULT TO REMOVE.

3 (N) PCW SKID PLAN VIEW DETAIL
M5.1 NTS

GENERAL NOTES

A. NO SHAW-WHITE COMPRESSION FITTINGS WITH ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.

B. ALL MANUAL AIR VENTS AND DRAINS ON ANY PCW/LOW SYSTEM NEED TO BE PLUMBED BACK TO SUMP OR POT-Y-HOLDING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.

C. INCLUDE ISOLATION VALVE(S) AND TEE(S) FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.

D. ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC SMT THROUGH SHOP DRAWING SUBMITTAL.

SLAC BUILDING INSPECTION OFFICE			
Construction Authorization (BFC)			
BY: cnahler			
DATE: 3/9/2028		PRSE#: 28-058	
Inspection Authorization	☒ Yes ☐ No	☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and EIR program requirements.			
The authorization of these plans shall not be construed to be a permit or approval for violation of any applicable codes, standards, or EIR programs.			
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.			

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STAMP:

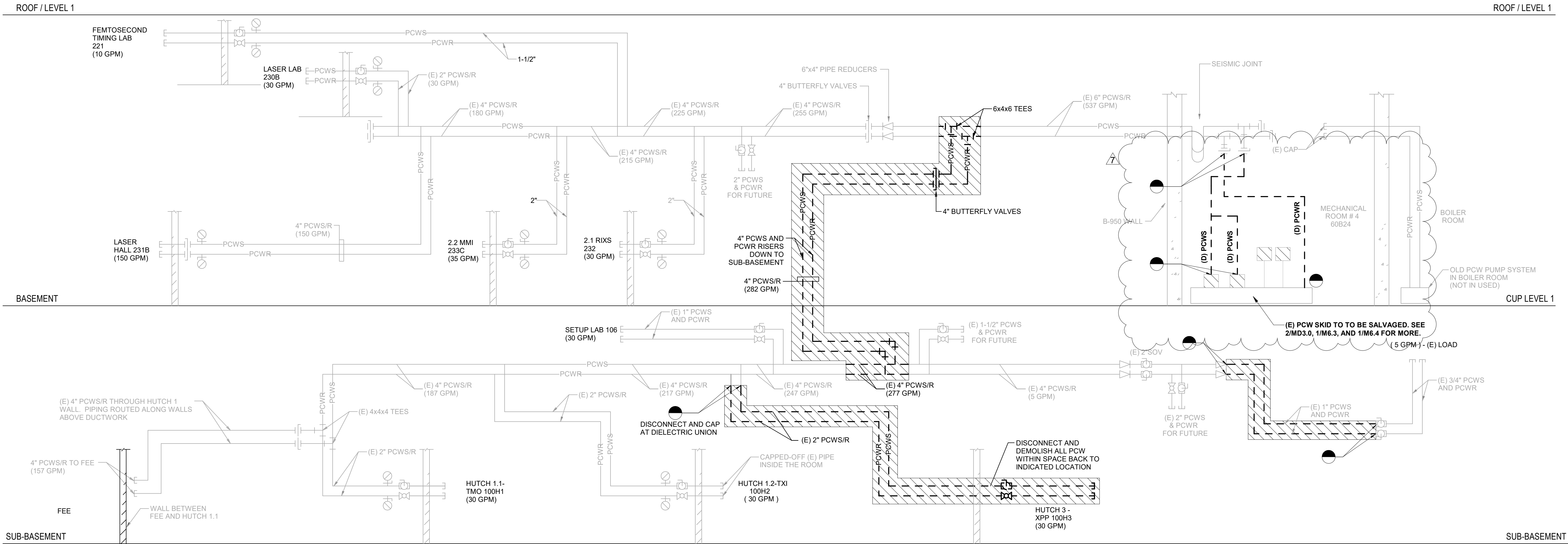


SCALE: As Indicated		DO NOT SCALE DRAWING		B0950.x		M5.1	
SLAC NATIONAL ACCELERATOR LABORATORY		U.S. DEPARTMENT OF ENERGY		B950/NEH LCLS-II-HE:EI MECHANICAL MECHANICAL DETAILS			
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF STANFORD UNIVERSITY / SLAC.		DATE: 02/28/2025 DES: 02/28/2025 CHK: 02/28/2025		APPROVALS: Approver		DRAWING NUMBER: B0950-M05-0342	
INSTR: GP DES: GP CHK: GP		DATE: 02/28/2025 DES: 02/28/2025 CHK: 02/28/2025		REVISION: 7		E	

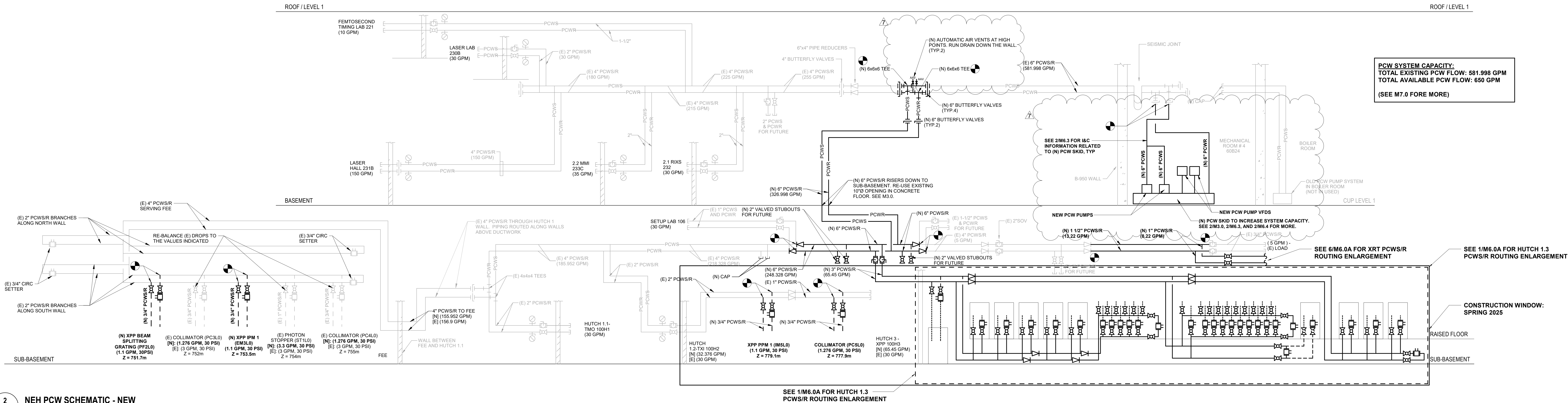
LCLS-II-HE:EI MECHANICAL DETAILS

B0950-M05-0342

REV	DESCRIPTION	DRN	CHK	APP	DATE
5	BULLETIN 3 - XRT SCOPE				25 OCT 28
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



1
M6.0
NEH PCW SCHEMATIC - DEMO
12" = 1'-0"



2
M6.0
NEH PCW SCHEMATIC - NEW
12" = 1'-0"

GENERAL NOTES

- A. NO SWAGE/COMPRESSOR FITTINGS WITH ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.
- B. ALL MANUAL AIR VENTS AND DRAINS ON ANY PCW/LOW SYSTEM NEED TO BE PLUMBED BACK TO SUMP OR POT-HOLDING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.
- C. INCLUDE ISOLATION VALVE(S) AND TEE(S) FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.
- D. ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC SMT THROUGH SHOP DRAWING SUBMITTAL.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (BFC)	
BY: ☒ cradler	DATE: 3/9/2028
REVIEWED: ☒ m	DATE: 3/9/2028
These plans shall be kept on the job site for all construction inspections.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

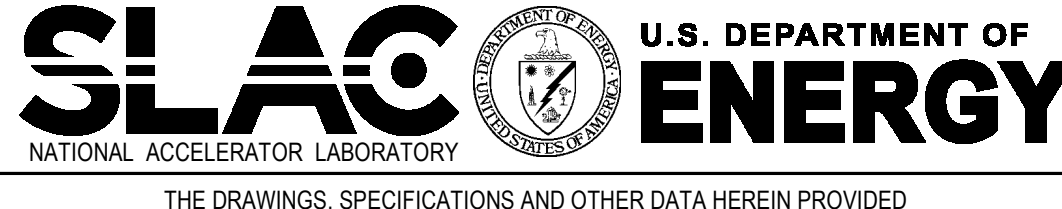


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SCALE: 12" = 1'-0" DO NOT SCALE DRAWING B0950.x



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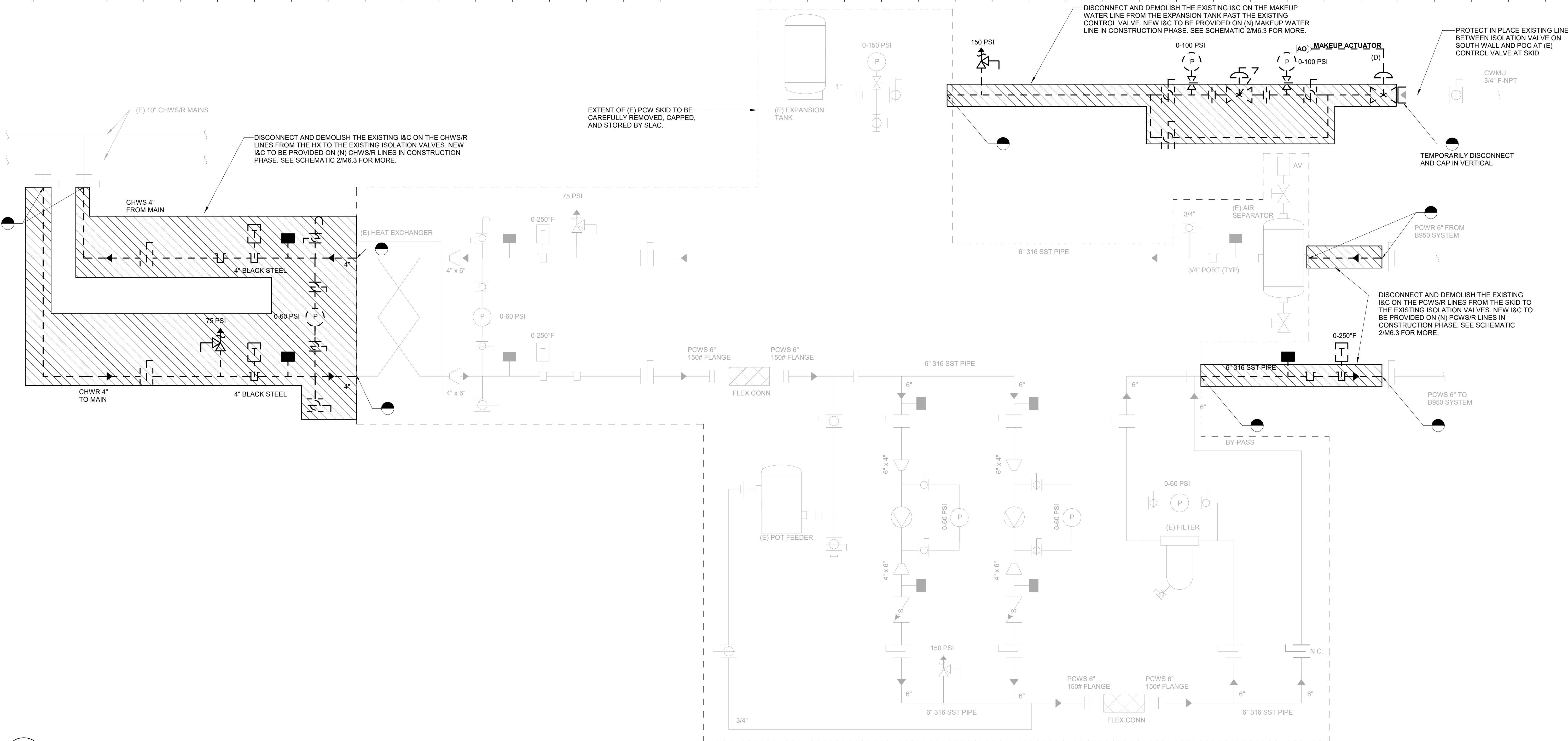
ENGR	GP	DATE	02/28/2025
DES	GP	DATE	02/28/2025
CHK	GP	DATE	02/28/2025

B950/NEH
LCLS-II-HE:EI
PROCESS
MECHANICAL SCHEMATIC - PCW

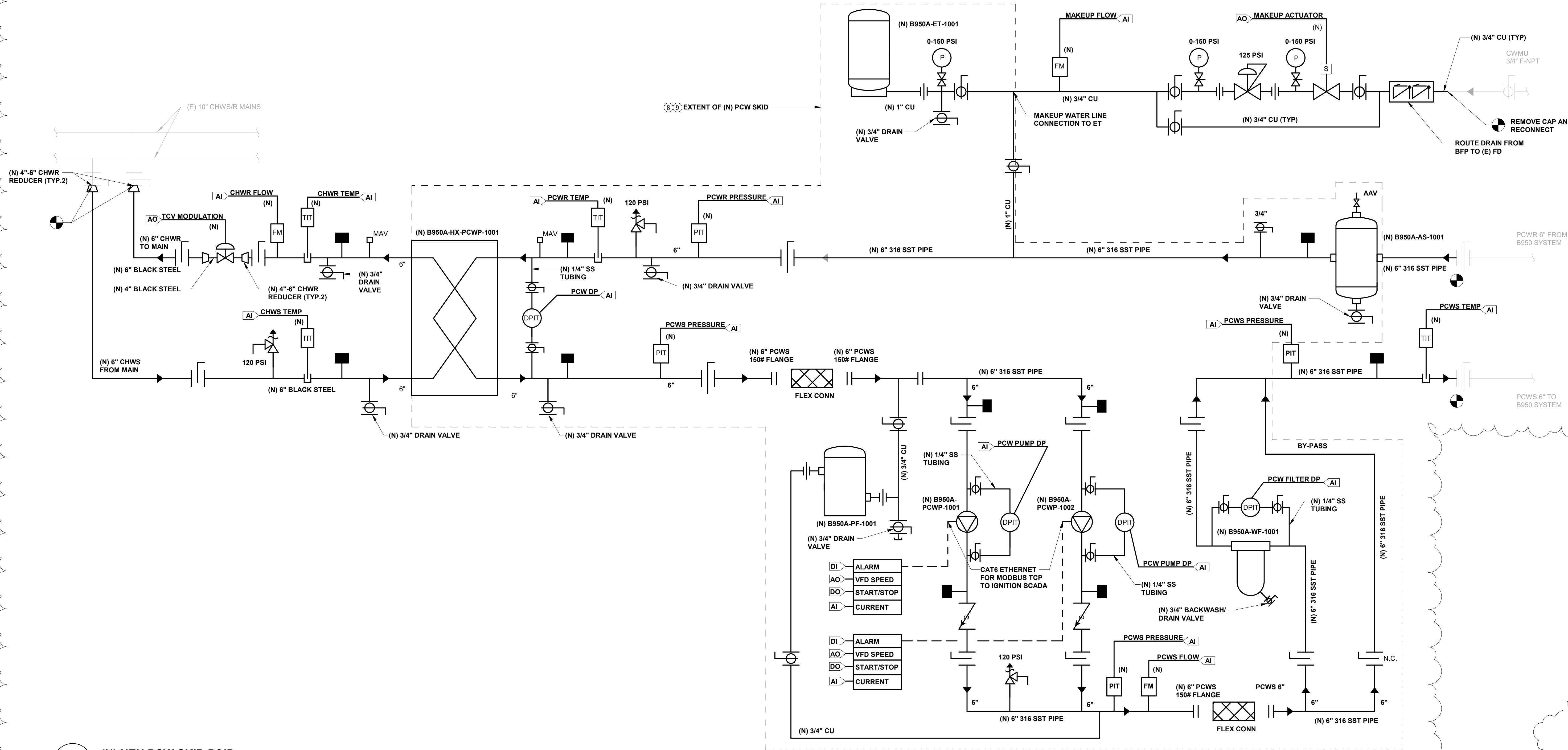
DRAWING NUMBER
B0950-M05-0336

REVISION
7
E

REV	DESCRIPTION	DRN	CHK	APP	DATE
E	RESPONSE TO BIO				25 MAR 26
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



1 (E) NEH PCW SKID P&ID
M6.3 NTS



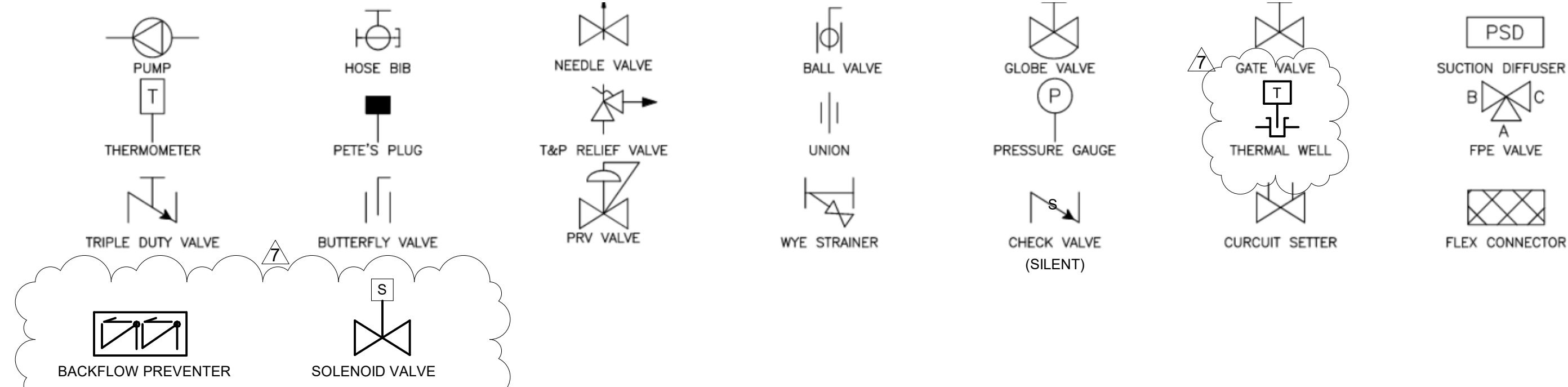
2 (N) NEH PCW SKID P&ID
M6.3 NTS

- SEQUENCE OF OPERATION:**
- 1) THE CONTROL SYSTEM MONITORS THE SYSTEM TEMPERATURES, FLOW AND PRESSURES.
 - 2) BOTH PUMPS SHALL BE RUNNING ALL THE TIME, 24 HOURS/DAY, 7 DAYS/WEEK.
 - 3) THE PLC SYSTEM CONTROLS THE PUMP SPEED TO MAINTAIN 42 PSIG (ADJUSTABLE) DIFFERENTIAL PRESSURE (SUPPLY VS RETURN PRESSURES).
 - 4) THE BMS SHALL MONITOR THE PUMP CURRENT.
 - 5) IF ONE PUMP FAILS, THE OTHER PUMP RAMP UP TO MAINTAIN PRESSURE.
 - 6) THE SUPPLY TEMPERATURE SENSOR AT THE PROCESS WATER LOOP MODULATES THE CONTROL VALVE AT THE CHILLED WATER SYSTEM TO MAINTAIN 63 °F (ADJUSTABLE) PROCESS WATER LOOP TEMPERATURE.
 - 7) THE MAKE-UP WATER SOLENOID VALVE IS NORMALLY OPEN. THE SOLENOID VALVE CLOSSES WHEN:
 - 1) THE MAKE-UP METER RUN MORE THE 15 SECONDS (ADJUSTABLE) AND
 - 2) THE SYSTEM RETURN PRESSURE DP IS LOWER THAN THE SETPOINT AND THE PUMP IS RUNNING 100% SPEED.
 - 8) THE BMS SHALL SEND OUT ALARM, IF ANY OF THE FOLLOWING CONDITIONS EXIST:
 - 1) THE PCW SUPPLY TEMPERATURE 0°F (ADJUSTABLE) ABOVE/BELW THE SETPOINT.
 - 2) DIFFERENTIAL PRESSURE 5 PSI (ADJUSTABLE) ABOVE/BELW THE SETPOINT.
 - 3) VFD FAILED.
 - 4) PUMP FAILED.
 - 5) THE MAKE-UP WATER RUN FOR MORE THAN 15 SECONDS (ADJUSTABLE).
 - 6) THE SOLENOID VALVE CLOSED
 - 7) SYSTEM PRESSURE IS ABOVE 90 PSIG (ADJ).

- NOTES:**
- 1) CURRENT ALC IO WILL BE DEMOTD BACK TO SOURCE, AND REWIRED TO NEW PLC PANEL.
 - 2) ALC CONTROL PANEL TO BE DEMOTD INCLUDING COMMUNICATION WIRES BACK TO UPSTREAM DEVICES. CURRENT 120VAC CIRCUIT CAN BE LEFT TO SERVE NEW PLC.
 - 3) ASSOCIATED WEBCRTL SCREEN & LOGIC TO BE DEMOTD.
 - 4) I&C CONTRACTOR TO INTEGRATE PLC SYSTEM INCLUDING: SEQUENCE OF OPERATIONS, CONTROLS DRAWINGS INCLUDING CONTROLS P&ID, POINTS LIST, PLC AND IO HARDWARE. EXCEPTION: WHITON SCADA DEVELOPMENT WILL BE IN-HOUSE BY SLAC SCADA ENGINEER.
 - 5) I&C CONTRACTOR TO COORDINATE INSTRUMENT TAGGING ACCORDING TO SLAC'S IMPLEMENTATION OF SAS.1.
 - 6) INSTALLATION TO INCLUDE NEW ENCLOSURE, 100% ALLEN BRADLEY COMPATIBLE PLC/IO, NEW SENSORS AS INDICATED, CONDUITS, INCLUDING CAT6 CABLEING TO NEAREST SLAC NETWORK SWITCH.
 - 7) ALL CONTROLS SHALL BE MIRRORRED INTO EPICS PROGRAM.
 - 8) ALL I&C COMPONENTS SHOWN WITHIN THE BOUND OF THE PCW SKID SHALL BE PROVIDED BY THE FACTORY. NOT FIELD INSTALLED.
 - 9) PCW SKID SHALL BE FACTORY PRESSURE TESTED WITH HYDROSTATIC PRESSURE TEST OF 1.5 OPERATING PRESSURE (180 PSI. SEE PRESSURE TEST TABLE ON MY 0 FOR MORE).
 - 10) SLAC TO RETAIN ALL REMOVED ALC DEVICES. COORDINATE WITH SLAC CONSTRUCTION/PROJECT MANAGER AS REQUIRED.

LEGEND:
(E) - EXISTING INSTRUMENTATION
(N) - NEW INSTRUMENTATION

P&ID SYMBOL LEGEND:



- GENERAL NOTES**
- A NO SNAKEBITE COMPRESSION FITTINGS WITH ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.
 - ALL MANUAL AIR VENTS AND DRAINS ON ANY PCWLLOW SYSTEM NEED TO BE PLUMBED BACK TO SUMP OR POLY HOLDING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.
 - INCLUDE ISOLATION VALVES (E) AND TEES) FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.
 - ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC SME THROUGH SHOP DRAWING SUBMITTAL.
 - PROVIDE DIELECTRIC UNIONS AT ALL DISSIMILAR MATERIAL CONNECTIONS.

SLAC BUILDING INSPECTION OFFICE Construction Authorization (BFC)	
BY: cmadler	DATE: 3/9/2018
REVIEWED: [Signature]	DATE: 3/9/2018
These plans shall be kept on the job site for all construction inspections.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

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STAMP:



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ENGR DES CHK	GP UP GP	DATE 02/26/2025 02/26/2025 02/26/2025
APPROVALS Approver		DATE

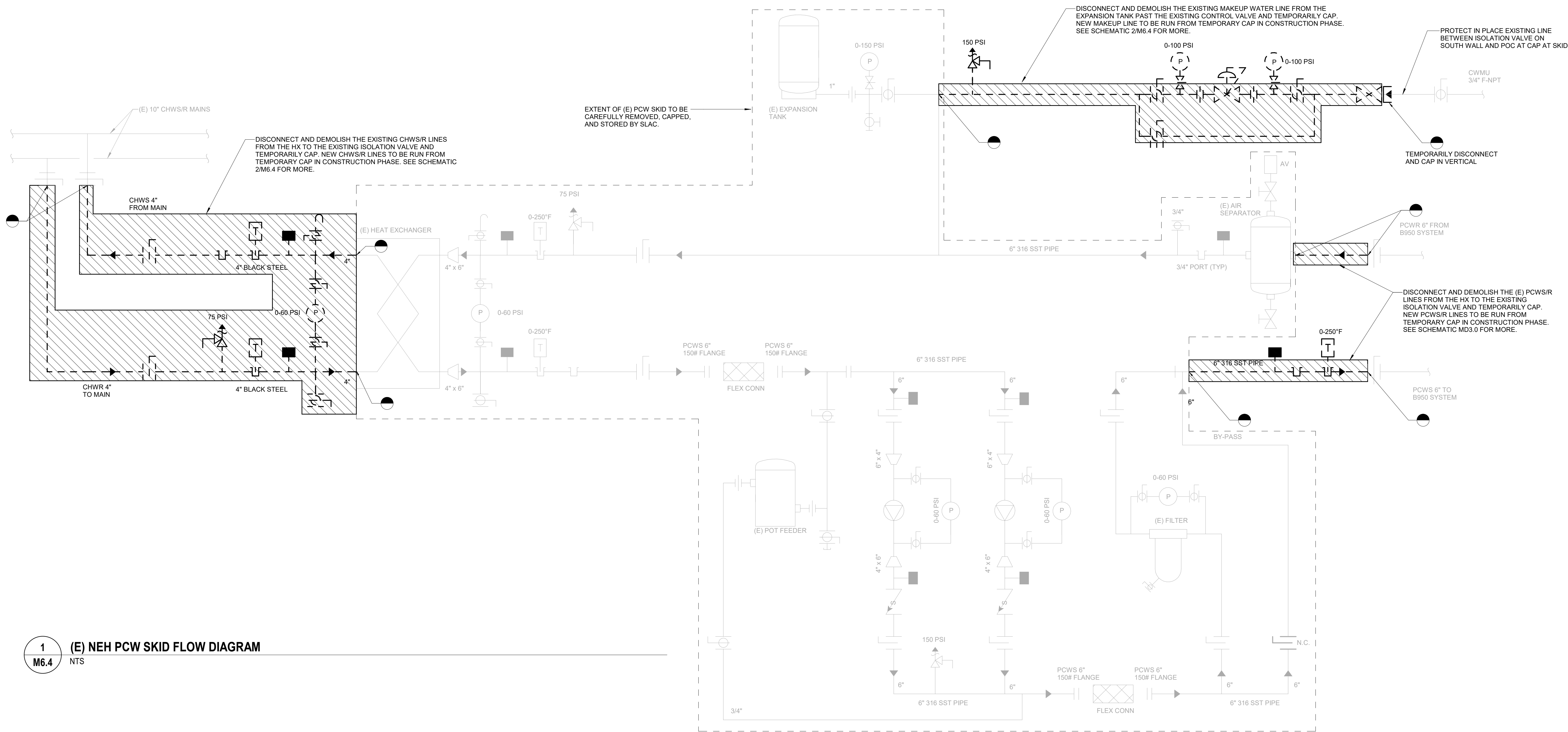
B950/NEH LCLS-II-HE:EI MECHANICAL MECHANICAL SCHEMATIC - PCW P&ID		M6.3	
DRAWING NUMBER B0950-M05-0340		REVISION 7	E

LCLS-II-HE:EI MECHANICAL SCHEMATIC - PCW P&ID

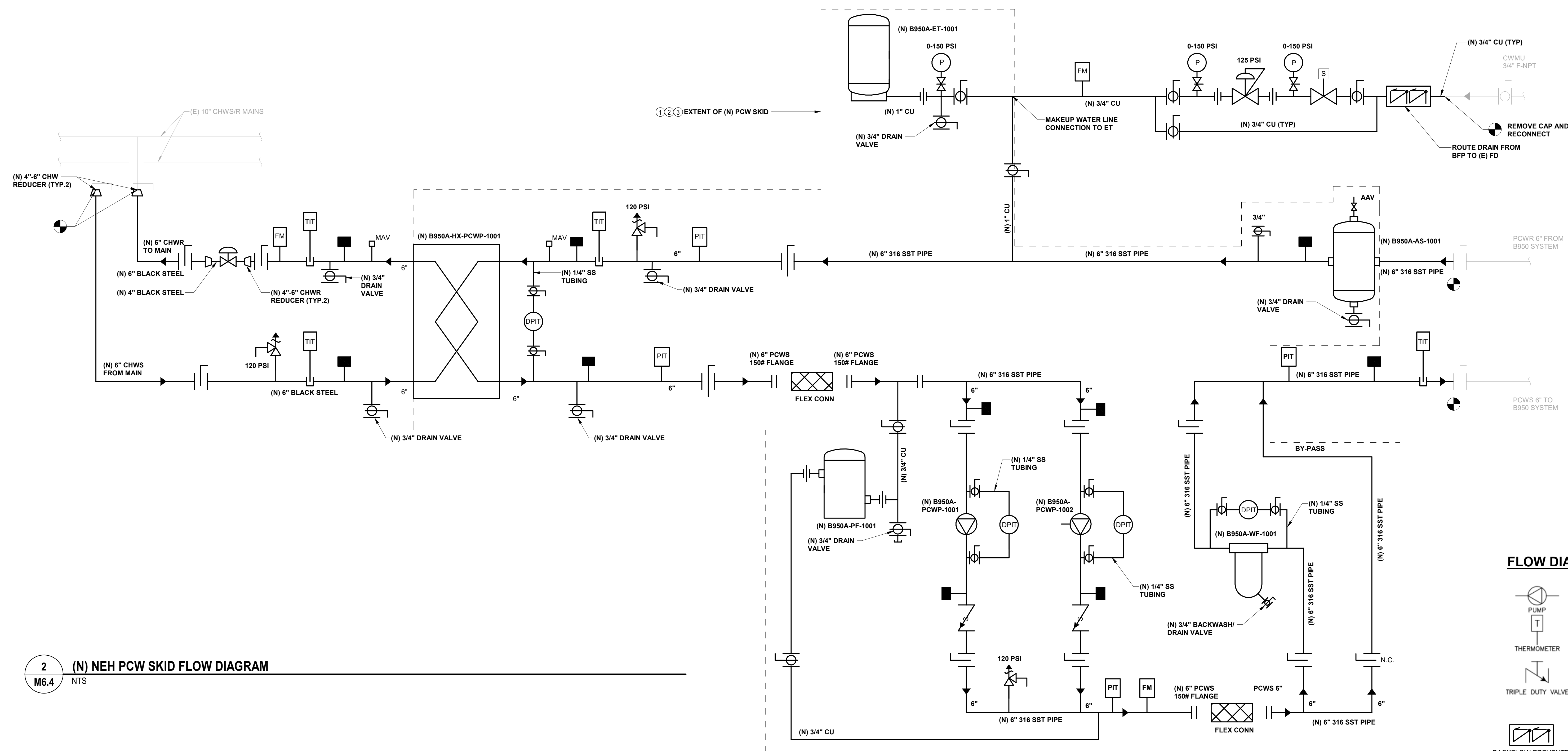
E

B0950-M05-0340

REV	DESCRIPTION	DRN	CHK	APP	DATE
E	RESPONSE TO BIO				25 MAR 26
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19



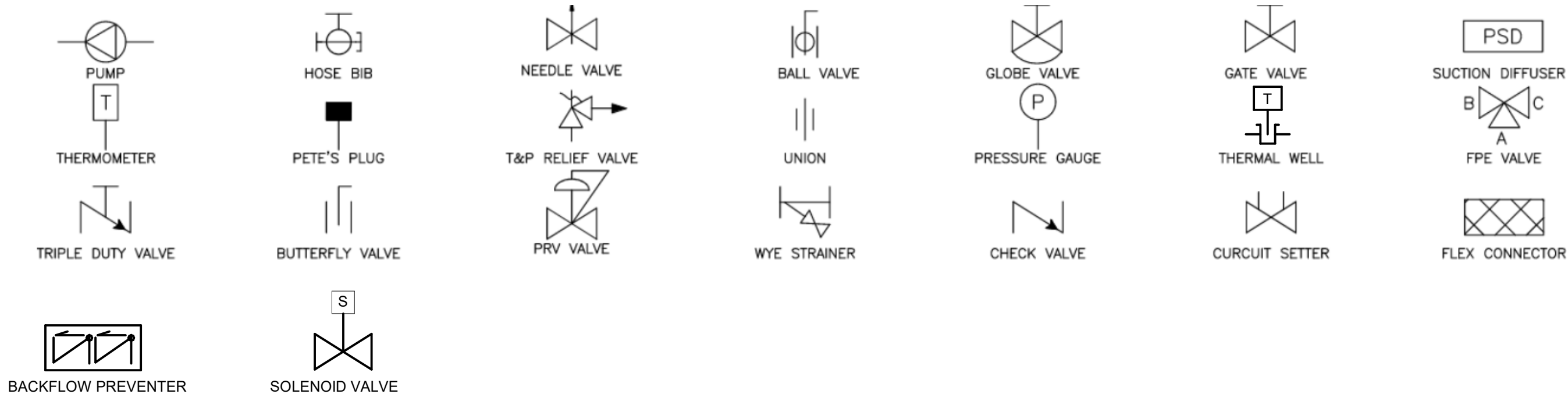
1
M6.4
(E) NEH PCW SKID FLOW DIAGRAM
NTS



2
M6.4
(N) NEH PCW SKID FLOW DIAGRAM
NTS

- NOTES:
1) SEE 2/M6.1 FOR I&C P&ID SCHEMATIC.
2) ALL I&C COMPONENTS SHOWN WITHIN THE BOUNDS OF THE PCW SKID SHALL BE PROVIDED BY THE FACTORY, NOT FIELD INSTALLED.
3) PCW SKID SHALL BE FACTORY PRESSURE TESTED WITH HYDROSTATIC PRESSURE TEST OF 1.5 OPERATING PRESSURE (180 PSI, SEE PRESSURE TEST TABLE ON M7.0 FOR MORE).
LEGEND:
(E) - EXISTING INSTRUMENTATION
(N) - NEW INSTRUMENTATION

FLOW DIAGRAM SYMBOL LEGEND:



GENERAL NOTES

- A. NO SWAGELOK COMPRESSION FITTINGS WITH ELASTOMER O-RINGS SHALL BE USED IN HOUSING OR ELSEWHERE IN SLAC CAMPUS PIPING. SEE PROCESS PIPING SPECIFICATIONS.
B. ALL MANUAL AIR VENTS AND DRAINS ON ANY PCW/LOW SYSTEM NEED TO BE PLUMBED BACK TO SUMP OR POLY HOLDING TANK THROUGH A COMMON HEADER. LOCATE ALL AIR VENTS AT HIGH POINTS IN THE SYSTEMS. LOCATE ALL DRAINS AT LOW POINTS IN THE SYSTEM.
C. INCLUDE ISOLATION VALVE(S) AND TEES FOR ALL NEW BRANCH PIPING TO FACILITATE ISOLATION AND TESTING.
D. ACTUAL PIPE ROUTING TO BE APPROVED BY SLAC SME THROUGH SHOP DRAWING SUBMITTAL.
E. PROVIDE DIELECTRIC UNIONS AT ALL DISSIMILAR MATERIAL CONNECTIONS.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (BFC)	
BY: cmadler	
DATE: 3/9/2026	PRSE: 261-058
Inspected/Authorized:	☒ Yes ☐ No
The submission of these plans shall not be construed to be a permit or approval.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

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ENGR	DATE	APPROVALS	DATE
DES	02/28/2025	Approved	02/28/2025
CHK	02/28/2025		
CHR	02/28/2025		

B950/NEH
LCLS-II-HE:EI
MECHANICAL
MECH SCHEMATIC - PCW FLOW DIAG

DRAWING NUMBER
B0950-M05-0447

REVISION
7

E

LCLS-II-HE:EI MECH SCHEMATIC - PCW FLOW DIAG

E

B0950-M05-0447

REV	DESCRIPTION	DRN	CHK	APP	DATE
5	BULLETIN 3 - XRT SCOPE				25 OCT 28
6	BULLETIN 4 - PCW SKID REPLACEMENT				25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION				26 JAN 19

DUCT INSTALLATION SCHEDULE

SYSTEM	MATERIAL	FITTINGS	JOINTS	NOTES
METAL DUCT	GALVANIZED SHEET METAL	PER SMACNA STANDARDS	PER SMACNA STANDARDS	(1)(2)(3)
FABRIC DUCT	NANOSOX	(4)	(4)	(4)(5)

NOTES:
1. DUCT TO BE INSTALLED AND TESTED TO MEET 2019 CALIFORNIA MECHANICAL CODE REQUIREMENTS.
2. SUPPLY AND RETURN AIR DUCT TO BE INSULATED WITH 1-1/2" THICK OWENS CORNING TYPE 75 FACED FIBERGLASS INSULATION.
3. FABRICATE AND INSTALL SHEET METAL DUCT PER LATEST SMACNA MANUAL.
4. FABRIC INLETS ARE TO BE CONNECTED TO METAL DUCT VIA ANCHOR PATCHES SUPPLIED BY MANUFACTURER. ZIP SCREW FASTENERS USED FOR SECURE INLET ANCHOR PATCHES ARE TO BE SUPPLIED BY CONTRACTOR.
5. REFER TO SPECIFICATION SECTION 23.37.16 FOR MORE INFORMATION ON THE FABRIC DUCT.

(N) PUMP SCHEDULE

MARK	MANUFACTURER AND MODEL NUMBER	LOCATION	SERVICE	TYPE	DESIGN FLOW (GPM)	SELECTION FLOW (GPM)	DESIGN HEAD (FT WG/PSI)	MAX FLOW (GPM)	MAX HEAD (FT WG/PSI)	MOTOR BHP	HP	RPM	ELECTRICAL V/PH/Hz	TOTAL OPERATING WEIGHT	REMARKS
B950A-PCWP-1001	Goulds 125SV-55BG7-4C60	MECH ROOM 4 60B24	NEHXPP	INLINE	581.9	650	277/120	677	300/130	73.9	75	3500	480/3/60	1386	(1)-(9)
B950A-PCWP-1002	Goulds 125SV-55BG7-4C60	MECH ROOM 4 60B24	NEHXPP	INLINE	581.9	650	277/120	677	300/130	73.9	75	3500	480/3/60	1386	(1)-(9)

NOTES:
(1) N+1 PUMPS
(2) VFD'S SUPPORTING PUMPS SHALL BE NEW. SEE ADJACENT VFD SCHEDULE.
(3) THE PUMP INTO NEW COMPACT LOGIC PLC CONTROL PANEL FOR EACH PUMP. MOVING CONTROLS FROM ALC TO PLC.
(4) PROVIDE WITH PREMIUM EFFICIENCY, VFD COMPATIBLE, TFC MOTOR WITH EXTERNAL AGESIS MOTOR SHAFT GROUNDING. INTERNAL SHAFT GROUNDING IS NOT ACCEPTABLE.
(5) PROVIDE ALL CONTROL DEVICES TO CONTROL PANEL.
(6) PUMP SHALL HAVE 316 SS CONSTRUCTION.
(7) PUMP SEAL FACES SHOULD BE SILICON CARBON WITH VITON SEAL RINGS.
(8) EQUIPMENT SHALL BE PROCURED BY SLAC.
(9) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

PRESSURE TESTING REQUIREMENTS

TYPE OF SYSTEM	TEST MEDIUM	TEST PRESSURE	DESIGN PRESSURE	TEST PROCEDURE	CODE REFERENCE
PCW	WATER	180 PSI	120 PSI	30 MIN	ASME B31.3 SS 345.5 CMC 1405
CHW	WATER	120 PSI	80 PSI	30 MIN	ASME B31.3 SS 345.5 CMC 1405
MAKEUP WATER	WATER	120 PSI	80 PSI	30 MIN	ASME B31.3 SS 345.5 CMC 1405
CA	AIR OR A NON-TOXIC, NON-FLAMMABLE GAS	180 PSI	150 PSI	FIRST LEAK CHECK PERFORMED (SMALLER OF 1.5 TIMES DESIGN PRESSURE AND 25 PSIG) RAISE PRESSURE GRADUALLY IN STEPS, 30 MINUTES (MINIMUM) AT TEST PRESSURE. PRESSURE SHALL BE REDUCED TO DESIGN PRESSURE BEFORE EXAMINING FOR LEAKS	ASME B31.3 SS 345.5 CMC 1405

NOTES:
(1) PRESSURE TESTING OF SYSTEMS SHALL BE CONSISTENT WITH SLAC ENVIRONMENT, SAFETY AND HEALTH DIVISION CHAPTER 14: PRESSURE TESTING.
(2) ALL PRESSURE TESTING WILL REQUIRE A SLAC APPROVED PRESSURE TESTING PLAN. COORDINATE WITH SLAC PROJECT CM.
(3) SLAC PROJECT TEAM SHALL PUT TOGETHER A CONTINGENCY PLAN SHOULD ANY SYSTEMS RUN INTO ISSUES WITH PRESSURIZATION DURING PRESSURE TESTS.
(4) ALL TAPS OFF EXISTING SYSTEMS SHALL BE PROVIDED WITH ISOLATION VALVES ON BOTH ENDS OF THE CONNECTION TO EXISTING SYSTEM. IN ADDITION, PROVIDE HIGH POINT VENTS AND LOW POINT DRAINS IN ORDER FOR TESTING OF NEW SYSTEM TO BE COMPLETELY ISOLATED FROM EXISTING SYSTEM.

GRILLE SCHEDULE

MARK NUMBER	MAX AIRFLOW (CFM)	DESIGN AIRFLOW (CFM)	AIR VELOCITY (FPM)	FACE SIZE IN (WxH)	CONNECTION SIZE, IN (WxH)	MAX STATIC PRESS. DROP (IN WG)	MAX NC	MOUNTING LOCATION	MATERIAL	FINISH	BASIS OF DESIGN	NOTES
RG-1	1555	1410/1415	450	30x16	30x16	0.08	25	DUCT	STEEL	#26 WHITE	TITUS 350RL	(1)-(2)

NOTES:
1. PROVIDE VOLUME DAMPER IN DUCT PRIOR TO GRILLE CONNECTION.
2. AIRFLOW TO BE AS DESIGNATED ON THE HVAC DUCT ROUTING FLOOR PLAN.

REHEAT COIL SCHEDULES (RELOCATED)

ID TAG	ROOM NUMBER	ROOM NAME	MAXIMUM AIRFLOW CFM	CAPACITY MBH	AIR PD IN W.G.	EAT DEG F	LAT DEG F	FLOW GPM	WATER PD FT WG	EWT DEG F	LWT DEG F	PIPE CONN IN	COIL DIMENSION HxW (IN)	BASIS OF DESIGN	NOTE
B950-RHC-1002	100H3	HUTCH 1.3 XPP	6,050	126	0.10	55	73	9	16.3	180	155	1 1/4	24x48	HEATCRAFT	1-2

NOTES:
1. EXISTING REHEAT COIL, RELOCATED FROM DEMOLITION PHASE. AIRFLOW IS BASED ON AIR BALANCE REPORT. PRE-MEASUREMENT IS REQUIRED TO DETERMINE ACTUAL AIRFLOW
2. CLEAN REHEAT COIL PRIOR TO RE-INSTALLATION.

AIR HANDLING UNIT SCHEDULE (EXISTING)

MARK NUMBER	LOCATION	CAPACITY DESIGNED CFM	OUTSIDE (NOTE 3) CFM	FANS TYPE	NUMBER OF FANS	DESIGN TOTAL SP IN W.G.	BHP PER FAN BHP	MOTOR HP PER FAN HP	ELEC POWER V/Phz	FLA AMPS	MCA AMPS	COILS COOLING CAPACITY MBH	EAT	LAT	EWT	LWT	APD IN W.G.	WPD FT	FILTERS PRE FILTER	FINAL FILTER	WEIGHT LBS	BASIS OF DESIGN	NOTE
B950-AHU-3001	MECH RM	60,000	5,445	DIRECT DRIVE PLENUM	10	9.36	12.54	15	460/3/60	146	164	1,195	68.5/58.2	48.6/48.5	44	55.5	1.03	15.26	MERV 8	99.97 HEPA		TEMTROL	1, 2, 3

NOTES:
1. LISTED ONLY FOR REFERENCE AND TO INDICATE REVISED EQUIPMENT TAG AND MAKE-UP AIRFLOW.
2. NEW WORK IS LIMITED TO ADJUSTING MOTORIZED DAMPERS TO NEW AIRFLOWS INDICATED IN MOTORIZED DAMPER SCHEDULES
3. MAKE-UP AIR IS CONDITIONED AIR FROM MAU-950-1. THEREFORE NO ADDED COOLING LOAD TO EXISTING AHU CAPACITY.

HUTCH 1.3 PCW RISER UTILITY SCHEDULE					TOTAL FLOW: 65.45	
	DESCRIPTION	FLOW (GPM)	DP (PSI)	SERIES	POC BRANCH	BRANCH SUM
(N)	B950-100H3-R10	1.936	20		MAIN	1.936
(E)	B950-100H3-R04	4.4	20		LOOP 1	
(E)	B950-100H3-R05	4.4	20		LOOP 1	
(N)	B950-100H3-R06	4.4	20		LOOP 1	
(E)	B950-100H3-R07	4.4	20		LOOP 1	
(E)	B950-100H3-R08	4.4	20		LOOP 1	
(N)	XPP RE-ENTRANT WINDOW	1.65	30		LOOP 1	30.15
(N)	HE IPM (EM2L2)	1.1	30		LOOP 1	
(N)	HE WAVEFRONT SENSOR (PF2L2)	1.1	30		LOOP 1	
(N)	HE PPM (IM2L2)	1.1	30		LOOP 1	
(N)	HE PHOTON TERMINATOR (SF1L2)	2.2	30		LOOP 1	
(N)	CIRCUIT SETTER	1	-		LOOP 1	
(N)	LODCM TOWER 2 WATER CHILLER	1.1	30		LOOP 2	
(N)	PHOTON STOPPER (ST1L2)	3.3	30		LOOP 2	
(N)	HE POWER SLITS (SL1L2)	0.44	30		LOOP 2	
(N)	HE TRANSFOCATOR (RL1L2)	0.66	30		LOOP 2	
(N)	HE ATM (TM1L2)	1.1	30		LOOP 2	
(N)	HE IPM (EM1L2)	1.1	30		LOOP 2	19.16
(N)	HE SOLID ATTENUATOR (AT1L2)	0.66	30		LOOP 2	
(N)	HE PPM (IM1L2)	1.1	30		LOOP 2	
(N)	EXIT WINDOW (PF1L2)	1.1	30		LOOP 2	
(N)	LIB (LI1L2)	1.1	30		LOOP 2	
(N)	CIRCUIT SETTER	1	-		LOOP 2	
(N)	LASER CHILLER	5.5	30		LOOP 2	
(N)	CIRCUIT SETTER	1	-		LOOP 2	
(E)	B950-100H3-R01	4.4	20		LOOP 3	
(N)	B950-100H3-R02	4.4	20		LOOP 3	14.2
(N)	B950-100H3-R03	4.4	20		LOOP 3	
(N)	CIRCUIT SETTER	1	-		LOOP 3	

PIPE INSTALLATION SCHEDULE

SYSTEM	MATERIAL	INSULATION	K-VALUE (BTU IN/HR FT2 °F)	FITTINGS	JOINING METHOD	MAX HORIZONTAL SUPPORT SPACING (CMC 313.3)	HANGER ROD SIZING (CMC 313.6)	NOTES
PCW/S/R (PCW SYSTEM) <2" (2")	COPPER TYPE L	YES	0.24	COPPER	SOLDERED, BRAZED, OR THREADED	<1 1/2" 6 FT >2" 10 FT	PIPE 1/2"-4" ROD 3/8"	(1)-(6),(8)-(10)
PCW/S/R (PCW SYSTEM) (3" AND LARGER)	316L STAINLESS STEEL	YES	0.24	316L STAINLESS STEEL	WELDED	12 FT	PIPE 5"-8" ROD 1/2"	(1)-(6),(8)-(10)
CHW/S/R (CHW SYSTEM)	SCHEDULE 40 BLACK STEEL	YES	0.24	STEEL FITTINGS	WELDED	12 FT	PIPE 5"-8" ROD 1/2"	(1)-(6),(8)-(10)
COMPRESSED AIR	COPPER TYPE L	NO	-	COPPER	SOLDERED, BRAZED, OR THREADED	<1 1/2" 6 FT >2" 10 FT	PIPE 1/2"-4" ROD 3/8"	(1)-(3),(5),(8)-(10)
HEATING HOT WATER (SUPPLY/RETURN)	COPPER TYPE L	YES	0.24	COPPER	SOLDERED, BRAZED, OR THREADED	<1 1/2" 6 FT >2" 10 FT	PIPE 1/2"-4" ROD 3/8"	(2)-(10)
MAKEUP WATER	COPPER TYPE L	NO	-	COPPER	SOLDERED, BRAZED, OR THREADED	<1 1/2" 6 FT	PIPE 1/2"-4" ROD 3/8"	(1)-(3),(5),(8)-(10)

NOTES:
(1) REFERENCE SPECIFICATIONS FOR ADDITIONAL INFORMATION. IF INFORMATION IN SCHEDULE DIFFERS FROM SPECIFICATIONS, INFORMATION IN SPECIFICATION SHALL GOVERN. (PROCESS PIPING SPEC)
(2) PRESSFIT NOT ALLOWED ON ANY SYSTEM.
(3) NO USE OF TEFLON MATERIAL (TAPE, VALVE SEAT, ETC.) IN THE RADIATION AREA.
(4) CELLULAR GLASS INSULATION SHALL BE COMPLIANT WITH ASTM C552. FLEXIBLE ELASTOMERIC INSULATION SHALL BE COMPLIANT WITH ASTM C534/C533M.
(5) SUPPORT SPACING VALUES FROM 2022 CPC TABLE 313.3
(6) PIPING TO BE INSULATED WITH 1-1/2" THICK FIBERGLASS INSULATION.
(7) ALL HEATING HOT WATER VALVES AND ACCESSORIES SHALL BE RATED AT 240°F
(8) REFERENCE SPECIFICATIONS FOR ADDITIONAL INFORMATION. IF INFORMATION IN SCHEDULE DIFFERS FROM SPECIFICATIONS, INFORMATION IN SPECIFICATION SHALL GOVERN. (HYDRONIC PIPING SPEC)
(9) PER CMC 313.6, PIPING SUPPORTED VIA HANGERS WITH SIZES 1/2" - 4" SHALL USE A 3/8" ROD, AND PIPING SUPPORTED VIA HANGERS WITH SIZES 5" - 8" SHALL USE A 1/2" ROD.
(10) PIPING SHALL BE LABELED FOR DIRECTION OF FLOW AND CONTENTS OF PIPE IN ACCORDANCE WITH ASME B31.3 (PROCESS PIPING) AND ASME/ANSI A13.1.

(N) HEAT EXCHANGER SCHEDULE

MARK	MANUFACTURER AND MODEL NUMBER	LOCATION	SERVICE	DESIGN CAPACITY (MBH)	MAX CAPACITY (MBH)	HIGH TEMP WATER SIDE					LOW TEMP WATER SIDE					DESIGN PLATES (QTY)	MAX PLATES (QTY)	TOTAL OPERATING WEIGHT	REMARKS		
						DESIGN GPM	MAX GPM	EWTF (°F)	LTWTF (°F)	PD/PSI	FLUID	DESIGN GPM	MAX GPM	EWTF (°F)	LTWTF (°F)					PD/PSI	FLUID
B950A-HX-PCWP-1001	BELL & COSSETT AP47	B950A MECHANICAL ROOM	B950A	3462	3905	581.9	650	74°	62°	4.97	PCW	349.12	389.16	45	65	1.92	CHW	186	228	1648 LBS	(1)-(3)



NOTES:

(1) PROVIDE WITH DOUBLE WALL CONSTRUCTION, 316 SS CONSTRUCTION, AND EPDM GASKETS.

(2) EQUIPMENT SHALL BE PROCURED BY SLAC.

(3) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

(N) VFD SCHEDULE

MARK	MANUFACTURER AND MODEL NUMBER	HP	FLA	VOLTS	WEIGHT	ENCLOSURE	REMARKS
B950A-VFD-PCWP-1001	PUMP SMART ABB ACH580-PCR-096A-4	75	96	480	359 LBS	NEMA 3R	(1)-(5)
B950A-VFD-PCWP-1002	PUMP SMART ABB ACH580-PCR-096A-4	75	96	480	359 LBS	NEMA 3R	(1)-(5)

NOTES:
(1) VFD SHALL BE PROVIDED WITH INTEGRAL LOCKABLE DISCONNECT AND BE SUITABLE FOR OUTDOOR INSTALLATION. DISCONNECT ALSO TO BE PROVIDED BETWEEN POWER SOURCE AND VFD. SEE ELECTRICAL DRAWINGS FOR LOCATION AND MORE INFORMATION.
(2) PROVIDE NEMA 3R ENCLOSURE.
(3) VFD'S SHALL BE DOWNLOADED WITH THE SOFTWARE ABB DRIVE COMPOSER TO KEEP PARAMETERS CONSISTENT THROUGHOUT DEVICES. COORDINATE WITH SLAC FOR DOWNLOAD AND SETUP OF PROGRAM.
(4) VFD TO BE PROVIDED BY MECHANICAL CONTRACTOR.
(5) EQUIPMENT TO BE PROCURED BY SLAC.

(N) COMPRESSED AIR REGULATOR

MARK	MANUFACTURER AND MODEL NUMBER	LOCATION	OUTLET PRESSURE	DIAPHRAGM	INLET/OUTLET SIZE	WALL MOUNTABLE	SHIPPING WEIGHT	REMARKS
(N) REG-1	CONOFLW GH30-XTHMXSG	HUTCH 1.3	125 PSIG	BUNA "N" (WITH RELIEF NO BLEED)	1/4"	YES	2 LB	1-4

NOTES:
(1) RANGE 0-125 PSI, MAX SUPPLY 250 PSI.
(2) 316 STAINLESS STEEL BODY AND INTERNALS.
(3) ADJUSTMENT VIA HANDWHEEL.
(4) PROVIDE GAUGE DOWNSTREAM OF REGULATOR.

PCW AIR SEPARATOR SCHEDULE

MARK	MFR AND MODEL NUMBER	LOCATION	SERVICE	INLET CONNECTION	OUTLET CONNECTION	TANK VOLUME	MAX CAPACITY	MAX PRESSURE	MAX TEMP	DIMENSIONS	TANK WEIGHT (DRY)	TANK WEIGHT (WETTED)	NOTES
(N) B950A-AS-1001	ROLAIRTRLO RL-6FB	B950A MECH ROOM	AIR SEPARATOR	6" FLANGE	6" FLANGE	34 GAL	850 GPM	125 PSIG	350°F	44-116"H x 21-1/4"D	280 LBS	564 LBS	(1)-(6)

NOTES:
(1) EQUIPMENT SHALL BE PROCURED BY SLAC.
(2) INSTALL PER MANUFACTURER RECOMMENDATIONS.
(3) SEE IM 3 FOR I&C P&ID.
(4) SHELL SHALL BE CARBON STEEL. PROVIDE DIELECTRIC UNIONS AT ALL DISSIMILAR MATERIAL CONNECTIONS.
(5) SHALL BE CONSTRUCTED TO ASME SECTION VIII, DIV. 1.
(6) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

PCW POT FEEDER SCHEDULE

MARK	MFR AND MODEL NUMBER	LOCATION	SERVICE	INLET CONNECTION	OUTLET CONNECTION	TANK VOLUME	MAX PRESSURE	MAX TEMP	DIMENSIONS	TANK WEIGHT (DRY)	TANK WEIGHT (WETTED)	NOTES
(N) B950A-PF-1001	JL WINGERT CO 90D	B950A MECH ROOM	POT FEEDER	3/4" FNPT	3/4" FNPT	5.79 GAL	200 PSIG	200°F	21-18"H x 10"D	33 LBS	81.37 LBS	(1)-(5)

NOTES:
(1) EQUIPMENT SHALL BE PROCURED BY SLAC.
(2) INSTALL PER MANUFACTURER RECOMMENDATIONS.
(3) SEE IM 3 FOR I&C P&ID.
(4) BODY SHALL BE CARBON STEEL. PROVIDE DIELECTRIC UNIONS AT ALL DISSIMILAR MATERIAL CONNECTIONS.
(5) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

PCW EXPANSION TANK SCHEDULE

MARK	MFR AND MODEL NUMBER	PART NUMBER	LOCATION	SERVICE	TOP CONNECTION	BOTTOM CONNECTION	TANK VOLUME	MAX PRESSURE	MAX TEMP	DIMENSIONS	TANK WEIGHT (DRY)	TANK WEIGHT (WETTED)	NOTES
(N) B950A-ET-1001	WESELS NLA-85	22010085	B950A MECH ROOM	EXPANSION TANK	3/4" NPT	1" NPT (CAPPED)	23 GAL	125 PSIG	240°F	37"H x 16"D	90 LBS	281.82 LBS	(1)-(8)

NOTES:
(1) EQUIPMENT SHALL BE PROCURED BY SLAC.
(2) INSTALL PER MANUFACTURER RECOMMENDATIONS.
(3) SEE IM 3 FOR I&C P&ID.
(4) BOTTOM CONNECTION (1" NPT) SHALL BE CAPPED.
(5) CHARGING VALVE CONNECTION SHALL BE 0.352"-2.32NC FOR CHARGING SYSTEM.
(6) TANK SHALL BE BLADDER TYPE AND ASME RATED.
(7) PROVIDE WITH SIGHT GLASS OPTION.
(8) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

PCW FILTER SCHEDULE

MARK	MFR AND MODEL NUMBER (HOUSING)	MFR AND MODEL NUMBER (CARTRIDGES)	QTY OF CARTRIDGES	FLOW CAPACITY		FILTER	LOCATION	SERVICE	FILTER HOUSING DIMENSIONS	FILTER HOUSING WEIGHT	FILTER CARTRIDGE DIMENSIONS	FILTER CARTRIDGE WEIGHT	TOTAL FILTER WEIGHT	NOTES
				PER CARTRIDGE	TOTAL									
(N) B950A-WF-1001	SHELLCO SJCH-170 FILTER HOUSING	SHELLCO SJCH-170-20 FILTER CARTRIDGES	4	150 GPM	600 GPM	20 MICRON	B950A MECH ROOM	WATER FILTER	47-1/2"H x 12-1/2"D	91 LBS	30-3/4"H x 7-3/4"D	8 LBS EA	123 LBS	(1)-(5)

NOTES: (1) EQUIPMENT SHALL BE PROCURED BY SLAC.
(2) INSTALL PER MANUFACTURER RECOMMENDATIONS.
(3) SEE IM 3 FOR I&C P&ID INCLUDING BACKWASH DRAIN VALVE.
(4) FILTERS CAN EXCEED LISTED CAPACITY OF FLOW PER MANUFACTURER CORRESPONDENCE.
(5) VENDOR MUST PROVIDE LISTING BY A NRTL FIELD INSPECTION BODY UNLESS THE INTEGRATING VENDOR IS A NRTL-QUALIFIED ASSEMBLY SHOP. ALL COMPONENTS MUST BE NRTL LISTED/LABELED AS APPROPRIATE FOR APPLICATION.

PCW RISER DIAGRAM SCHEDULE - HUTCH 1.3

M7.0

NEH PCW FLOW DEMAND

M7.0

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (BFC)	
BY: ☐ creator	
DATE: 3/9/2028	PRSE: 26-058
Reviewed/Authorized: ☒ Yes ☐ No	
These plans shall be kept on the job site for all construction inspections.	

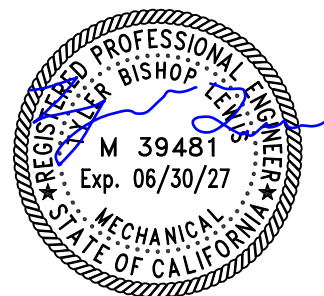
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SCALE: 1/2" = 1'-0"		DO NOT SCALE DRAWING		B0950.x	
				U.S. DEPARTMENT OF ENERGY	
NATIONAL ACCELERATOR LABORATORY					
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF STANFORD UNIVERSITY / SLAC.					
ENGR	GP	DATE	APPROVALS	DATE	
DES	GP	02/26/2025	Approved		
CHK	GP	02/26/2025			

REVISION	7	E
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(LCLS II - HE-B) MECHANICAL SCHEDULES

ON DRAWINGS FOR THE ENGINEERS & SLAC REVIEW WILL BE REQUIRED AS FOLLOWS:

1. MIX DESIGNS.
2. REINFORCING STEEL.
3. STRUCTURAL STEEL AND MISCELLANEOUS METALS;

ERIFICATION SHALL NOT PROCEED UNTIL SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED BY THE ENGINEER, BY SLAC, AND BY THE BUILDING INSPECTION OFFICE.

STRUCTURAL SPECIFICATIONS

SHOP DRAWINGS FOR THE ENGINEERS & SLAC REVIEW WILL BE REQUIRED AS FOLLOWS

1. MIX DESIGNS;
2. REINFORCING STEEL;
3. STRUCTURAL STEEL AND MISCELLANEOUS METALS;
- FABRICATION SHALL NOT PROCEED UNTIL SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED BY THE ENGINEER, BY SLAC, AND BY THE BUILDING INSPECTION OFFICE.

GOVERNING CODE: 2019 CALIFORNIA BUILDING CODE

SHEET LIST			
SHEET NUMBER	SHEET NAME	Sheet Organization	Discipline
S0.1	STRUCTURAL SPECIFICATIONS	1	03 STRUCTURAL
S0.2	SPECIAL INSTRUCTIONS	2	03 STRUCTURAL
S02.1	SUB-BSMT - ENL DEMO FLOOR PLAN	3	03 STRUCTURAL
S2.1	SUB-BSMT - ENL FLOOR PLAN	4	03 STRUCTURAL
S2.2	SUB-BSMT - ENL RCP	5	03 STRUCTURAL
S2.3	SUB-BSMT - ENL RCP-APP WALL	6	03 STRUCTURAL
S3.1	BSMT - POW SKID	7	03 STRUCTURAL
S5.0	STRUCTURAL DETAILS	8	03 STRUCTURAL
S5.1	STRUCTURAL DETAILS	9	03 STRUCTURAL
S5.2	STRUCTURAL DETAILS	10	03 STRUCTURAL

SHEET LIST			
SHEET NUMBER	SHEET NAME	Sheet Organization	Discipline
S0.1	STRUCTURAL SPECIFICATIONS	1	03 STRUCTURAL
S0.2	SPECIAL INSTRUCTIONS	2	03 STRUCTURAL
S02.1	SUB-BSMT - ENL. DEMO FLOOR PLAN	3	03 STRUCTURAL
S2.1	SUB-BSMT - ENL. FLOOR PLAN	4	03 STRUCTURAL
S2.2	SUB-BSMT - ENL. RCP	5	03 STRUCTURAL
S2.3	SUB-BSMT - ENL. RCP-APP WALL	6	03 STRUCTURAL
S3.1	BSMT - POW SKID	7	03 STRUCTURAL
S5.0	STRUCTURAL DETAILS	8	03 STRUCTURAL
S5.1	STRUCTURAL DETAILS	9	03 STRUCTURAL
S5.2	STRUCTURAL DETAILS	10	03 STRUCTURAL

POWDER ACTUATED FASTENERS

REINFORCING STEEL
BARS FOR REINFORCING SHALL BE GRADE 60 DEFORMED BARS CONFORMING TO ASTM A615 INCLUDING SUPPLEMENT S1.

SELF-DRILLING FASTENERS
SELF-DRILLING FASTENERS SHALL BE "TRAXX/TEKS" AS MANUFACTURED BY ITW BUILDDEX, ITASCA, ILLINOIS
OR APPROVED EQUAL. USE LENGTH AND TYPE AS RECOMMENDED BY MANUFACTURER FOR INTENDED USE.

ARC WELDING SHALL BE PERFORMED BY A WELDER OR WELDER'S ATTENDANT TRAINED IN ACCORDANCE WITH THE CALIFORNIA BUILDING CODE TO PERFORM THE TYPE OF WORK REQUIRED. ALL WELDING SHALL BE IN ACCORDANCE WITH AWS WELDING CODE. WELDING ELECTRODES SHALL BE E70 SERIES. WELDING SHALL BE DONE IN AN APPROVED FABRICATOR'S SHOP IN ACCORDANCE WITH CBC SECTION 1705.2. SHOPS NOT APPROVED WILL BE CHARGED FOR ALL SHOP WELDING INSPECTION REQUIRED BY THE CALIFORNIA BUILDING CODE.

HIGH STRENGTH (H.S.) ANCHOR BOLTS SHALL BE GR 105 (MIN.) USE NUTS MEETING STRENGTH REQUIREMENTS OF BOLTS & ANCHOR BOLTS. HOLES IN STEEL SHALL BE STANDARD DIAMETER LISTED IN TABLE J3.3 OF AISC STEEL MANUAL U.N.O. HOLES IN COLUMN BASE PLATE FOR ANCHOR BOLTS MAY BE OVERSIZED PER TABLE 14-2. PROVIDE MINIMUM WASHERS FOR OVERSIZED HOLES AS REQUIRED BY TABLE 14-2 U.N.O.

ASTM F1554, GR 105, WHERE H.S. THREADED RODS SPECIFIED ON DRAWINGS.

EXPANSION ANCHORS SHALL BE KWIK BOLT 1/2" FOR ANCHORS IN CONCRETE AS MANUFACTURED BY HILL, INC. TULSA OKLAHOMA. PREPARATION OF HOLES AND INSTALLATION OF ANCHORS SHALL CONFORM TO MANUFACTURER' RECOMMENDATIONS AND ICC REPORT NO. ESR-4266. CONCRETE STRENGTH MUST BE AT 28 DAY STRENGTH IN ORDER TO DRILL & INSTALL.

EXPLOY ANCHOR & DOWELS SHALL BE INSTALLED USING "HILTI HIT-HY 200 V3" ADHESIVE ANCHOR SYSTEM" AS MANUFACTURED BY HILTI, INC., TULSA, OKLAHOMA. PREPARATION OF HOLES SHALL CONFORM TO MANUFACTURER'S RECOMMENDATIONS. ANCHORS AND RODS SHALL BE INSTALLED PER MANUFACTURER'S RECOMMENDATIONS AND ICC REPORT NO. ESR-4888. THREADED RODS SHALL BE GRADE "105 888-1 CLASS 5.5", ASTM 193, B7 OR "ASTM F593 316-SS", UNLESS NOTED OTHERWISE ON THE DRAWINGS. REINFORCEMENT BARS SHALL BE "ASTM A 615" GRADE 60 MIN. UNLESS NOTED OTHERWISE ON THE DRAWINGS. CONCRETE STRENGTH MUST BE AT 28 DAY STRENGTH IN ORDER TO DRILL AND INSTALL.

EPOXY INSERTS SHALL BE INSTALLED USING "HLTI HIS-A" OR "HLTI HIS-RN" WITH "HLTI HIT-HY 200 V3" ADHESIVE ANCHOR
 SYSTEM AS MANUFACTURED BY HLTI, INC., TULSA, OKLAHOMA. PREPARATION OF HOLES SHALL CONFORM TO MANUFACTURER'S
 RECOMMENDATIONS. INSERTS AND CAP SCREWS SHALL BE INSTALLED PER MANUFACTURER'S RECOMMENDATIONS AND
 ICC REPORT NO. ESR-4968. CAP SCREWS SHALL BE "SAE J429 GRADE 5" FOR USE WITH "HIS-A" OR "ASTM A193 GRADE B8M" FOR USE
 WITH "HIS-RN". UNLESS NOTED OTHERWISE ON THE DRAWINGS, CONCRETE STRENGTH MUST BE AT 28 DAY STRENGTH IN ORDER
 TO DRILL AND INSERT.

UNISTRUT CHANNELS AND FITTINGS SHALL BE THOSE MANUFACTURED BY UNISTRUT CORPORATION, WAYNE, MICHIGAN OR APPROVED EQUAL. INSTALLATION SHALL CONFORM TO MANUFACTURER'S RECOMMENDATIONS.

ALL STRUCTURAL STEEL AND MISCELLANEOUS IRON SHALL RECEIVE SHOP PRIME COAT AND FINISH COAT PER ARCH, EXCEPT AS FOLLOWS:

- ALL STEEL EXPOSED TO WEATHER SHALL BE GALVANIZED UNLESS NOTED OTHERWISE.
- CONTACT SURFACES IN SLIP CRITICAL CONNECTIONS SHALL BE FREE OF OIL, PAINT, LACQUER OR OTHER COATING (CLEAN MILL SCALE). NO OVERSPRAY IS PERMITTED. THESE CONNECTIONS SHALL BE PRIMED AFTER TIGHTENING.
- STEEL WHICH WILL RECEIVE SPRAY ON FIRE PROOFING MATERIAL.

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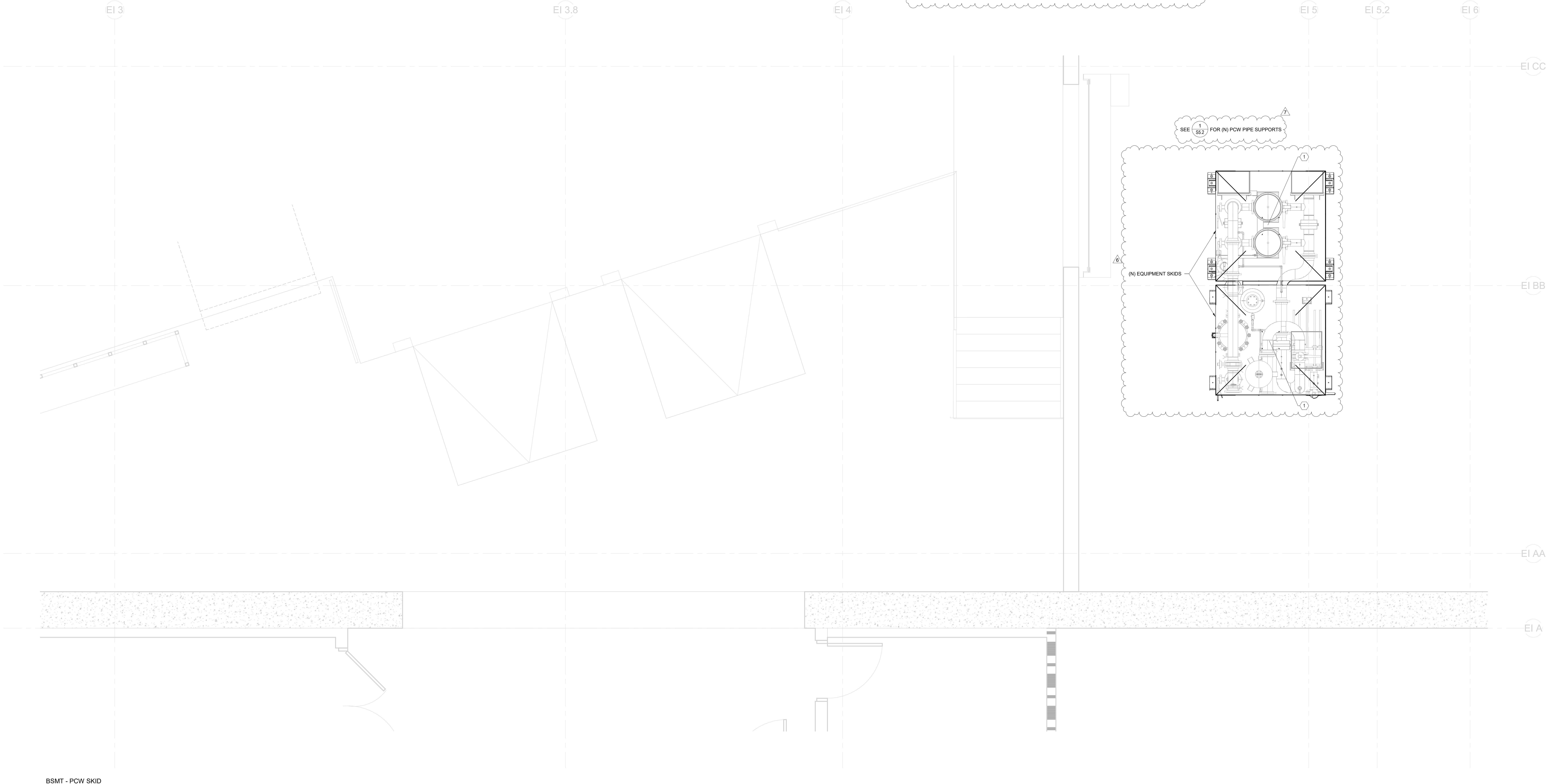
LOCAL ME-STRUCTURAL SPECIFICATIONS

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

REV	DESCRIPTION	DRN	CHK	APP	DATE
2	RFI-004 - XPP LIGHTING	SMP	DJS		25 JULY 28
6	BULLETIN 4 - PCW SKID REPLACEMENT	SMP	DJS		25 DEC 03
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION	SMP	DJS		26 JAN 19

EQUIPMENT SCHEDULE					
EQUIP. I.D.	DESCRIPTION	MAX. WEIGHT (LBS.)	MAX. HEIGHT (IN.)	MIN. WIDTH (IN.)	LENGTH (IN.)
1	PCW SKID	8,940	82.7	72	147

1. CONTRACTOR TO VERIFY. ANY DISCREPANCIES SHALL BE BROUGHT TO ENGINEER'S ATTENTION.



LEGEND

- ⊙ INDICATES EQUIPMENT ID - SEE PLAN
- ⊖ INDICATES KEYED SHEET NOTE - SEE PLAN
- T.O.S. = 12'-1 1/2" DENOTES TOP OF STEEL ELEVATION AT COL. OR RIDGE
- DENOTES DIRECTION OF ROOF SLOPE
- ⊠ INDICATES (N) OR (E) EQUIPMENT - SEE PLAN AND M.E.P. DWGS.

GENERAL NOTES

- (E) DENOTES EXISTING CONSTRUCTION
(N) DENOTES NEW CONSTRUCTION
(D) DENOTES EXISTING CONSTRUCTION TO BE DEMOLISHED
(R) DENOTES RELOCATED OR REUSED CONSTRUCTION
- 2'-0"± DENOTES EXISTING DIMENSION
4'-0"± DENOTES DIMENSION OR WEIGHT PROVIDED BY OTHERS
- CONTRACTOR TO VERIFY
- SEE STRUCTURAL SPECIFICATIONS ON SHEET S0.1.
- CONTRACTOR SHALL VERIFY ALL CONTROLLING EXISTING CONDITIONS AND DIMENSIONS PRIOR TO BEGINNING CONSTRUCTION AND/OR ORDERING MATERIALS. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER IMMEDIATELY.
- WEIGHTS OF EQUIPMENT SHOWN ON THE DRAWINGS ARE THE MAXIMUM WEIGHT ALLOWED INCLUDING ATTACHED ACCESSORIES.
- THE SIZE, LOCATION AND ORIENTATION OF ALL EQUIPMENT, SHALL BE VERIFIED BY THE CONTRACTOR. ANY CONFLICTS SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER IMMEDIATELY.
- FOR LIGHTING, SEE ELEC. DWGS.
- SEE DETAILS S6.1 AND S6.1 FOR SUPPORT RFI-004

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (BFC)

BY: **crader**

DATE: 3/20/2025 PERM: 36-2025

Personal Authorization ☒ Yes ☐ No

These plans have been reviewed for compliance with applicable building codes and standards, and I am a duly licensed professional engineer in the State of California.

THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

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STAMP: **PASE**

REGISTERED PROFESSIONAL ENGINEER
STRUCTURAL
S. BRUCE
S629

SCALE: As Indicated DO NOT SCALE DRAWING **B0950.2024.03**

SLAC
NATIONAL ACCELERATOR LABORATORY

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DATE: 03/26/2025 APPROVALS: **Approver** DATE: 03/26/2025

ENGR: DJS DES: DJS CHR: DJS

B950
LCLS-II-HE:EI
STRUCTURAL
BSMT - PCW SKID

DRAWING NUMBER: **B0950-SB1-0317**

REVISION: **7**

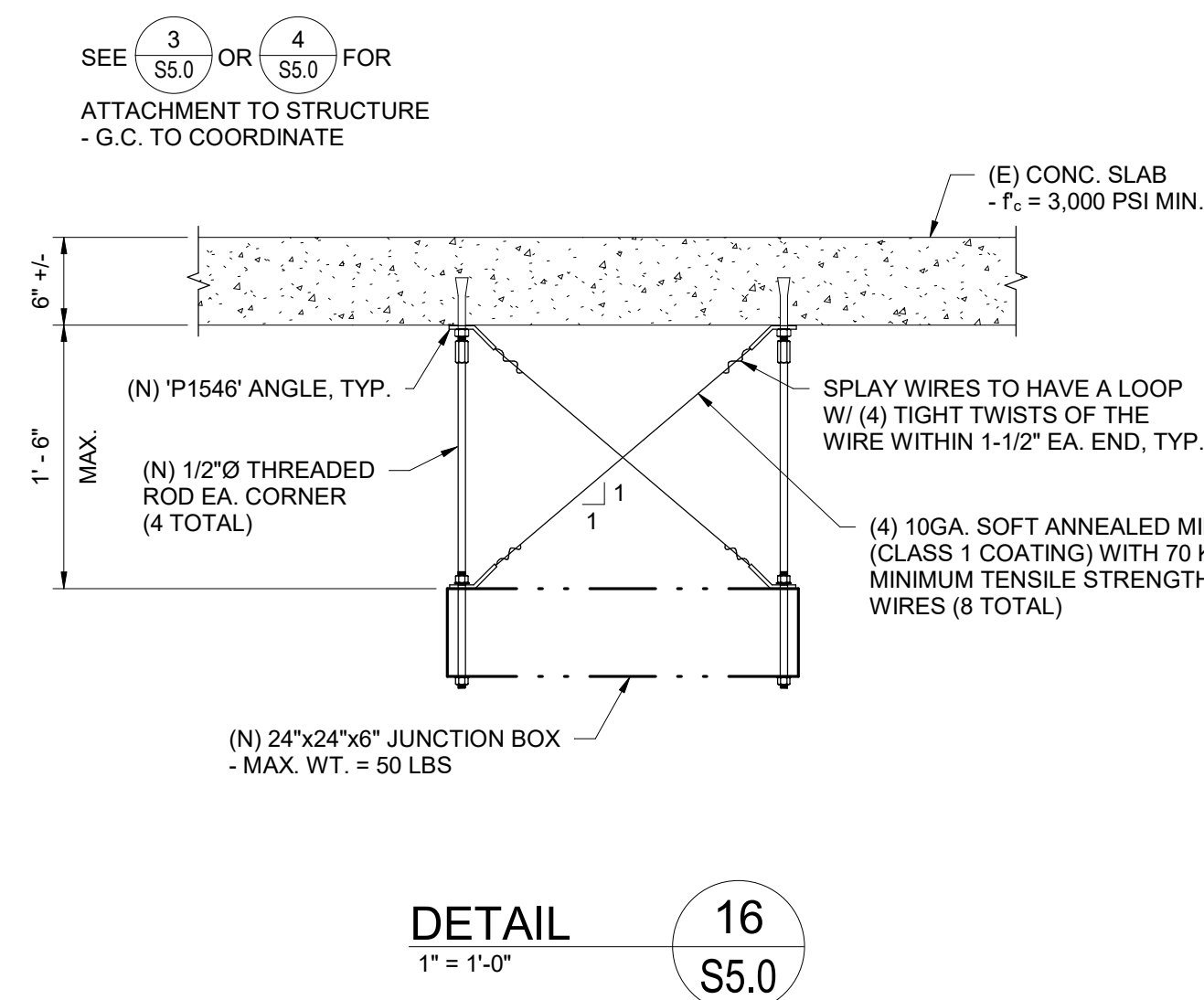
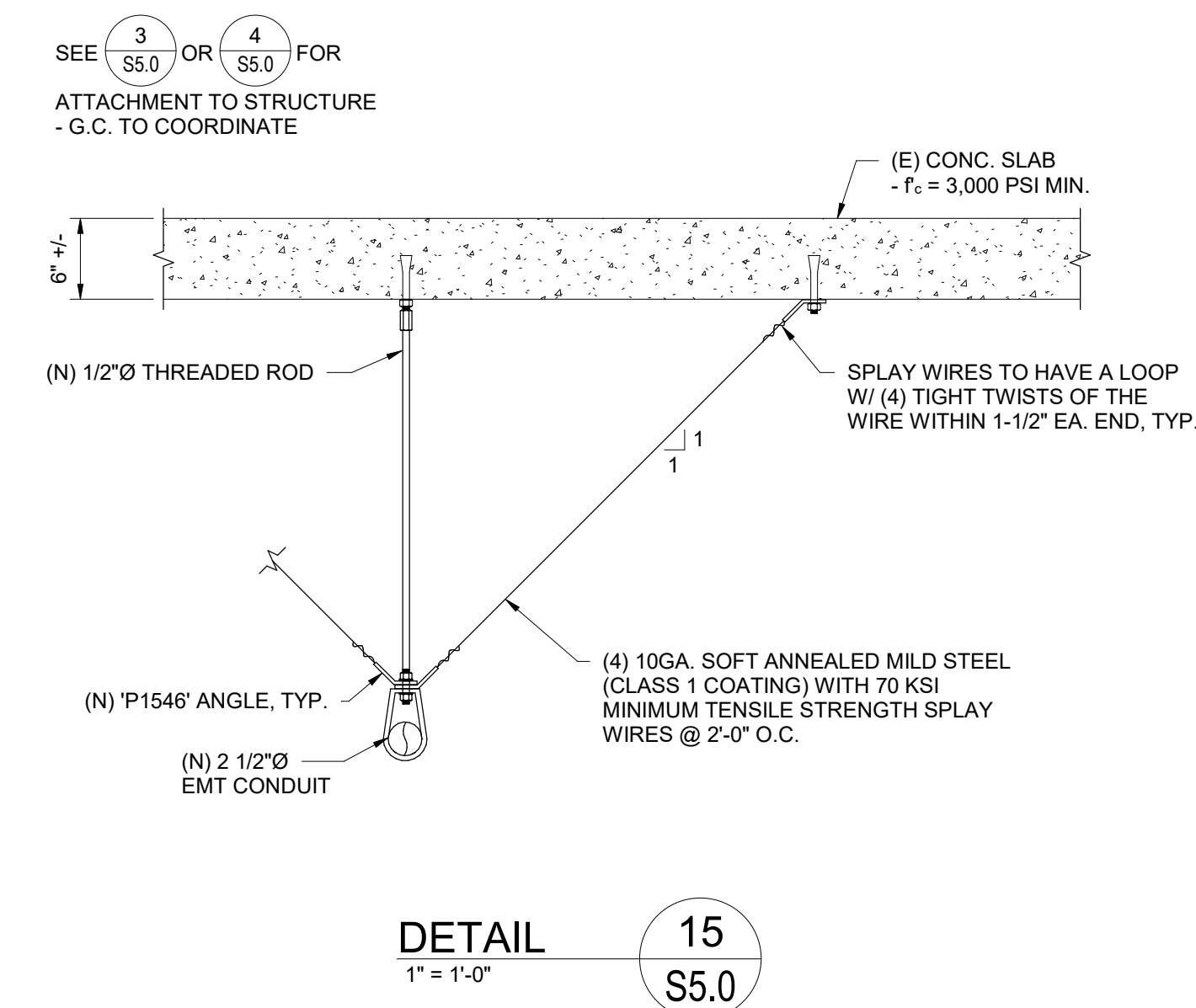
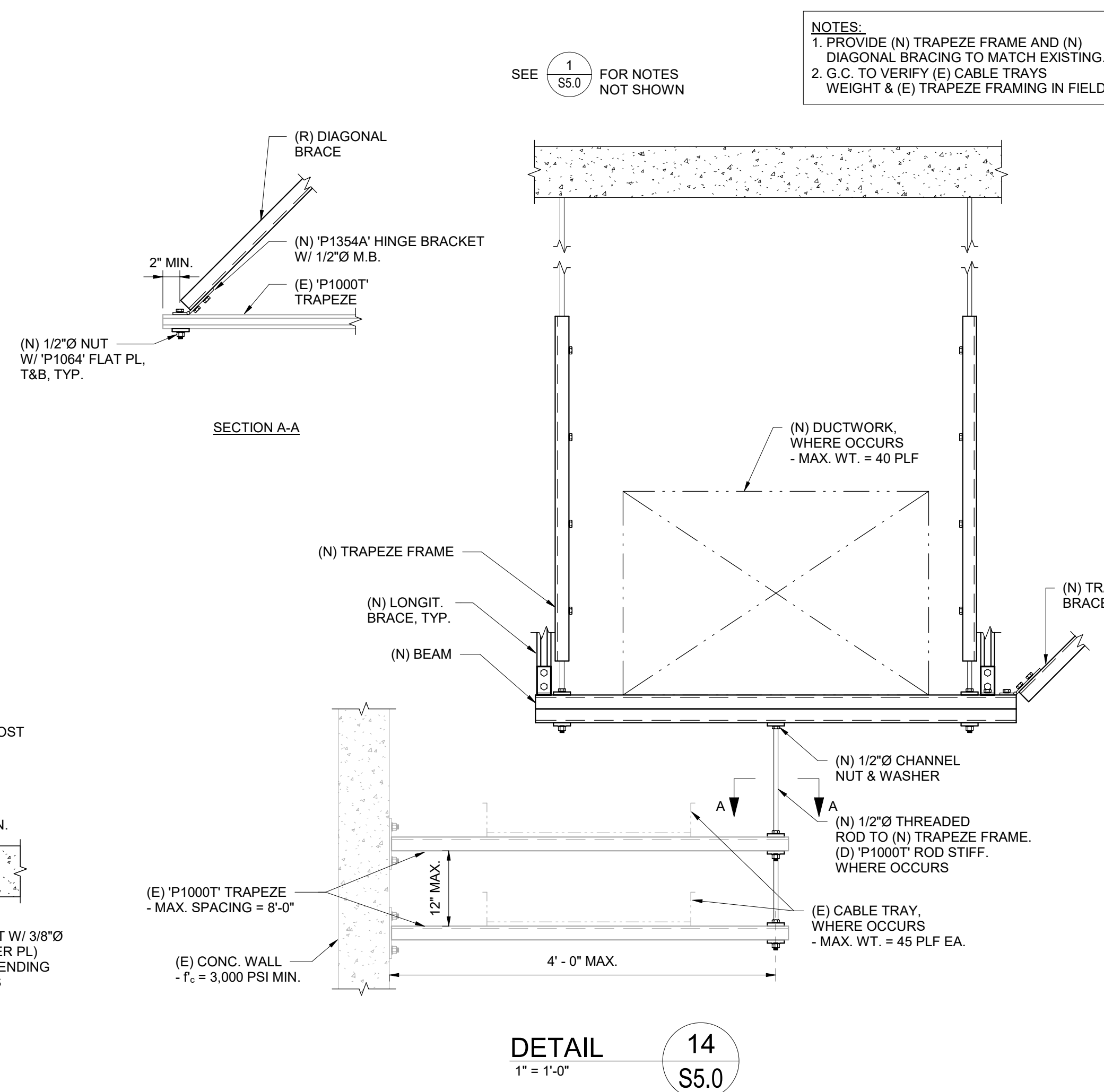
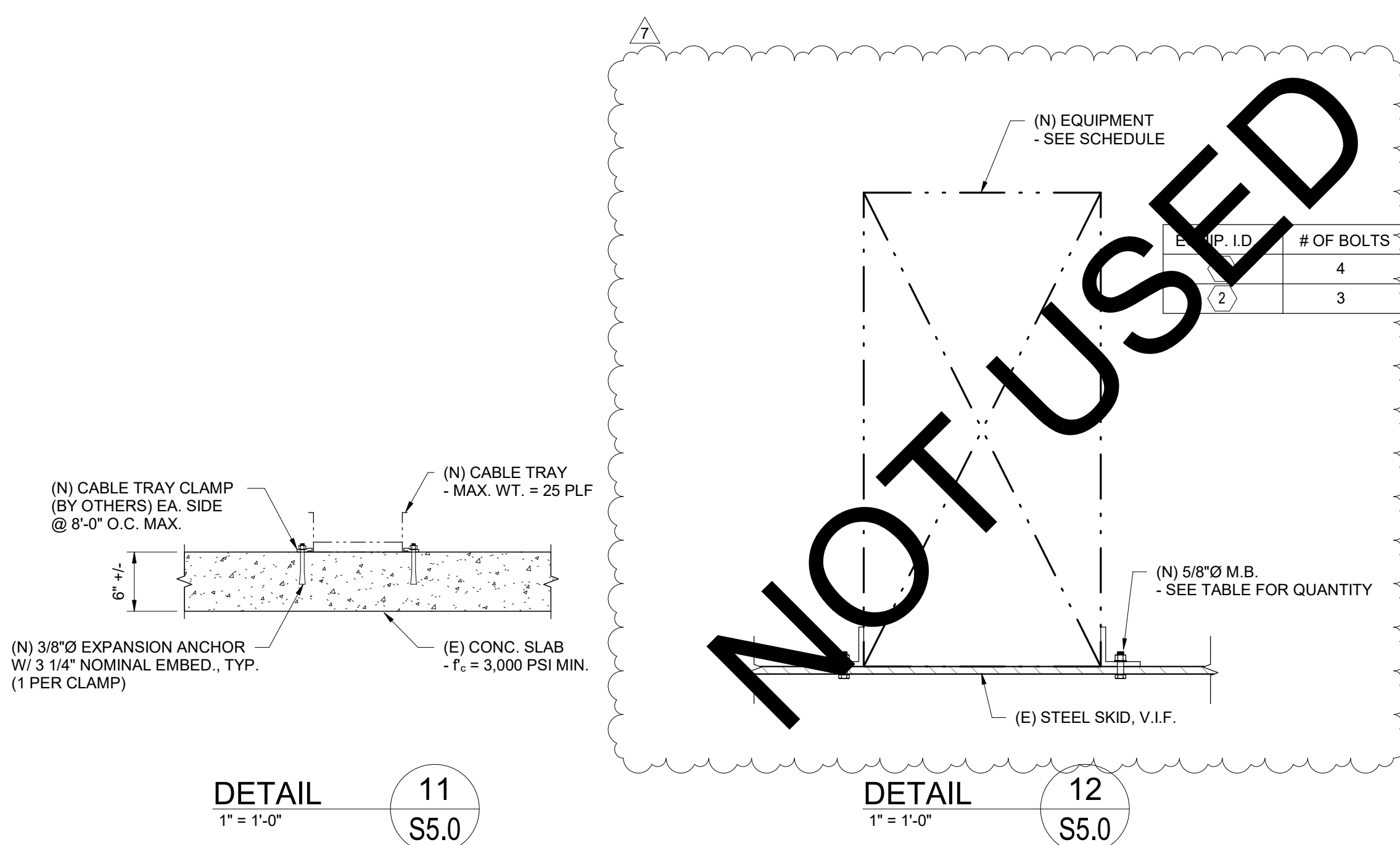
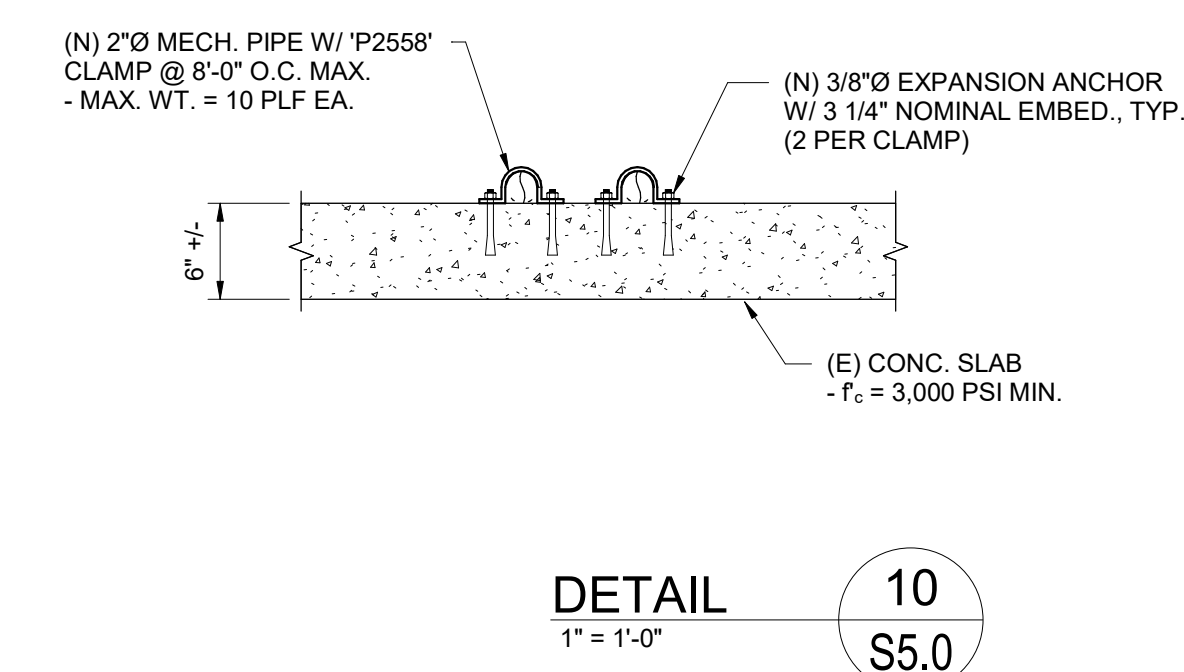
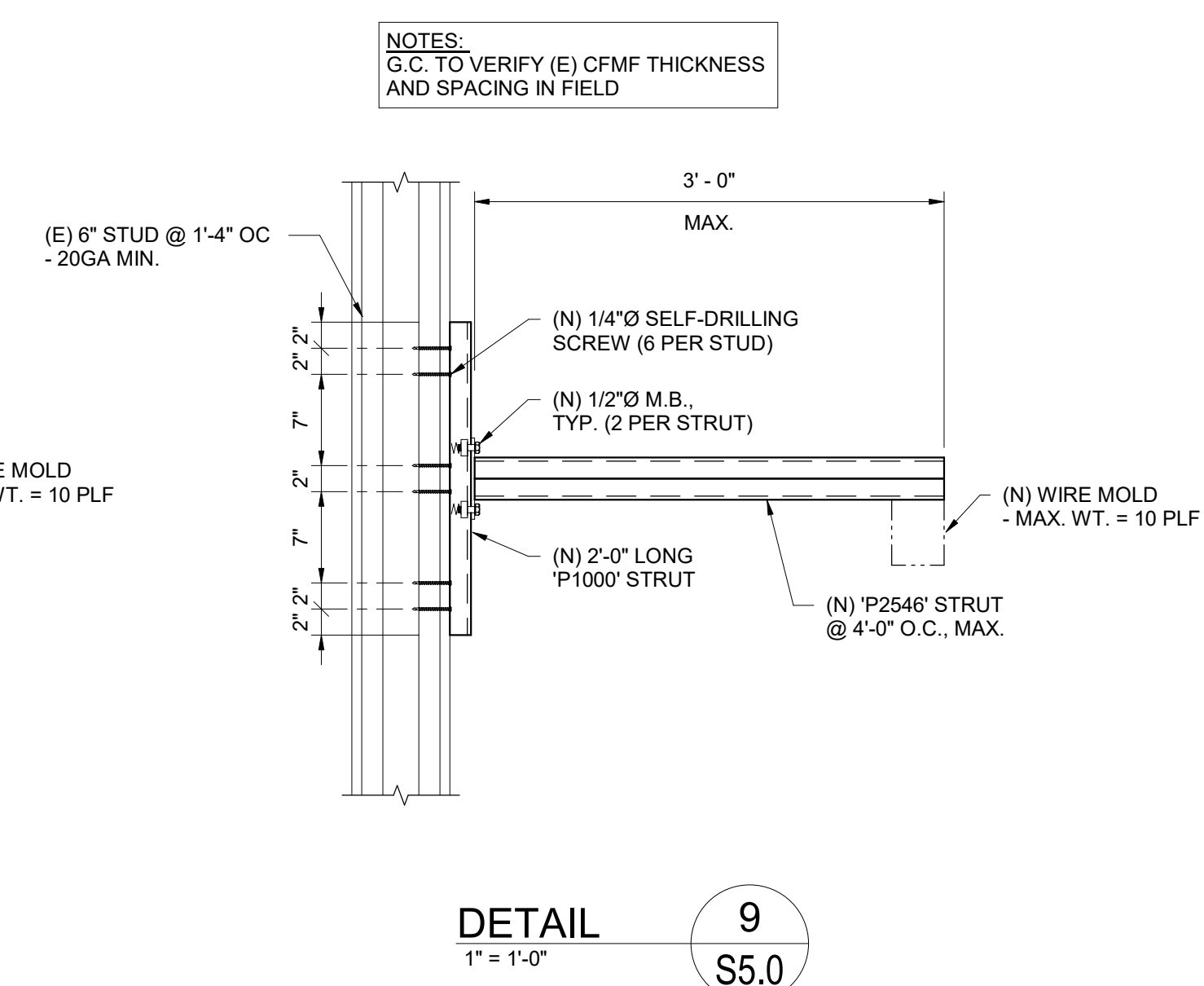
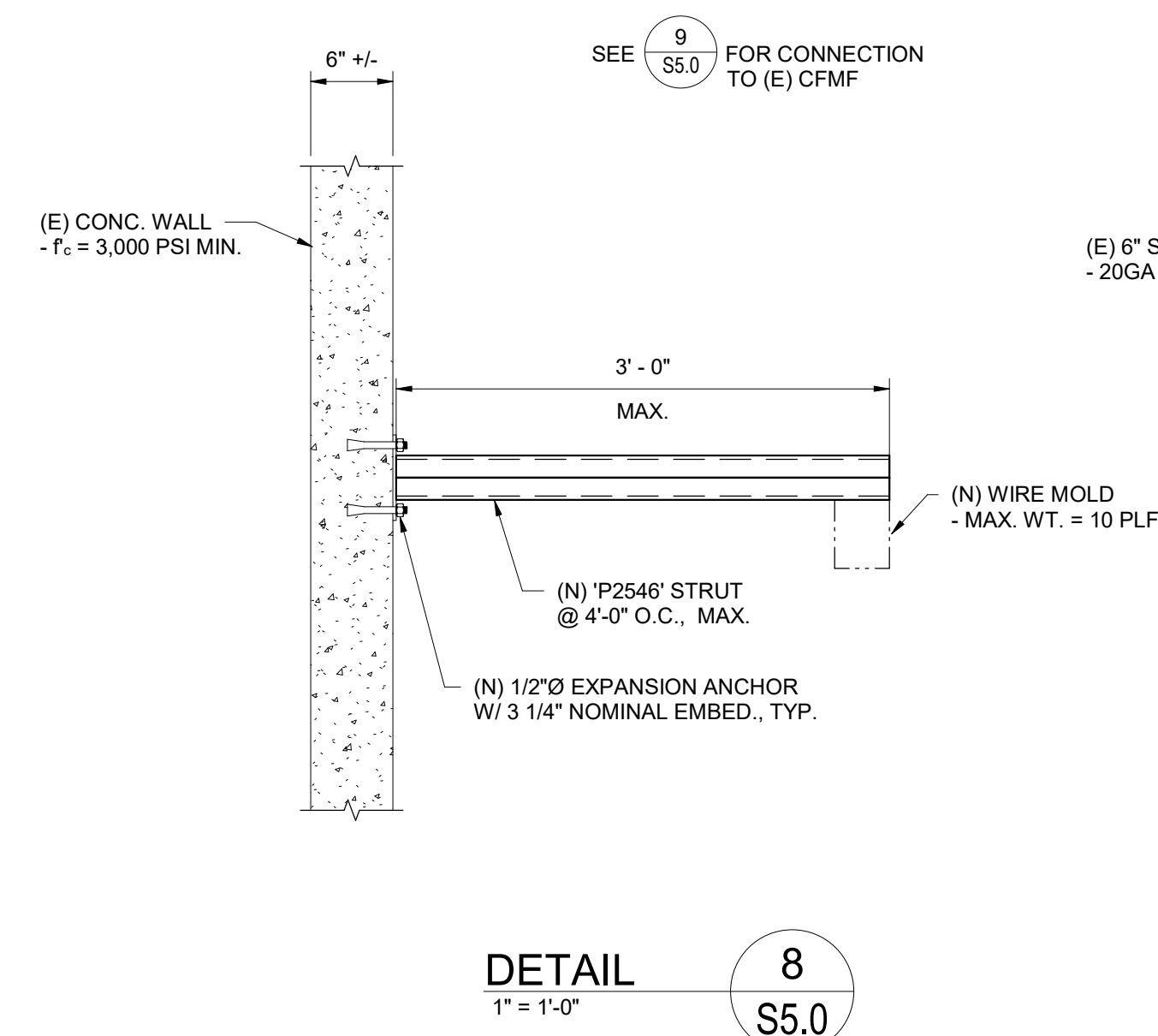
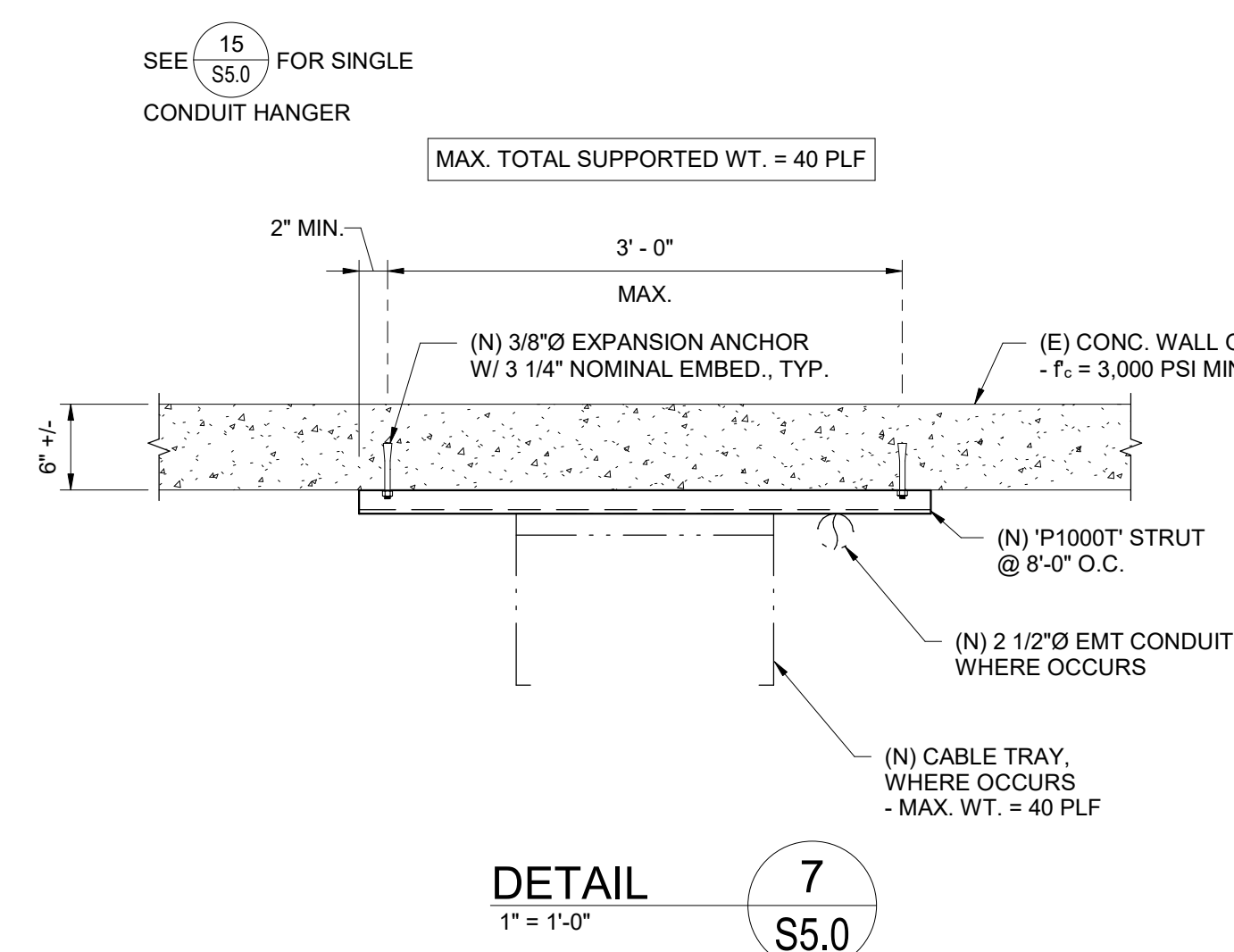
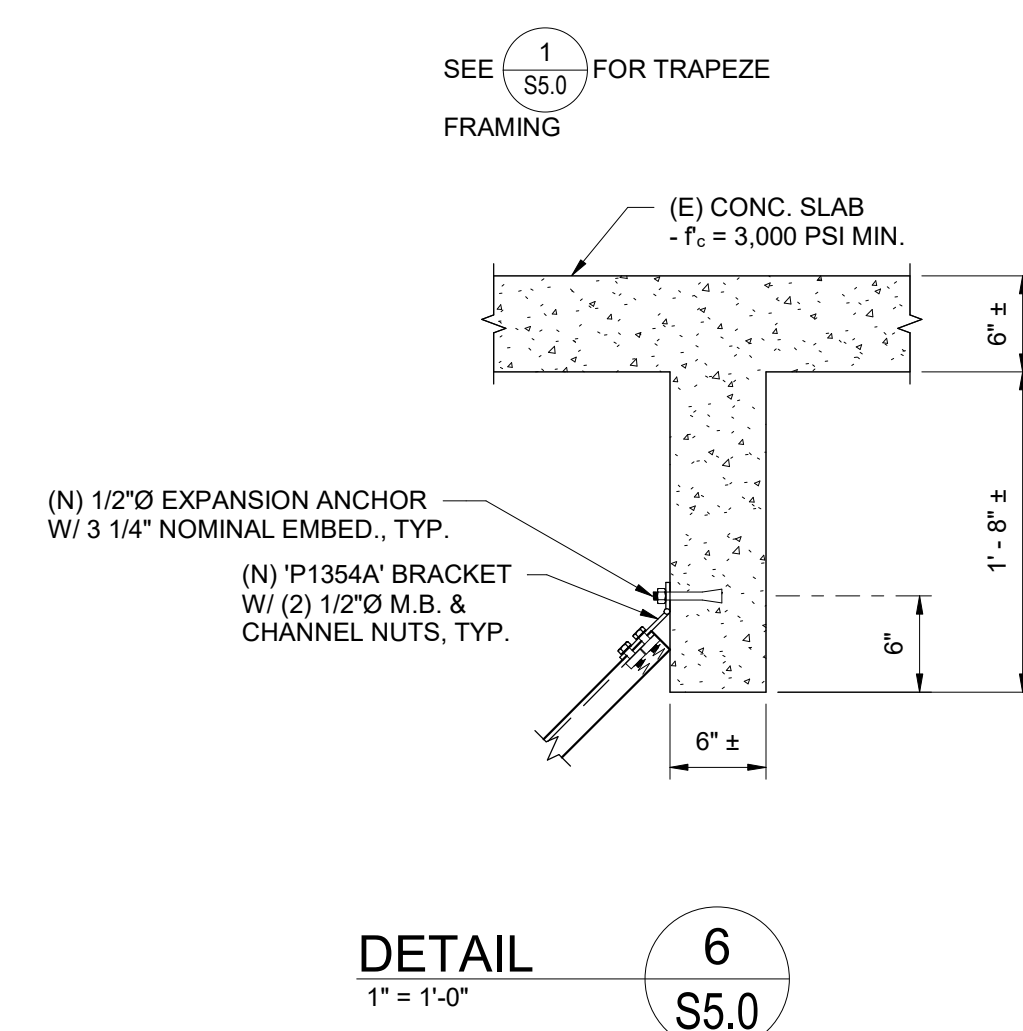
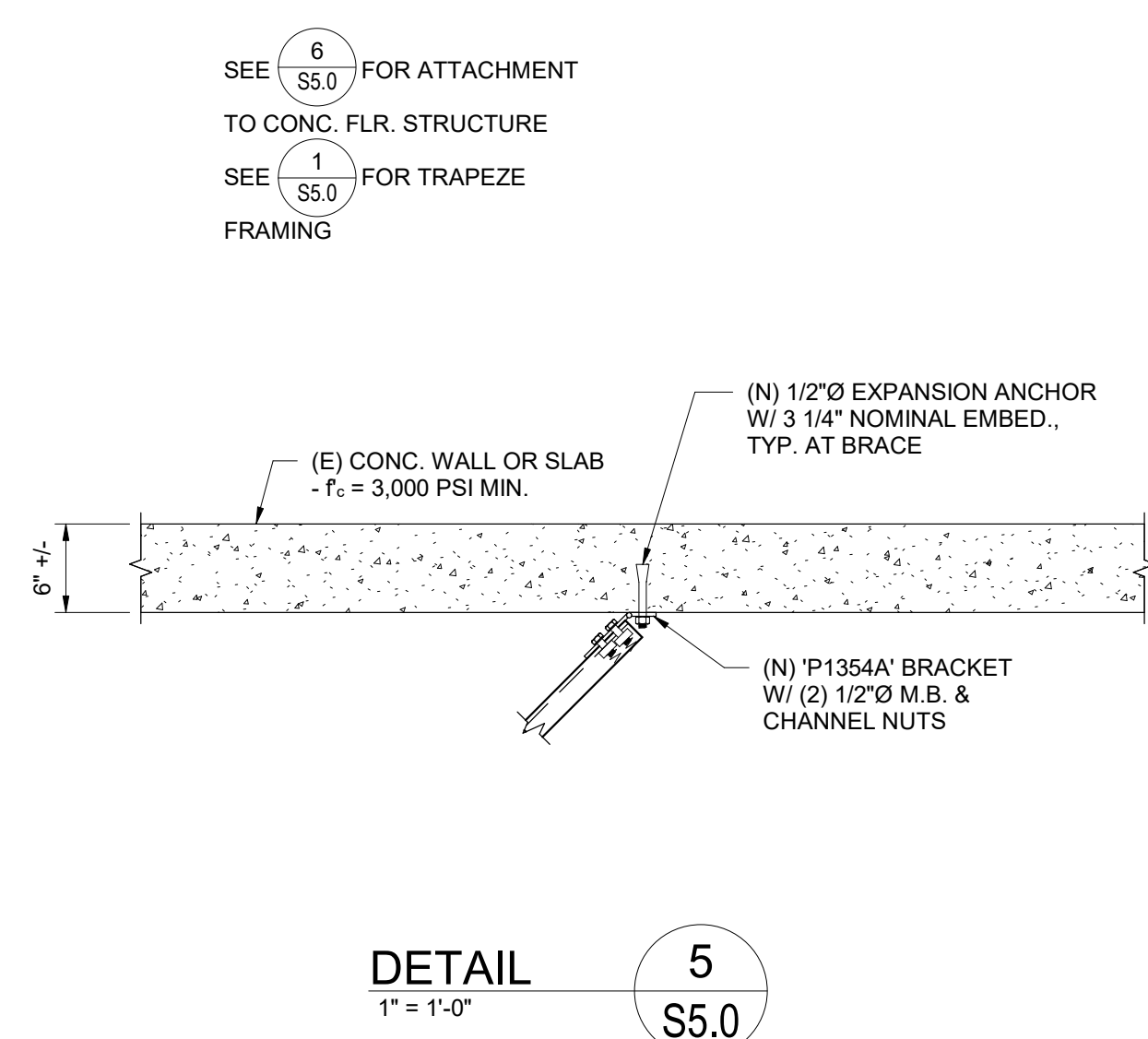
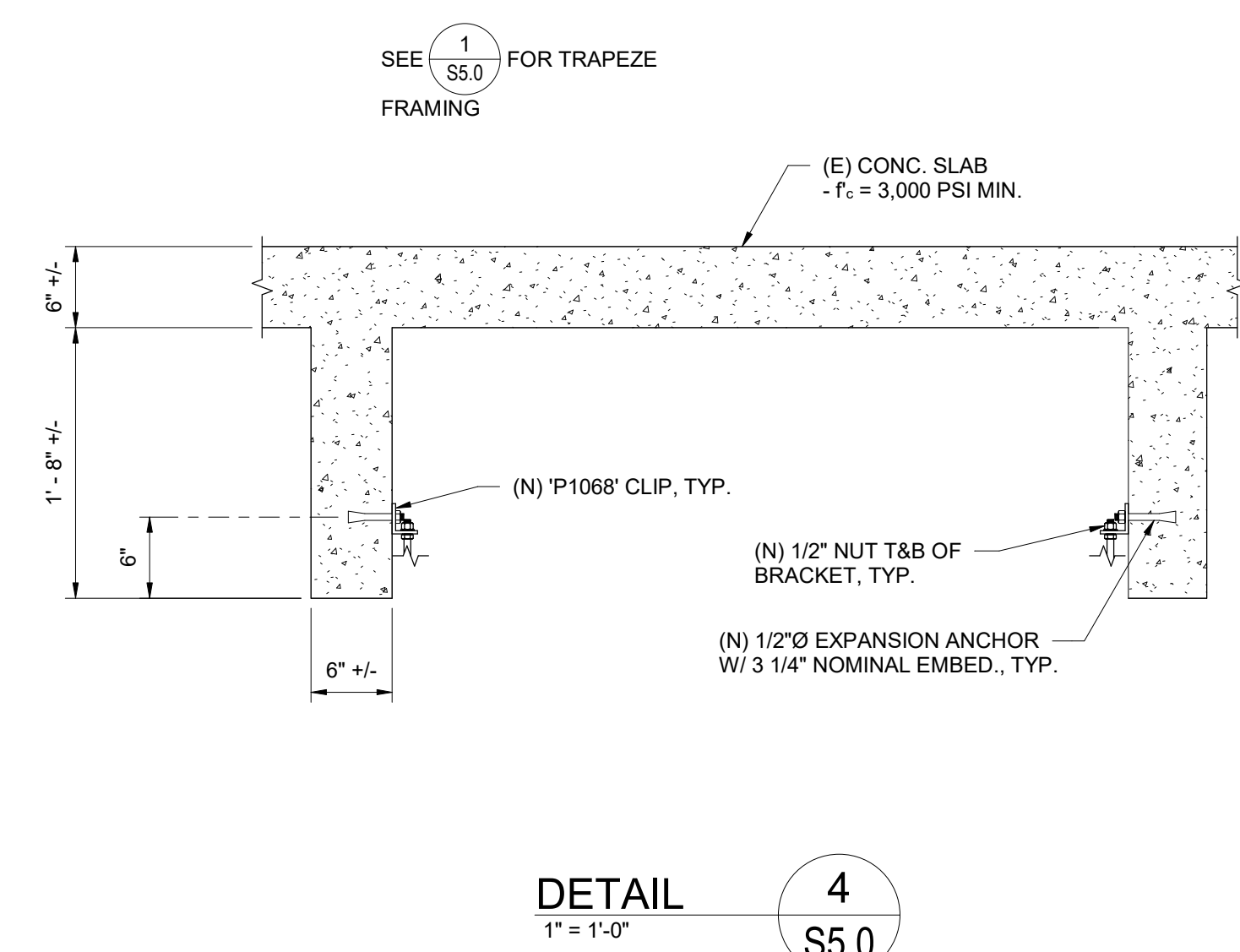
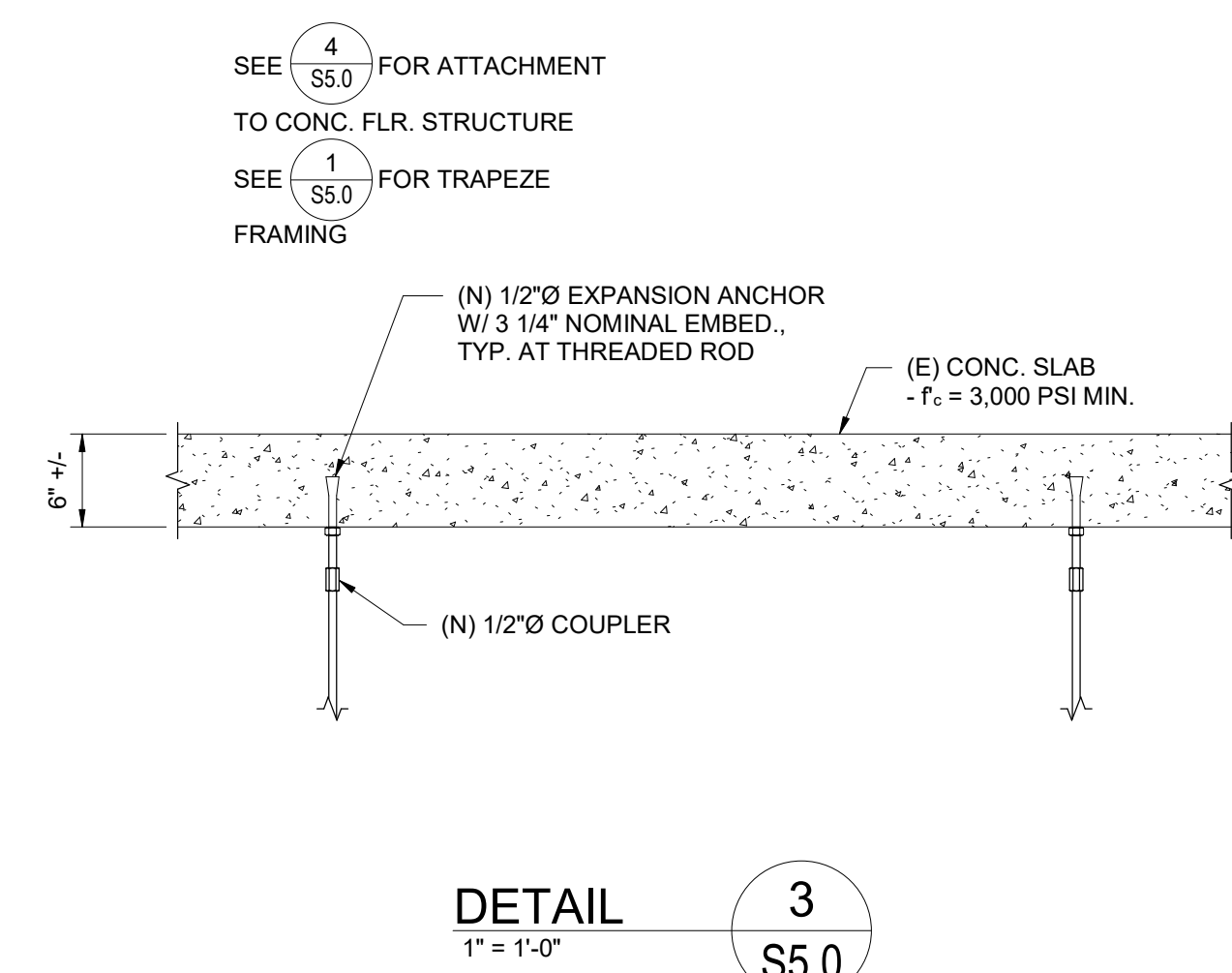
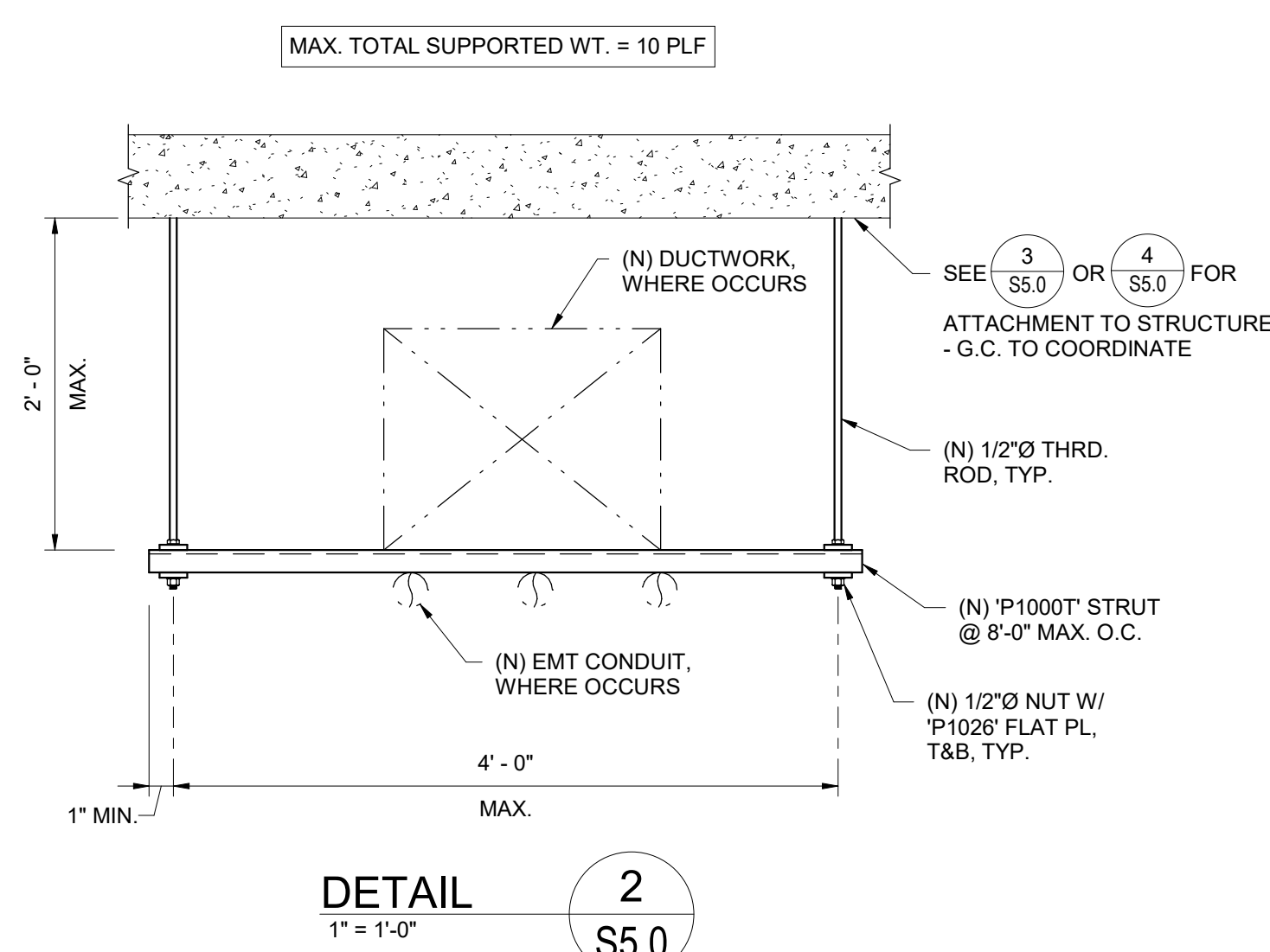
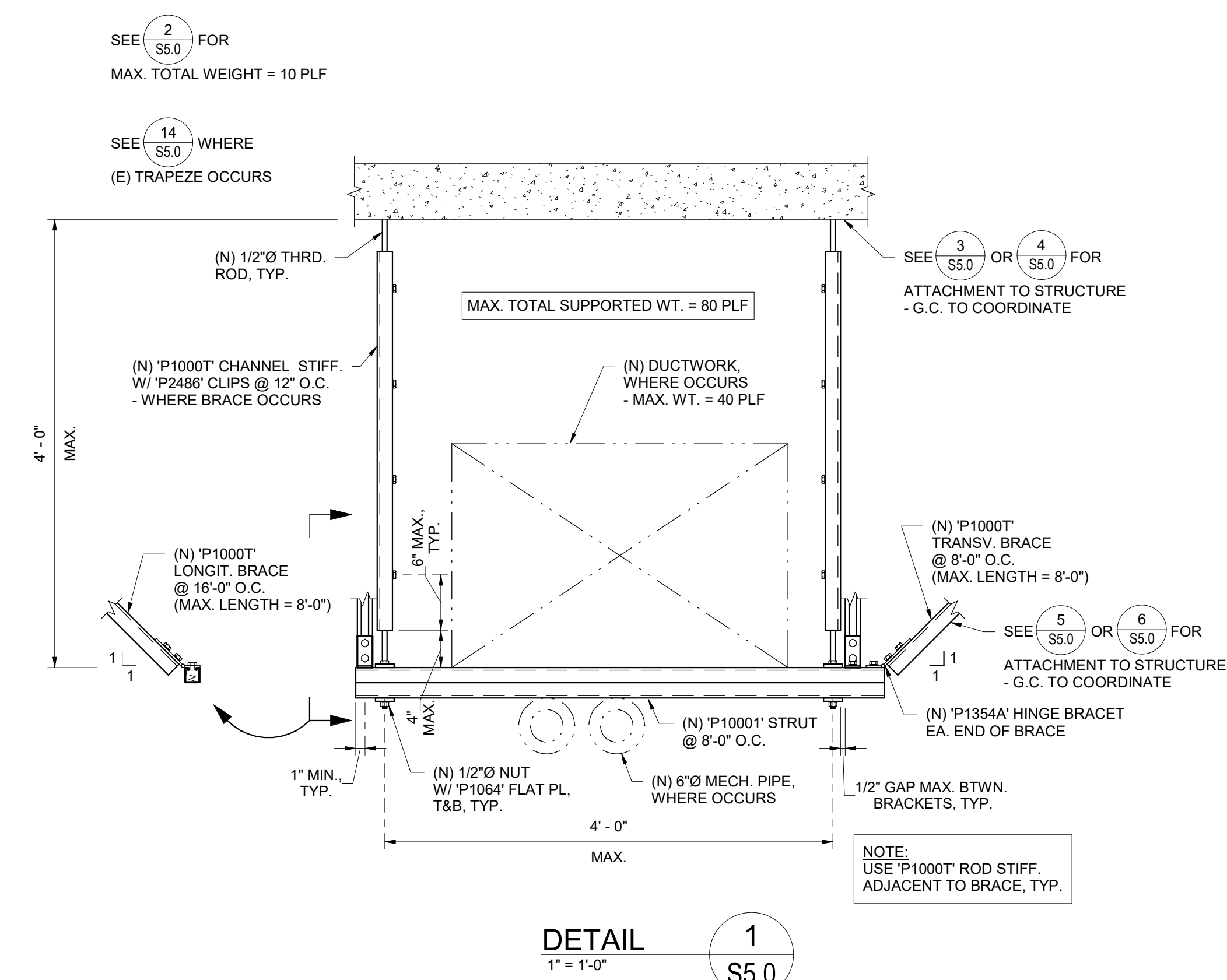
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GRAPHIC SCALE: 1/64" = 1'

0 64 128 196 256 320

KEY PLAN

REV	DESCRIPTION	DRN	CHK	APP	DATE
C	100% DESIGN REVIEW	SMP	DJS		25 FEB 28
D	ISSUE FOR BIO REVIEW	SMP	DJS		25 MAR 10
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION	SMP	DJS		25 DEC 03



SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 3/9/2026	PRSE: 26-058
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or ESH programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

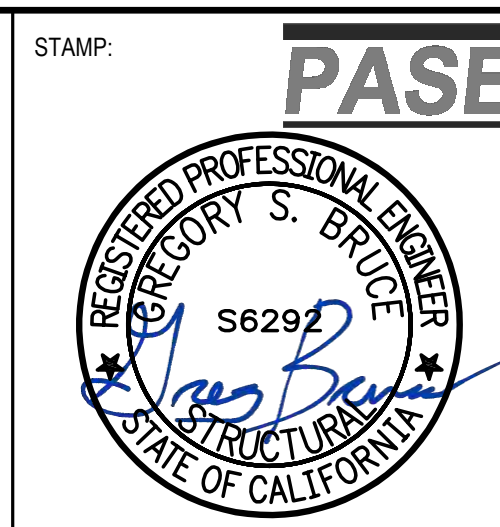
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ENGR	DJS	DATE	APPROVAL/S	DATE
DES	DJS	03/26/2005	Approver	
QAPP	DJS	03/26/2005		
		03/26/2005		



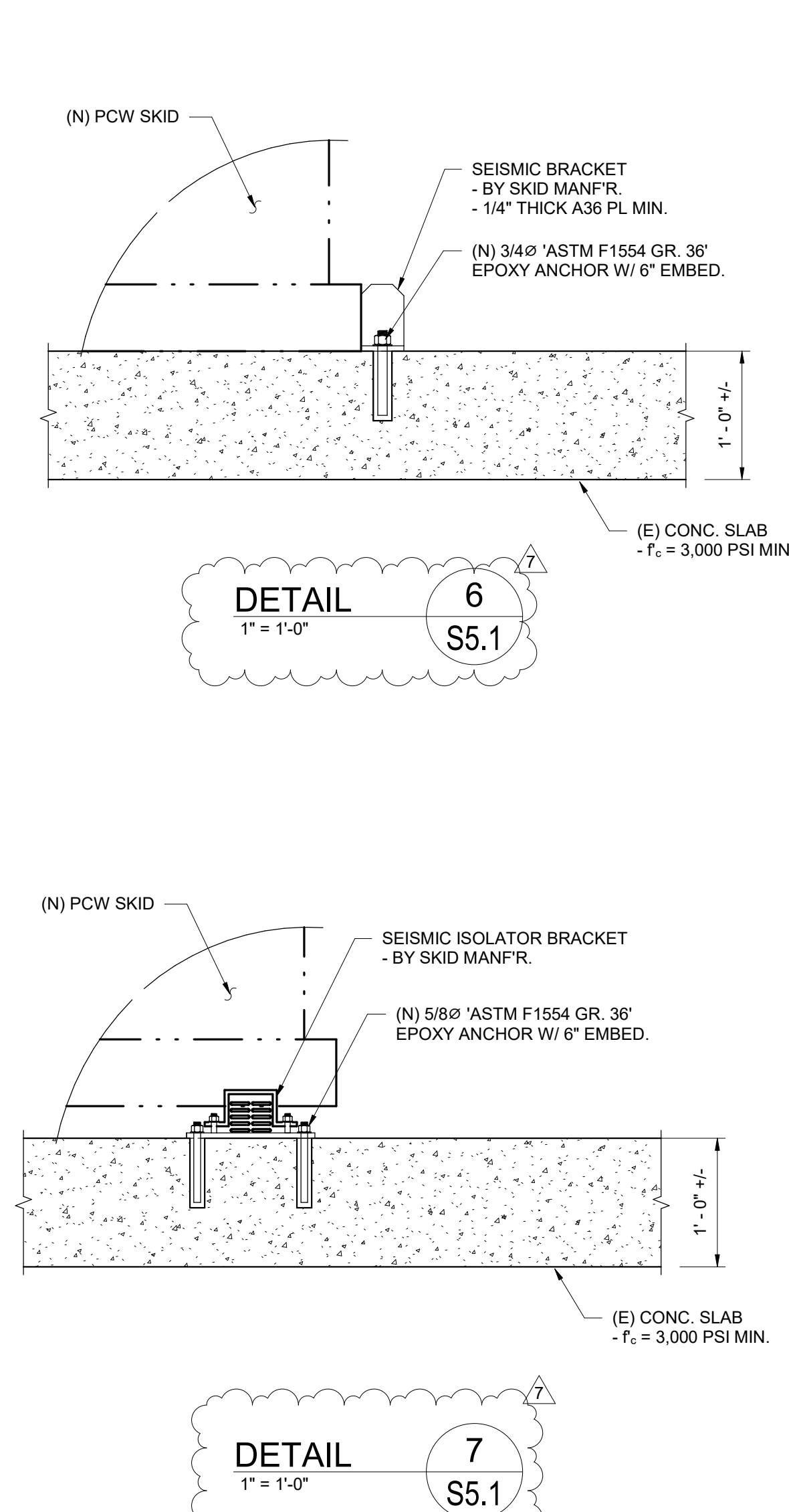
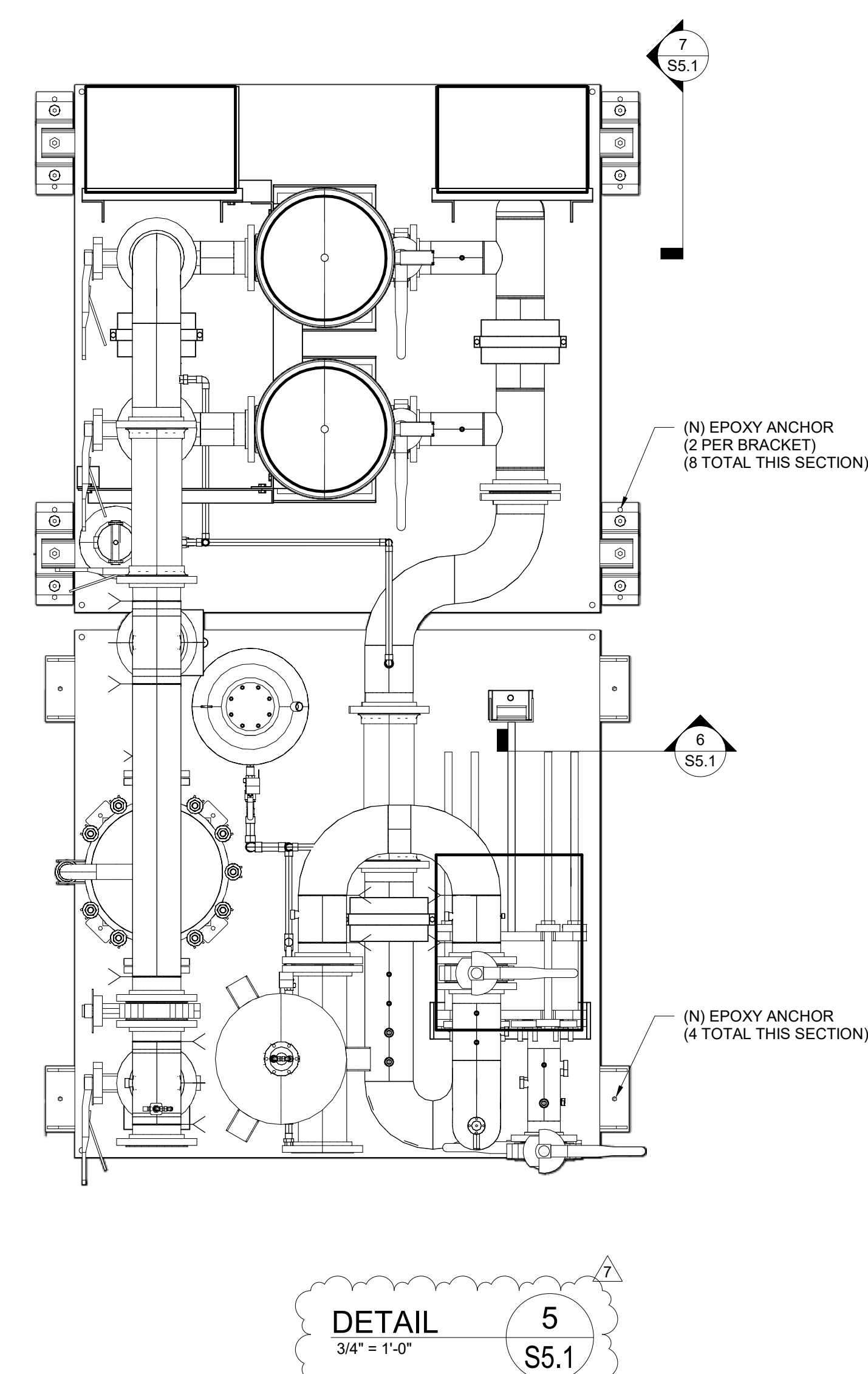
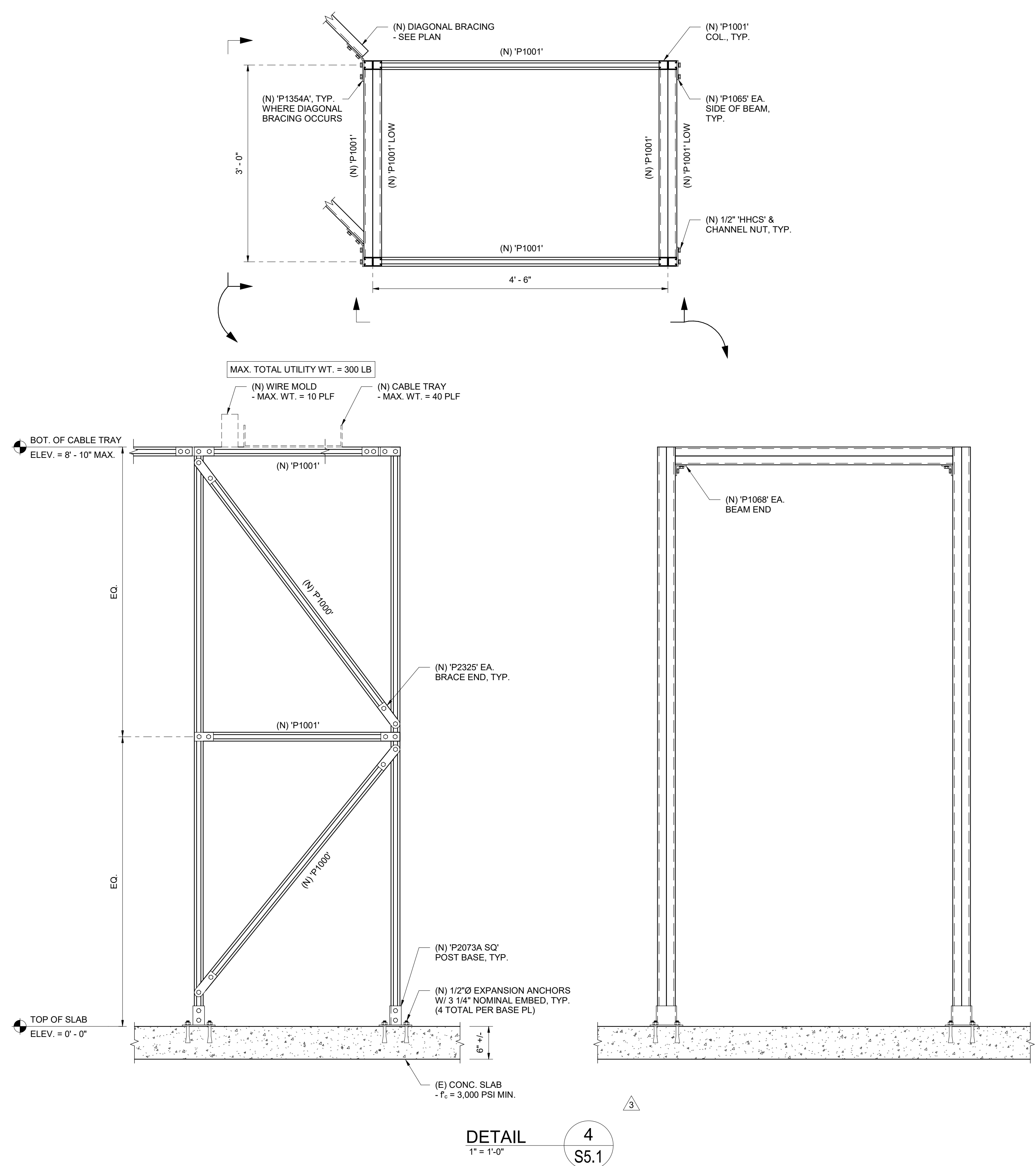
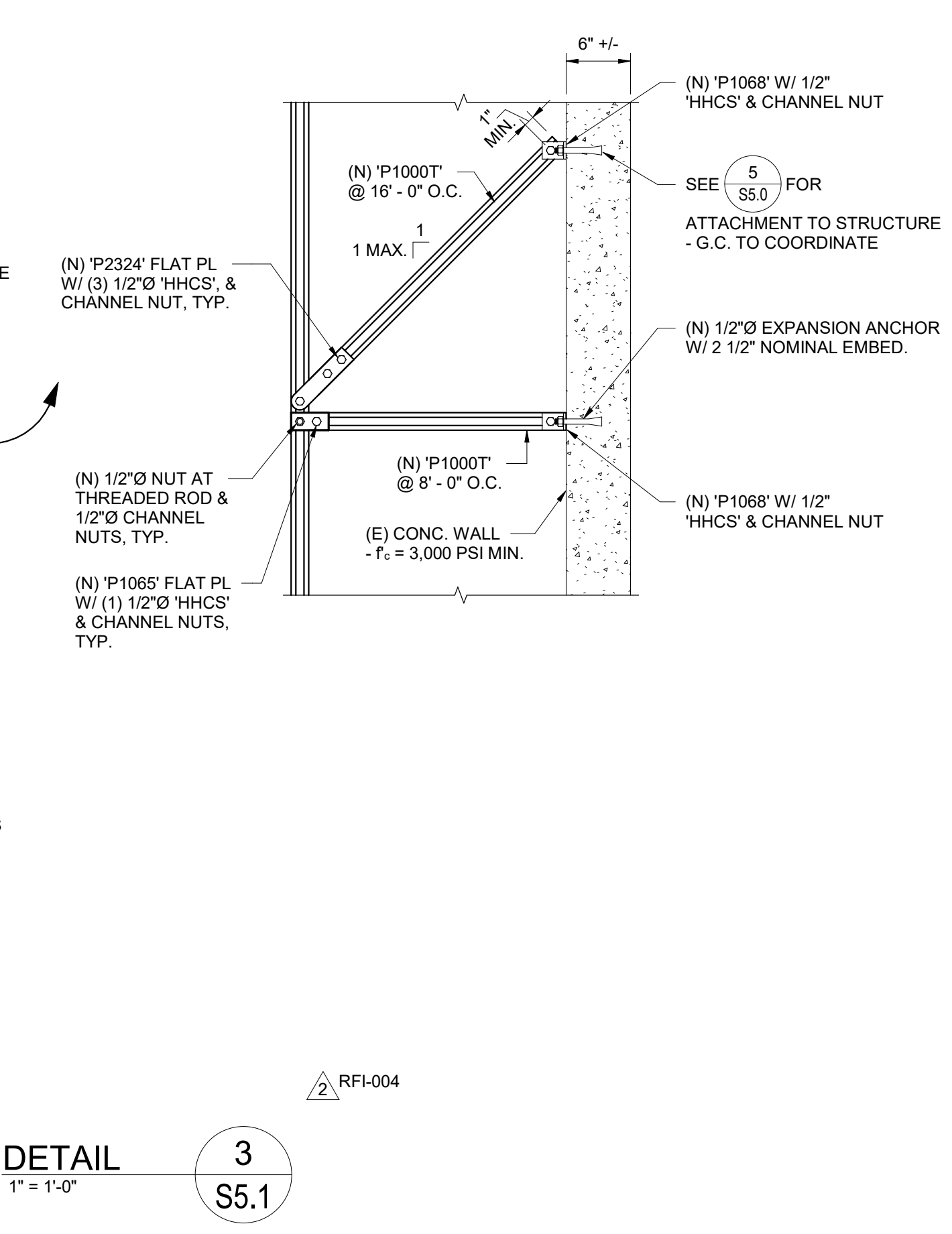
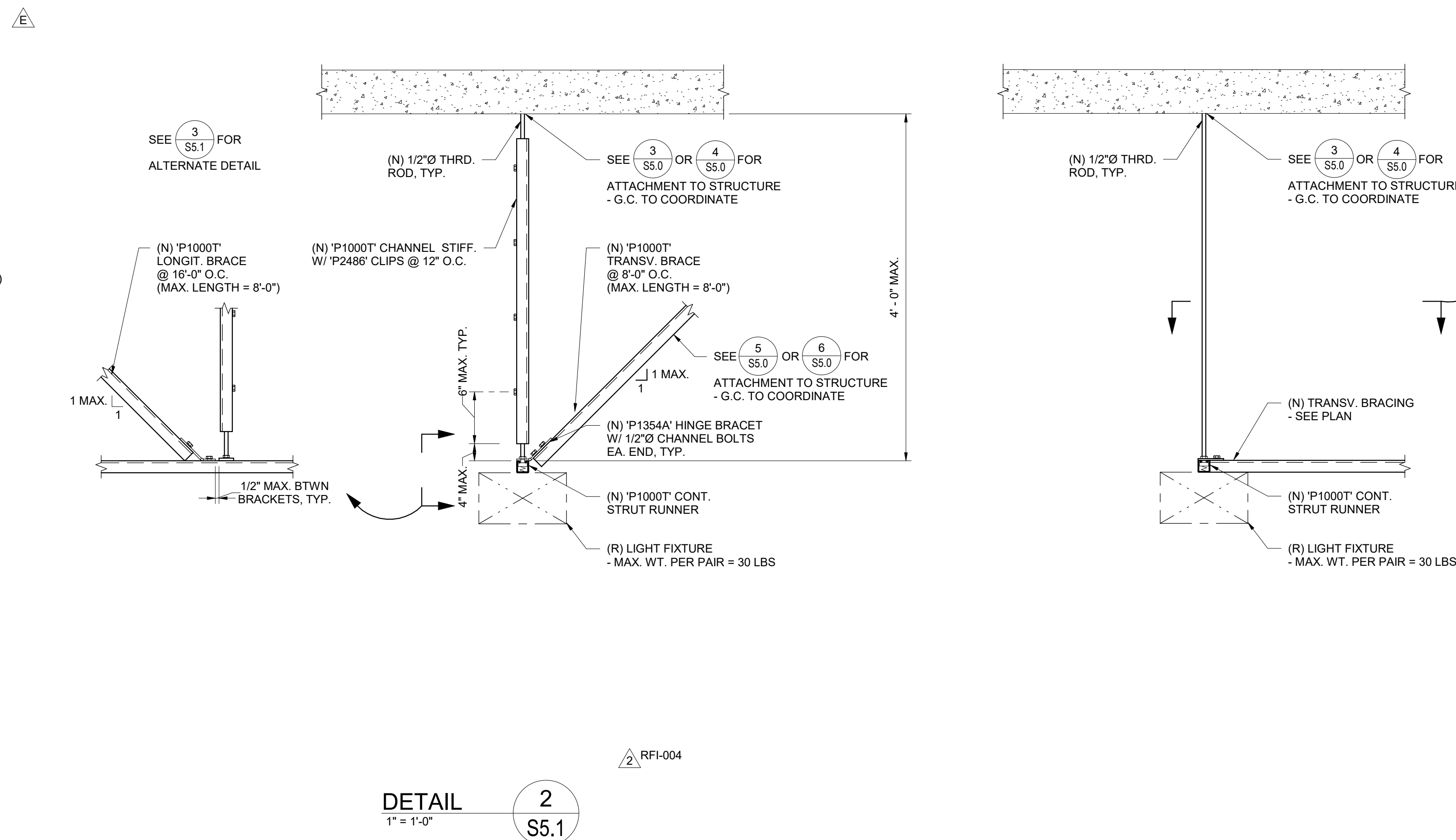
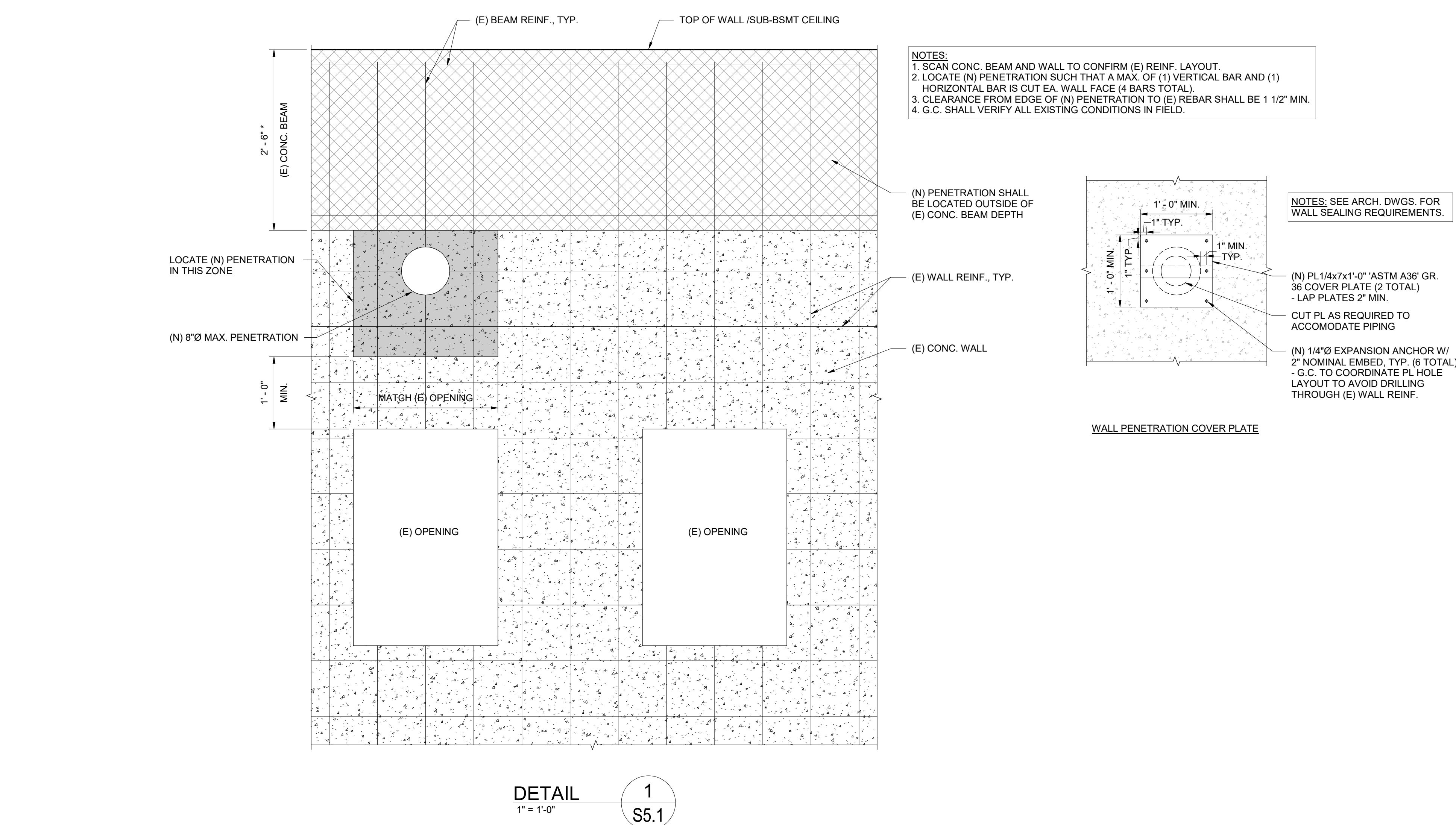
U.S. DEPARTMENT OF
ENERGY

OTHER DATA HEREIN PROVIDED

	B950 LCLS-II-HE:EI STRUCTURAL STRUCTURAL DETAILS
	DRAWING NUMBER B0950-S05-0318

REVISION	
6	E

REV	DESCRIPTION	DRN	CHK	APP	DATE
2	RFI-004 - XPP LIGHTING	SMP	DJS		25 JULY 28
3	BCS RACK ADDITION	SMP	DJS		25 NOV 19
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION	SMP	DJS		25 DEC 03



SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 3/9/2026	PRSE: 26-058
Revised Authorization	☒ No ☐ Yes
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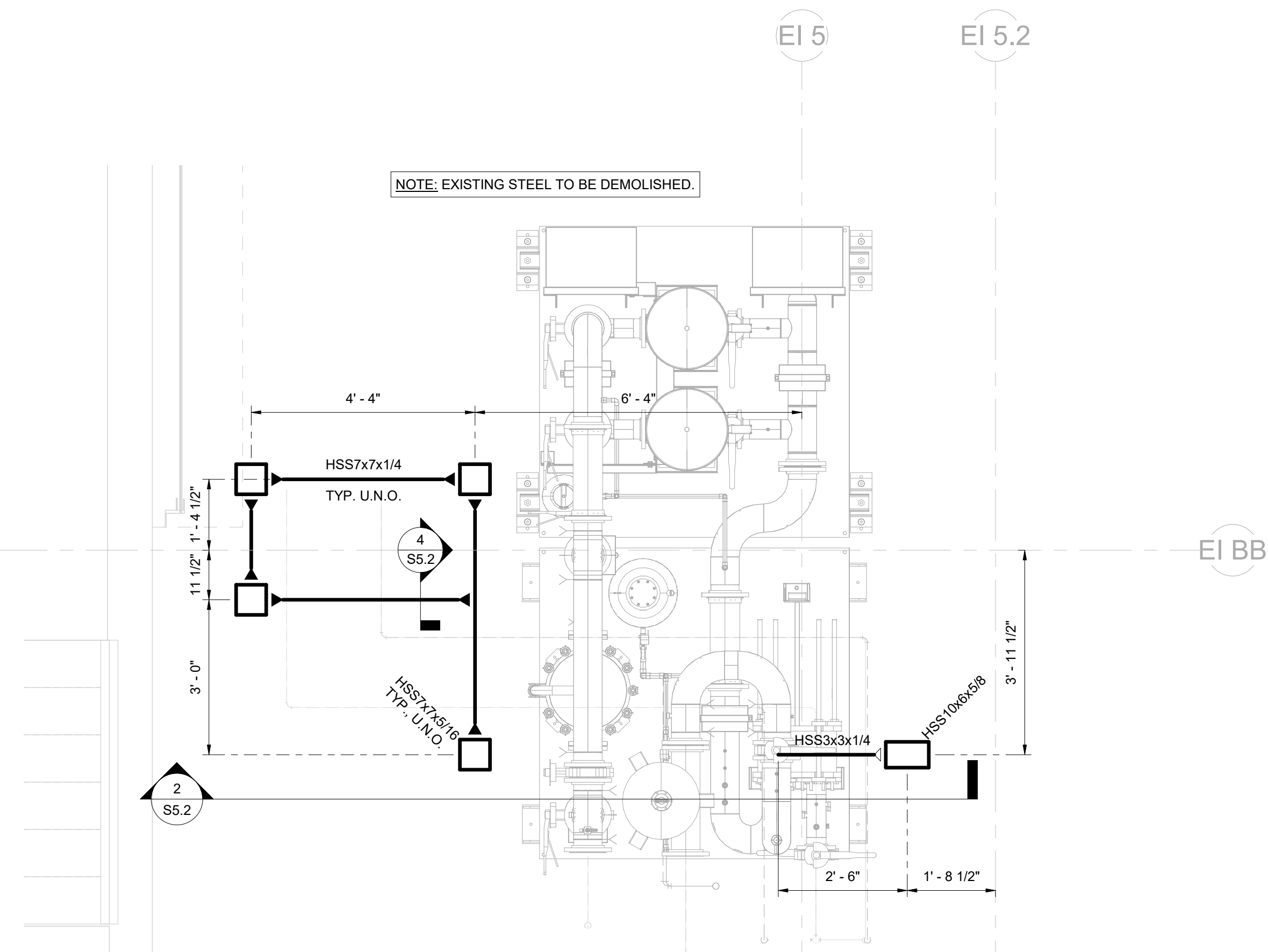


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ENGR	DJS	DATE	APPROVALS	DATE	
DES	DJS	03/06/2025	Approver		
CHKR	DJS	03/06/2025			

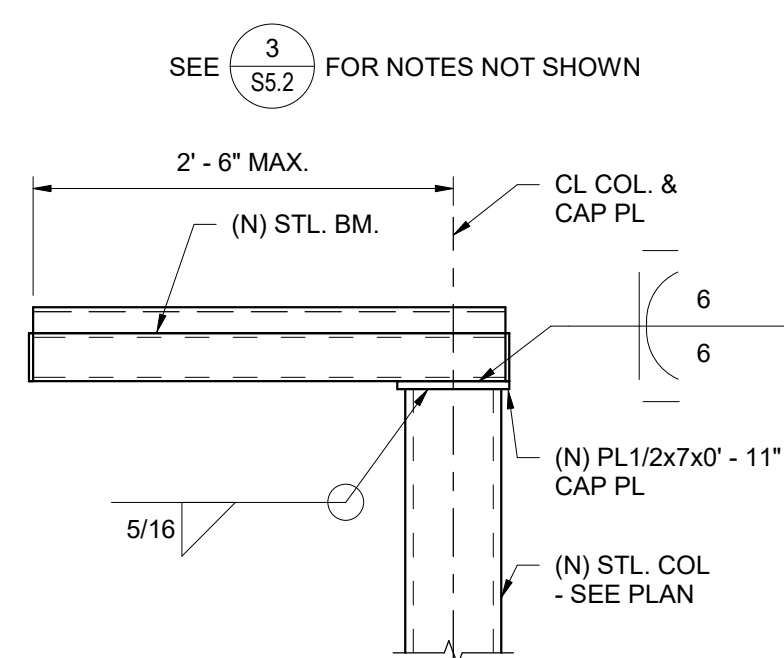
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REVISION	
6	E

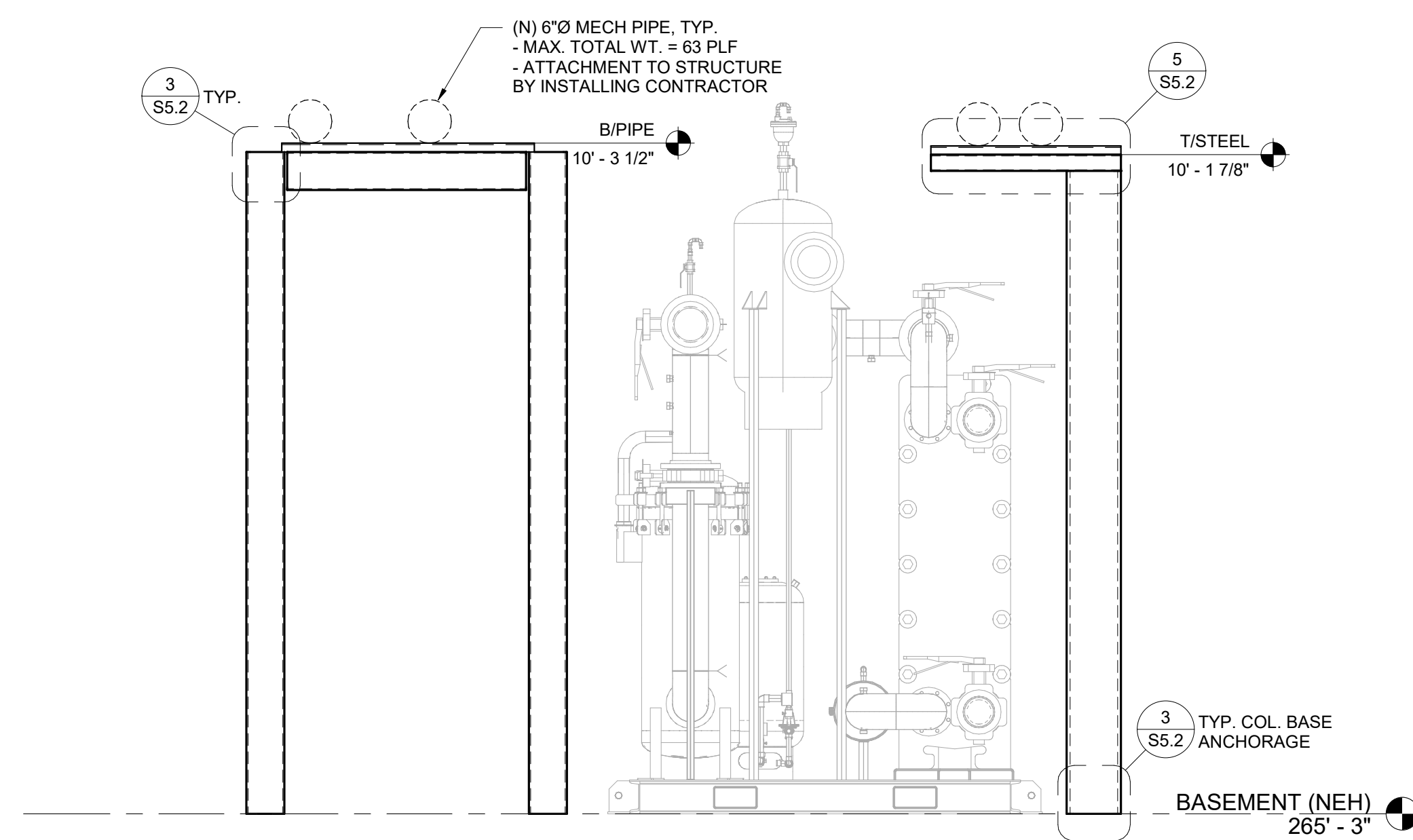
REV	DESCRIPTION	DRN	CHK	APP	DATE
C	100% DESIGN REVIEW	SMP	DJS		25 FEB 28
D	ISSUE FOR BIO REVIEW	SMP	DJS		25 MAR 10
7	BULLETIN 4.2 - PCW SKID REPLACEMENT - INSTALLATION	SMP	DJS		26 JAN 19



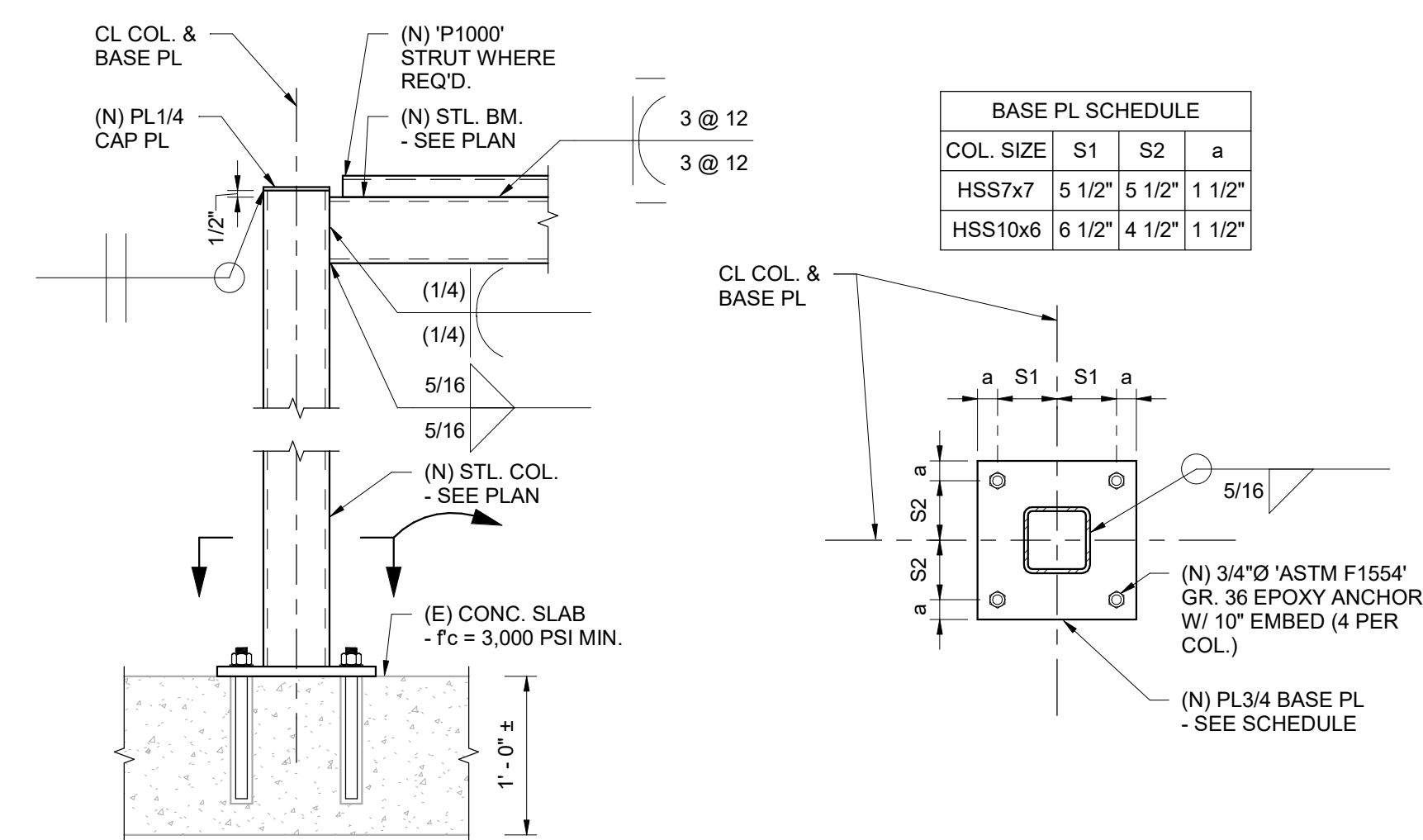
DETAIL 1
1/2" = 1'-0"



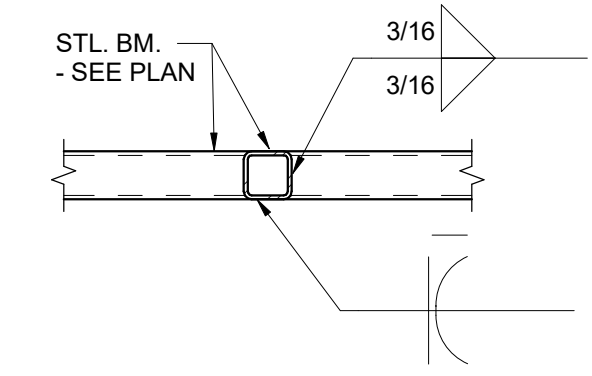
DETAIL 5
1" = 1'-0" S5.2



DETAIL 2
S5.2
1/2" = 1'-0"



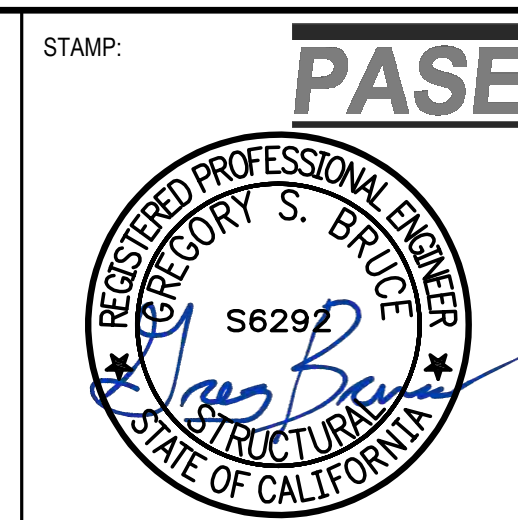
DETAIL 3
S5.2
1" = 1'-0"



DETAIL
1" = 1'-0"

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 3/3/2026	PRP#: 26-058
Revised Authorization	☒ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and IBC program requirements. The adoption of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or ESH programs.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.



SCALE: As indicated		DO NOT SCALE DRAWING		B0950.2024.03	
				U.S. DEPARTMENT OF ENERGY	
THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF STANFORD UNIVERSITY / SLAC					
		DATE		DATE	
ENGR	DJS	02/26/2025			
DES	DJS	03/26/2025			
CHDR	DJS	03/26/2025		APPROVALS 	

B950 LCLS-II-HE:EI STRUCTURAL STRUCTURAL DETAILS	
DRAWING NUMBER B0950-S05-0320	

S5.2

REVISION
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